TM Forum Standard

Resource Process Decompositions

GB922

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General Information

This document contains the text of the Business Process Framework for context and more information please refer to “GB991 Core Frameworks Concepts and Principles” guidebook.

Preface

eTOM Business Process Framework

The eTOM or Business Process Framework is a reference framework for categorizing all the business activities used by an enterprise involved in delivering on-line Information, Communications and Entertainment services. This is done through definition of each area of business activity, in the form of process components or Process Elements that can be decomposed to expose progressive detail. These process elements can then be positioned within a model to show organizational, functional and other relationships, and can be combined within process flows that trace activity paths through the business.

The eTOM framework can serve as the blueprint for standardizing and categorizing business activities (or process elements) that will help set direction and the starting point for development and integration of Business and Operations Support Systems (BSS and OSS respectively). An important additional application for the eTOM framework is that it helps to support and guide work by TM Forum members and others to develop their solutions. It provides an industry-standard reference point, when considering internal process reengineering needs, partnerships, alliances, and general working agreements with other enterprises, and for suppliers into such enterprises, the eTOM framework outlines potential boundaries of process solutions, and the required functions, inputs, and outputs that must be supported by process solutions.

Introduction

This document contains the main text of eTOM for this domain. It is comprised of decompositions and descriptions of the process elements that make out this domain.

The focus of the Business Process Framework is on the business processes used by service providers, the linkages between these processes, and the use of Market, Product, Customer, Service, Resource, Business Partner, and Enterprise Management related information by multiple processes.

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The decomposition starts with a top level process which is a Core Process that details the activity that takes place and describes all subsequent activities that occurs. Thereby this Core Process includes all of the information regarding the activity. Lower level processes also known as Task Processes are then describing a unique path of action and further detail the specific process. This is then repeated to lower level of granularly of up to seven levees.

At the start of each Level 2 process a diagram details all of its sub processes is placed. This diagram showers Level 3 process and the level 4 process below them in boxes where the same level boxes are side by side and the their child processes in side each box. Process that have further decomposition include another box in box diagram of level 5, 6 and 7 process if existing.

Process Associations and IDs

To aid understanding and navigation, each process described here has an associated indication.

Processes have been given a process ID throughout this document. The format of the numbering scheme work as follows:

.b.x.c.d.e where

b: This digit is used to identify the originator of the process It is set to 1 if the process is the TMF original one, to 2 otherwise so if a company chooses to extended the Business Process Framework it can do so keeping the original numbering and extend as needed.

x: Digit representing Domain

c: Digit representing Level 2 process

d: Digit representing Level 4 process

e: Digit representing Level 5 process

The unique identifier thus associated with individual processes can thus be used to assist in locating individual processes.

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Typographic Conventions

- A Business Process is represented by a *decorator* in form of an arrow located to the top right. A red dot decorator with the initial TMF indicates that this is a TM Forum Business Process. The color yellow background hints to the business layer.

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Glossary

Name	Description
ABE	Aggregate Business Entity
BE	Business Entity
eTOM	enhanced Telecom Operation Map (TMF Framework)
ODA	Open Digital Architecture
SID	Shared Information Data model (TMF Framework) is the previous name of Information Framework
TAM	Telecom Applications Map (TMF Framework) is the previous name of Application Framework

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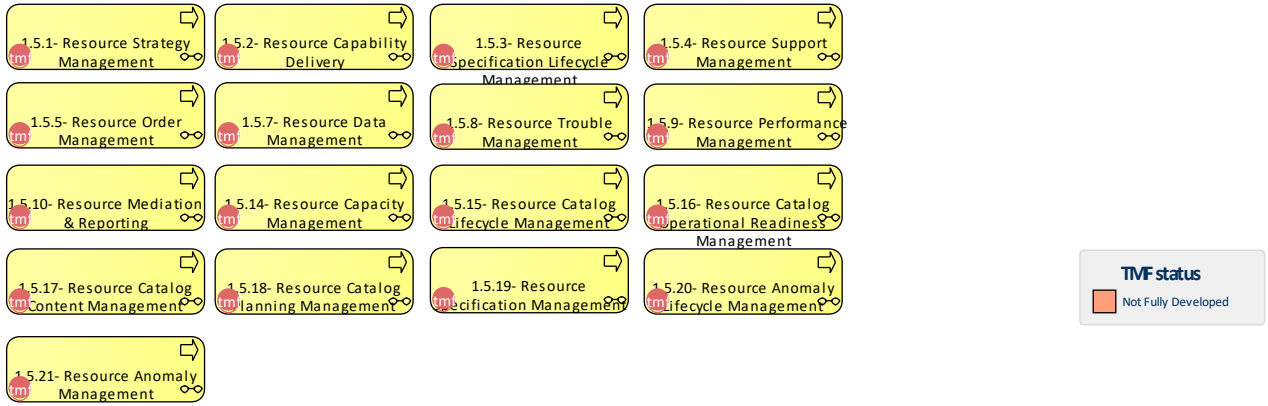
Resource Domain



Resource Domain Main Diagram

Resource Strategy Management

Process Identifier: 1.5.1

Hierarchy Level: 2

Brief Description:

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Develop resource strategies, policies and plans for the enterprise, based on the long-term business, market, product and service directions of the enterprise.

Extended Description:

Resource Strategy Management processes develop resource strategies, policies and plans for the enterprise, based on the long-term business, market, product and service directions of the enterprise These processes understand the capabilities of the existing enterprise infrastructure, capture the requirements on the infrastructure based on market, product and service strategies of the enterprise', manage the capabilities of the suppliers and partners to develop and deliver new resource capabilities, and define the way that new or enhanced infrastructure may be deployed.

Research & analysis is performed to determine resource targets as well as strategies to reach the defined targets. This research may be obtained from the external market and resource suppliers and partners, or developed internally through research programs and activities, or leverage of other internal knowledge, such as marketing data. A key input to the resource strategy arises from the enterprise's market, product portfolio and service strategy and forecasts. The latter may include AI based predictions towards resource requirements. A focus is placed on the expansion of the existing resource capabilities and the identification of new resource capabilities required.

These processes deliver and develop annual and multi-year resource plans in support of services, products and offers that include volume forecasts, negotiation for required levels of resources, gaining resource development and management as well as supply chain commitment and executive approval for the plans. In addition, the processes may also develop more short-term resource plans in order to optimize near-term balance between supply and demand of resources.

Forecasting of resource demand, and capturing of new opportunities, are both essential to ensure that the enterprise can construct and implement the resources necessary for the future needs of their customers and potential customers.

These processes define the resource implementation standards sought, key new resource capabilities required, resource support levels and approaches required, resource design elements to be developed, as well as resource cost parameters and targets. These processes also define the policies relating to technical resources and their implementation.

This process was renamed in 24.0 Old Name was Resource Strategy & Planning

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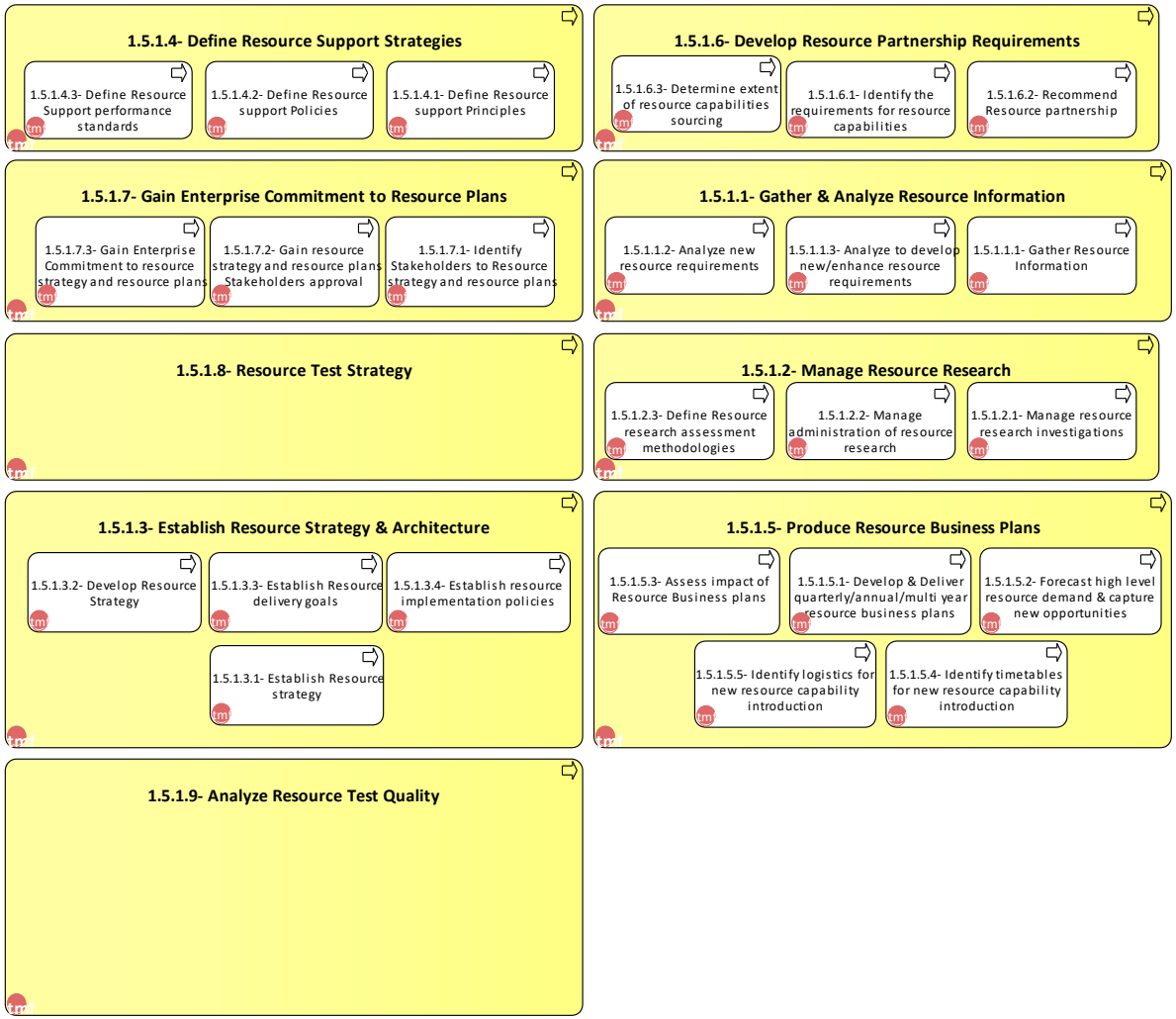

Figure 1: Resource Strategy Management Children

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Gather & Analyze Resource Information

Process Identifier: 1.5.1.1

Hierarchy Level: 3

Brief Description:

Research and analyze customer, technology, competitor and marketing information to identify new resource requirements and industry resource capabilities and availability

Extended Description:

The Gather & Analyze Resource Information processes research and analyze customer, technology, competitor, internal resource utilization trends and marketing information to identify new resource requirements and industry resource capabilities and availability. These processes undertake the necessary analysis to identify potential opportunities, compare current capabilities with the identified opportunities, and as a result of the analysis develop new resource requirements or enhancements to existing requirements. The new or enhanced resource requirements include an analysis of the customer value proposition.

These processes include the establishment and management of relationships with external providers of resource information, and the management of internal groups used for providing resource information.

A key source of input to this analysis is derived from the business, marketing, service and product strategy and planning processes.

In some cases these processes may leverage information through the use of advanced analytics tools or AI for fast, accurate and proactive forecasting.

Gather Resource Information

Process Identifier: 1.5.1.1.1

Hierarchy Level: 4

Brief Description:

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Establish and Management of external Providers and internal resources for providing resource information

Extended Description:

Gather Resource information processes focus on establishment and Management of relationships with external providers of resource information, and management of internal groups used for providing resource information. The gathering of resource information may be supported through automation and whereby resources can share their information with other resources through a common platform. In addition, relationship management with internal/external providers of resource information may leverage automation functions

Analyze new resource requirements

Process Identifier: 1.5.1.1.2

Hierarchy Level: 4

Brief Description:

Analyze to identify new resource requirements, resource capabilities and availability

Extended Description:

Analyze new resource requirements processes focus on research and analyze customer, technology, competitor, internal resource utilization trends and marketing information to identify new resource requirements and industry resource capabilities and availability. In some cases these processes may leverage information through the use of advanced analytics tools or AI for fast, accurate and proactive forecasting.

Analyze to develop new/enhance resource requirements

Process Identifier: 1.5.1.1.3

Hierarchy Level: 4

Brief Description:

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Develop new resource requirements, enhance to existing requirements

Extended Description:

Analyze to develop new/enhance resource requirements processes undertake the responsibility of necessary analysis to identify potential opportunities, compare current capabilities with the identified opportunities, and as result of these analysis develop new resource requirements or enhancements to existing requirements. Such analysis may include the use of predictive AI techniques to model future resource requirements. This process also include an analysis of the customer value proposition. Proactive resource planning between elements in the resource domain is a key in this process for fast and accurate development of resource requirements.

Orchestration between any elements within the Resource domain may be performed using the 1.8.5 Domain Orchestration process.

Manage Resource Research

Process Identifier: 1.5.1.2

Hierarchy Level: 3

Brief Description:

Manage internally driven research investigations and activities which are used to provide detailed technical assessment or investigation of new and emerging resource capabilities

Extended Description:

The Manage Resource Research processes manage internally driven research investigations and activities which are used to provide detailed technical assessment or investigation of new and emerging resource capabilities. The actual research activity may be totally internally delivered, or may be outsourced to external parties to perform the actual research activity. These processes manage the administration of the research activity, management of any research prioritization processes, define the assessment methodologies, administer any actual research contracts with external parties, and develop and manage any intellectual property and patent administration required for the resource research activity.

Note that the commercial arrangements with external parties are developed within the Party Offering Development & Retirement processes.

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Note that the actual legal arrangements, analysis and litigation surrounding intellectual property is managed within the Legal Management processes in the Enterprise Management process area.

Manage resource research investigations

Process Identifier: 1.5.1.2.1

Hierarchy Level: 4

Brief Description:

Manage internally driven resource research investigations and activities

Extended Description:

Manage resource research investigations processes manage internally driven research investigations and activities which are used to provide detailed assessment or investigation of new and emerging resource capabilities. The actual research activity may be totally internally delivered , or may be outsourced to external parties.

Manage administration of resource research

Process Identifier: 1.5.1.2.2

Hierarchy Level: 4

Brief Description:

Manage administration of resource research activity

Extended Description:

Manage administration of resource research processes are responsible for managing the administration of resource research activities, management of any research prioritization process, administer any actual research contracts with external parties, and develop and manage any intellectual property and patent

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administration required for the resource research activity. Actual commercial arrangements with external parties are managed with in Supply Chain Development and Management Processes. Actual legal arrangements and Intellectual property is managed with Legal Management Processes in Enterprise Management Process areas.

Define Resource research assessment methodologies

Process Identifier: 1.5.1.2.3

Hierarchy Level: 4

Brief Description:

Define Resource research assessment methodologies

Extended Description:

Not used for this process element

Establish Resource Strategy & Architecture

Process Identifier: 1.5.1.3

Hierarchy Level: 3

Brief Description:

Establish the resource strategies based on market trends, future products needs, technical capabilities, and addressing shortcomings in existing resource support

Extended Description:

The Establish Resource Strategy & Architecture processes proactively establish the resource strategies based on market trends, internal resource trends, future products needs, available and emerging resource capabilities, and addressing shortcomings in existing resource performance and operational

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support. These processes accept the changing resource requirements and develop appropriate resource strategies and strategic resource architectures based on the changing needs of the enterprise, and increasing expectations of the market. They establish realistic resource development and delivery goals for the enterprise and services to be delivered to the market. They establish implementation policies which act as a set of guidelines to be followed in the Resource Capability Delivery processes. They highlight shortcomings in existing resource performance and operational support arrangements and processes, including issues associated with resourcing and skill sets.

Major inputs to these processes are derived from the Gather & Analyze Resource Information and the Manage Resource Research processes. The processes may make use of AI based techniques to increase accuracy, agility and shorten TTM.

Establish Resource strategy

Process Identifier: 1.5.1.3.1

Hierarchy Level: 4

Brief Description:

Resource strategies based on market trends, internal resource utilization trends, highlight shortcomings of resource performance and operational support

Extended Description:

Establish resource strategy processes are responsible for proactively establishing the resource strategies based on markets trends, internal resource trends, future products needs, available and emerging resource capabilities, and addressing shortcomings in existing resource performance and operational support. These processes also highlight issue associated with resourcing and skill sets along with shortcomings in existing resource performance and operational support arrangements and processes. The processes may make of use of AI based techniques to increase accuracy, agility and shorten TTM.

Develop Resource Strategy

Process Identifier: 1.5.1.3.2

Hierarchy Level: 4

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Brief Description:

Accept changing resource requirements and develop resource strategies

Extended Description:

Develop resource Strategy processes, accept the changing of resource requirements and develop appropriate resource strategies and strategic resource architectures based on the changing needs of the enterprise, and increasing expectations of the markets. Develop resource strategy may perform in-parallel across all required resources proactively.

Today in digital ecosystem, Digital resource architecture map could help to facilitate this process.

Establish Resource delivery goals

Process Identifier: 1.5.1.3.3

Hierarchy Level: 4

Brief Description:

Establish realistic resource development and delivery goals for the enterprise and services to be delivered

Extended Description:

Establish Resource delivery goals processes focus on establishing realistic resource development and delivery goals for the enterprise and services to be delivered to market

Establish resource implementation policies

Process Identifier: 1.5.1.3.4

Hierarchy Level: 4

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Brief Description:

Establish implementation policies which act as a set of guidelines to be followed in the Resource Capability Delivery processes

Extended Description:

Not used for this process element

Define Resource Support Strategies

Process Identifier: 1.5.1.4

Hierarchy Level: 3

Brief Description:

Define the principles, policies and performance standards for the operational organization providing resource support

Extended Description:

The Define Resource Support Strategies processes define the principles, policies and performance standards for the operational organization providing resource support. Over time requirements for service levels and quality on existing resources change and operational challenges are introduced by new resources. These processes define the targets, and operational requirements, in this changing environment. The required performance standards are driven by community and regulatory expectations, internal process and technology capabilities and competitive offerings. A major input to these processes arises from the Gather & Analyze Resource Information processes, and from the product portfolio, resource strategy and market strategy processes. These processes also provide the feedback point from analyses undertaken in the Resource Management & Operations processes to determine bottlenecks and problems associated with existing resource operational support activities.

Define Resource support Principles

Process Identifier: 1.5.1.4.1

Hierarchy Level: 4

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Brief Description:

Define resource support principles for the operational organization providing resource support.

Extended Description:

Define Resource Support Principles processes define principles for service levels and service quality on existing resources or new resources

Define Resource support Policies

Process Identifier: 1.5.1.4.2

Hierarchy Level: 4

Brief Description:

Define resource support policies for the operational organization providing resource support.

Extended Description:

Define Resource Support Policies processes define policies for service levels and service quality on existing resources or new resources

Define Resource Support performance standards

Process Identifier: 1.5.1.4.3

Hierarchy Level: 4

Brief Description:

Define resource support performance standards for the operational organization providing resource support.

Extended Description:

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These processes define the targets, and operational requirements, in this changing environment. The required performance standards are driven by community and regulatory expectations, internal process and technology capabilities and competitive offerings. These processes also provide the feedback point from analyses undertaken in the resource Management & Operations process to determine bottlenecks and problems associated with existing resource operational activities. A major input to these processes arises from the Gather & Analyze Service Information processes, and from the product portfolio, resource strategy and market strategy processes.

Produce Resource Business Plans

Process Identifier: 1.5.1.5

Hierarchy Level: 3

Brief Description:

Develop and deliver quarterly, annual and multi-year resource plans in support of services, products and offers

Extended Description:

The Produce Resource Business Plans processes develop and deliver quarterly, annual and multi-year resource plans in support of services, products and offers that include volume forecasts, negotiation for required levels of resources and budgets, gaining resource development and management as well as supply chain commitment and executive approval for the plans. Volume forecasts may be based on near real-time information and use AI forecasting techniques.

Forecasting of aggregate or high level resource demand, and capturing of new opportunities, are both essential to ensure that budgets are assigned which allow the enterprise to implement the resources capabilities and capacity necessary for the future needs of their customers and potential customers.

The processes assess the impacts that new or modified resource infrastructure will cause on the installed infrastructure and workforce, and establish the functions and benefits that new or modified resource will provide to users. Assessment of resources impact to the new or modified resource structure may benefit from being performed parallel across all required resources proactively. These processes identify the timetables and logistics for new resource capability introduction.

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Develop & Deliver quarterly/annual/multi year resource business plans

Process Identifier: 1.5.1.5.1

Hierarchy Level: 4

Brief Description:

Develop and deliver quarterly, annual and multi-year resource plans, negotiations, approvals , resource development and management

Extended Description:

Develop & Deliver quarterly /annual/multi year resource business plans focus on developing and delivering quarterly, annual and multi-year resource plans in support of services, products and offers that include volume forecasts, negotiation for required levels of resources and budgets, gaining resource development and management as well as supply chain commitment and executive approval for the plans.

Forecast high level resource demand & capture new opportunities

Process Identifier: 1.5.1.5.2

Hierarchy Level: 4

Brief Description:

Forecast resource demand and capture new opportunities for future needs of new/existing customers

Extended Description:

Forecast Resource demand & capture new opportunities processes ensures that budgets are assigned which allow the enterprise to implement the resource capabilities and capacity necessary for the future needs of their customers and potential customers. This may use AI techniques to simulate different scenarios for more accurate predictions.

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Assess impact of Resource Business plans

Process Identifier: 1.5.1.5.3

Hierarchy Level: 4

Brief Description:

Assess the impacts of new and modified resource infrastructure on systems and establish functions and benefits

Extended Description:

Asses impact of Resource business plan processes assess the impacts that new or modified resource infrastructure will cause on the installed infrastructure and workforce, and establish the functions and benefits that new or modified resource will provide to users. Assessment of resources impact on the new or modified resource structure may benefit from being performed parallel across all required resources proactively. It may perform as a automated process.

Today in digital ecosystem, Digital resource architecture map could help to facilitate this process.

Identify timetables for new resource capability introduction

Process Identifier: 1.5.1.5.4

Hierarchy Level: 4

Brief Description:

Identify timetables for new resource capability introduction

Extended Description:

Not used for this process element

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Identify logistics for new resource capability introduction

Process Identifier: 1.5.1.5.5

Hierarchy Level: 4

Brief Description:

Identify logistics for new resource capability introduction

Extended Description:

Not used for this process element

Develop Resource Partnership Requirements

Process Identifier: 1.5.1.6

Hierarchy Level: 3

Brief Description:

Identify the requirements for resource capabilities to be sourced from partners or suppliers, and any resource capabilities to be delivered internally to the enterprise.

Extended Description:

The Develop Resource Partnership Requirements processes identify the requirements for resource capabilities sourced from partners or suppliers, and any resource capabilities to be delivered internally to the enterprise. The processes determine the extent to which resource capabilities need to be sourced externally or delivered from internal capabilities, they scan external availability, and make recommendations about the specific external partners or suppliers who should be used to provide the resource capabilities.

Note that the processes used to actually source any external suppliers occur with the Party Offering Development & Retirement processes.

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Identify the requirements for resource capabilities

Process Identifier: 1.5.1.6.1

Hierarchy Level: 4

Brief Description:

Identify the requirements for resource capabilities sourced from partners or suppliers, and the resource capabilities to be delivered internally to enterprise

Extended Description:

Not used for this process element

Recommend Resource partnership

Process Identifier: 1.5.1.6.2

Hierarchy Level: 4

Brief Description:

Scan external availability for resource capabilities and recommend external partners/Suppliers

Extended Description:

Recommend Resource partnership processes focus on scanning external availability of sourcing resource capabilities and make recommendations about specific external partners/suppliers who should be used to provide the resource capabilities

Determine extent of resource capabilities sourcing

Process Identifier: 1.5.1.6.3

Hierarchy Level: 4

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Brief Description:

Determine the extent to which resource capabilities needs to be sourced externally or delivered from internal capabilities

Extended Description:

Not used for this process element

Gain Enterprise Commitment to Resource Plans

Process Identifier: 1.5.1.7

Hierarchy Level: 3

Brief Description:

Gains enterprise commitment to the resource strategy and business plans.

Extended Description:

The Gain Commitment to Resource Plans processes gains enterprise commitment to the resource strategy and business plans. These processes include all aspects of identification of stakeholders and negotiation to gain stakeholder approval.

Identify Stakeholders to Resource strategy and resource plans

Process Identifier: 1.5.1.7.1

Hierarchy Level: 4

Brief Description:

Identify stakeholders of resource strategy and resource plans

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Extended Description:

Not used for this process element

Gain resource strategy and resource plans Stakeholders approval

Process Identifier: 1.5.1.7.2

Hierarchy Level: 4

Brief Description:

Negotiate & gain stakeholder approval for resource strategy and resource Plans

Extended Description:

Not used for this process element

Gain Enterprise Commitment to resource strategy and resource plans

Process Identifier: 1.5.1.7.3

Hierarchy Level: 4

Brief Description:

Gain Enterprise commitment to the resource strategy and resource plans

Extended Description:

Not used for this process element

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Resource Test Strategy

Process Identifier: 1.5.1.8

Hierarchy Level: 3

Brief Description:

Resource Test Strategy develops the strategies of the enterprise for Resource Test.

Extended Description:

Resource Test Strategy develops the strategies of the enterprise for Resource Test.

This process is in charge of identifying types of Resource Test to be conducted according to different context (i.e. business activities) for types of Resources.

The different context for Resource Test are:

- Resource Development & Retirement to qualify the capacity to deliver Resources before validating a new ResourceSpecification
- Resource Provisioning to test the Resource before closing the ResourceOrderItem
- Resource Performance Management to test Resource Quality
- Resource Trouble Management to test Resource functioning
- Resource Test Management to conduct tests not specific to a customer's product

Analyze Resource Test Quality

Process Identifier: 1.5.1.9

Hierarchy Level: 3

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Brief Description:

Resource Test Quality Analysis processes perform quality analysis of resource testing processes, in order to continuously fine-tune and improve them.

Extended Description:

Resource Test Quality Analysis processes perform quality analysis of resource testing processes, in order to continuously fine-tune and improve them.

Resource Test Quality Analysis processes provide an offline analysis of the performance of resource testing processes to ensure their continuous fitness for purpose, and eventually propose modifications or enhancements to the current resource test specifications (including: resource testing methods, quota, authorized users, recommendations and course of action if needed).

The analysis is done using statistical resource test usage data to evaluate resource tests performance and required improvements if needed.

Following the analysis, typical course of action can be:

- No changes needed currently,
- Creation of additional resource tests,
- Improving existing resource tests,
- Removing resource tests that are no longer relevant.

Resource Capability Delivery

Process Identifier: 1.5.2

Hierarchy Level: 2

Brief Description:

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Use the capability definition or requirements to deploy new and/or enhanced technologies and associated resources.

Extended Description:

Resource Capability Delivery processes use the capability definition or requirements to deploy new and/or enhanced technologies and associated resources.

The objectives of these processes is to ensure that network, application and computing resources are deployed, according to the plans set by Resource Development. They deliver the physical resource capabilities necessary for the ongoing operations, and long term well-being of the enterprise, and ensure the basis on which all resources and services will be built. Using automation may enhance resource capability delivery.

Responsibilities of the Resource Capability Delivery processes include, but are not limited to:

- Planning resource supply logistics (warehousing, transport, etc.)
- Planning the Resource Installation
- Contracting and Directing the Resource Construction where needed
- Verifying the Resource Installation
- Handover the Resource Capability to Operations through interactions with the Manage Resource Class Configuration

Logical network configurations (such as resource elements integration) are as important to the network resources as the physical aspects. All aspects must be planned and considered in the design and implementation of the network, including infrastructure owned by the enterprise, and by supplier/partners, other physical resources and logical elements. Logical network configuration may be designed digitally in order to make sure all the resources are aligned and planned proactively.

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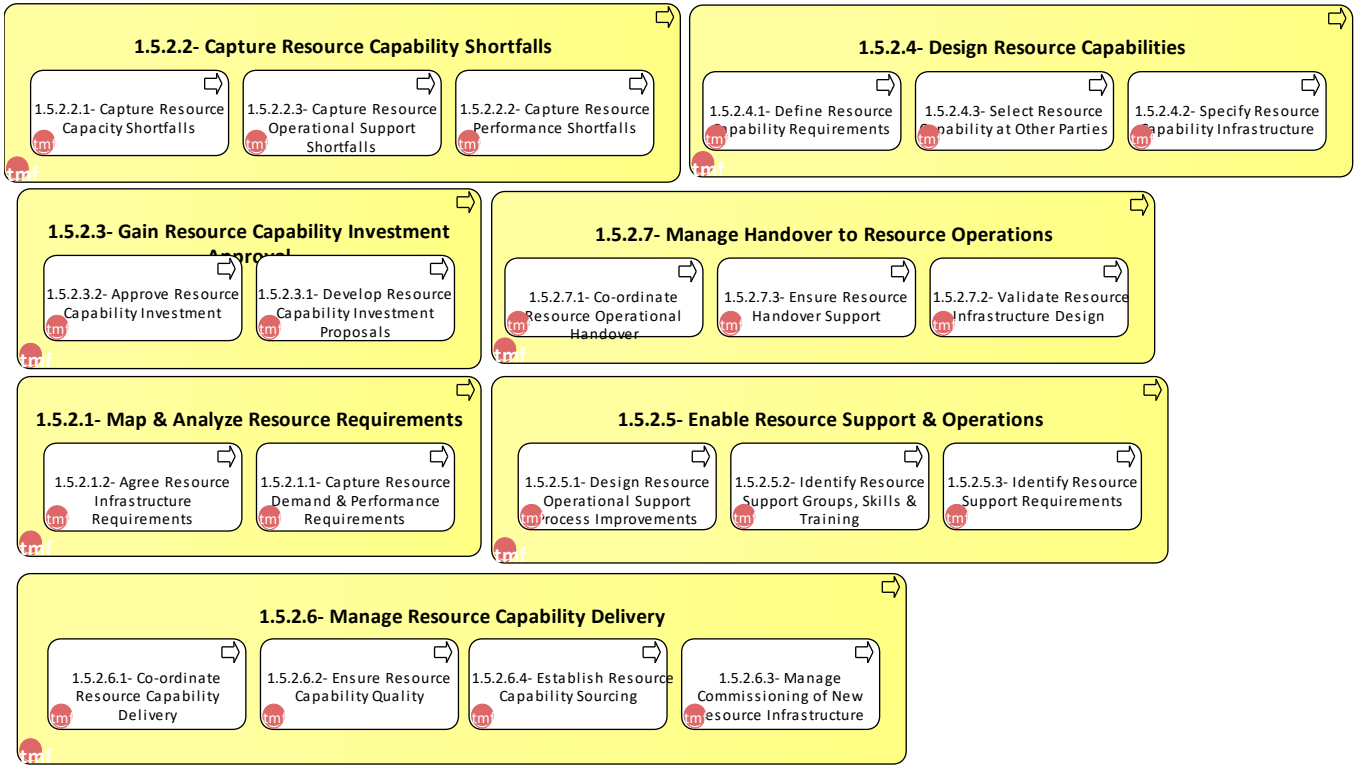

Figure 2: Resource Capability Delivery Children

Map & Analyze Resource Requirements

Process Identifier: 1.5.2.1

Hierarchy Level: 3

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Brief Description:

Define the detailed resource infrastructure requirements to support the service capabilities required by the enterprise

Extended Description:

The Map & Analyze Resource Requirements processes define the detailed resource infrastructure requirements to guide the definition of service capabilities required by the enterprise. The processes provide detailed analysis of new resource requirements linked to relevant geographic distributions which for example in mobile networks may be as small as a cell level. These processes also establish the detailed performance requirements. These processes take the forecast information available from the Produce Resource Business Plans and requirements information from the Map & Analyze Service Requirements processes, as well as resource infrastructure requirements developed by the Develop Detailed Resource Specifications processes, to establish detailed views of anticipated resource demand and performance requirements. These views may be displayed on a console or visual platform and shared by relevant resource owners.

These processes manage the capacity planning for the resource infrastructure, and identify capacity requirements based on service forecasts and appropriate resource related metrics, i.e., transaction volumes, storage requirements, traffic volumes, port availabilities, etc. These forecasts may be based on near real-time information and use AI forecasting techniques.

The processes include any interaction with cross-enterprise and cross-domain coordination and management functions to ensure that the demand distributions capture the needs of relevant stakeholders. Coordination and management function may use automated processes to improve efficiency and shorten TTM.

Agree Resource Infrastructure Requirements

Process Identifier: 1.5.2.1.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

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These processes obtain agreement to the detailed resource infrastructure requirements to guide the definition of the service capabilities required by the enterprise. The processes include any interaction with cross-enterprise and cross-domain coordination and management functions to ensure that the demand distributions capture the needs of relevant stakeholders. Coordination and management function may use automated processes to improve efficiency and shorten TTM.

Capture Resource Demand & Performance Requirements

Process Identifier: 1.5.2.1.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes provide detailed analysis of new resource requirements linked to relevant geographic distributions which may be as small as Cell level for mobile networks. These processes also establish the detailed performance requirements. These processes take the forecast information available from the Produce Resource Business Plans and requirements information from the Map & Analyze Service Requirements processes, as well as resource infrastructure requirements developed by the Develop Detailed Resource Specifications processes, to establish detailed views of anticipated resource demand and performance requirements. These views may be displayed on a console or visual platform and shared by relevant resource owners.

These processes manage the capacity planning for the resource infrastructure, and identify capacity requirements based on service forecasts and appropriate resource related metrics, i.e., transaction volumes, storage requirements, traffic volumes, port availabilities, etc. These forecasts may be based on near real-time information and use AI forecasting techniques.

Capture Resource Capability Shortfalls

Process Identifier: 1.5.2.2

Hierarchy Level: 3

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Brief Description:

Identify specific or imminent resource capacity, resource performance and/or resource operational support shortfalls

Extended Description:

The Capture Resource Capability Shortfalls processes identify specific or imminent resource capacity, resource performance and/or resource operational support shortfalls. These processes take information available from the Resource Management & Operations processes to establish detailed views of anticipated resource and shortfalls and support process issues. These views may be displayed on a console or visual platform and shared by relevant resource owners.

Resource capability shortfalls may be captured via an automated process. It may process information using advanced technologies such as AI.

Capture Resource Capacity Shortfalls

Process Identifier: 1.5.2.2.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes identify specific or imminent resource capacity shortfalls. These processes take information available from the Resource Management & Operations processes to establish detailed views of anticipated resource capacity issues. These views may be displayed on a console or visual platform and shared by relevant resource owners.

Resource capability shortfalls may be captured via an automated process. It may process information using advanced technologies such as AI.

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Capture Resource Operational Support Shortfalls

Process Identifier: 1.5.2.2.3

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes identify specific or imminent resource operational support shortfalls. These processes take information available from the Resource Management & Operations processes to establish detailed views of anticipated resource operational support issues. These views may be displayed on a console or visual platform and shared by relevant resource owners.

Resource capability shortfalls may be captured via an automated process. It may process information using advanced technologies such as AI.

Capture Resource Performance Shortfalls

Process Identifier: 1.5.2.2.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes identify specific or imminent resource performance shortfalls. These processes take information available from the Resource Management & Operations processes to establish detailed views of anticipated resource performance issues. These views may be displayed on a console or visual platform and shared by relevant resource owners.

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Resource capability shortfalls may be captured via an automated process. It may process information using advanced technologies such as AI.

Gain Resource Capability Investment Approval

Process Identifier: 1.5.2.3

Hierarchy Level: 3

Brief Description:

Capture all activities required to develop and gain necessary approval for investment proposals to develop and deliver the required resource capabilities

Extended Description:

The Gain Resource Capability Investment Approval processes capture all activities required to develop and gain necessary approval for investment proposals to develop and deliver the required resource capabilities, including identification of potential other parties. These processes take the input from the Map & Analyze Resource Requirements, the Capture Resource Capability Shortfalls and the Map & Analyze Service Requirements processes to develop and gain approval for any business proposals arising. In some cases the business proposal may require the creation and approval of a formal business case, in other cases the business proposal approval may be delegated to local management. In any event the cost estimates for delivering the resource infrastructure, including costs for materials (equipment and tools), labor and training are part of the investment proposal.

The rules and procedures outlining the necessary approval process to be used are also part of these processes.

The processes include any interaction with cross-enterprise and cross-domain coordination and management functions to ensure that the demand distributions capture the needs of relevant stakeholders. Coordination and management function may use automated processes to improve efficiency and shorten TTM.

Approve Resource Capability Investment

Process Identifier: 1.5.2.3.2

Hierarchy Level: 4

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Brief Description:

Not used for this process element.

Extended Description:

These processes capture all activities required to gain necessary approval for business proposals to develop and deliver the required resource capabilities, including identification of potential other parties.

The processes include any interaction with cross-enterprise and cross-domain coordination and management functions to ensure that the demand distributions capture the needs of relevant stakeholders. Coordination and management function may use automated processes to improve efficiency and shorten TTM.

Develop Resource Capability Investment Proposals

Process Identifier: 1.5.2.3.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes capture all activities required to develop business proposals to develop and deliver the required resource capabilities, including identification of potential other parties. These processes take the input from the Map & Analyze Resource Requirements, the Capture Resource Capability Shortfalls and the Map & Analyze Service Requirements processes to develop and gain approval for any business proposals arising. In some cases the business proposal may require the creation and approval of a formal business case, in other cases the business proposal approval may be delegated to local management. In any event the cost estimates for delivering the resource infrastructure, including costs for materials (equipment and tools), labor and training are part of the investment proposal.

The rules and procedures outlining the necessary approval process to be used are also part of these processes.

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Design Resource Capabilities

Process Identifier: 1.5.2.4

Hierarchy Level: 3

Brief Description:

Manage the design of the resource infrastructure to meet the requirements in any approved investment proposals.

Extended Description:

The Design Resource Capabilities processes manage the design of the resource infrastructure to meet the requirements in any approved investment proposals. These processes ensure the collation and coordination of requirements from all approved investment proposals, assess the most appropriate resource infrastructure, develop the tactical/solution architecture and design specifications to be used to build or source the necessary resource infrastructure components, and select the most appropriate resource infrastructure other parties to support the resource requirements. A key element of the overall design is the integration approach between the existing legacy resource infrastructure and any proposed new resource infrastructure. This integration design is managed within the architecture and specification processes.

Note that the actual management of the sourcing process is handled within the Party Offering Development & Retirement processes.

Define Resource Capability Requirements

Process Identifier: 1.5.2.4.1

Hierarchy Level: 4

Brief Description:

These processes ensure the collation and coordination of requirements from all approved investment proposals.

Extended Description:

Not used for this process element

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Select Resource Capability at Other Parties

Process Identifier: 1.5.2.4.3

Hierarchy Level: 4

Brief Description:

These processes select the most appropriate resource infrastructure at other parties to support the identified resource capability requirements.

Extended Description:

Not used for this process element

Specify Resource Capability Infrastructure

Process Identifier: 1.5.2.4.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes assess the most appropriate resource infrastructure, and develop the tactical/solution architecture and design specifications to be used to build or source the necessary resource infrastructure components, to meet the identified resource capability requirements. A key element of the overall design is the integration approach between the existing legacy resource infrastructure and any proposed new resource infrastructure. This integration design is managed within the architecture and specification processes.

Note that the actual management of the sourcing process is handled within the Supply Chain Development & Management processes.

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Specify Service Capability Infrastructure

Process Identifier: 1.4.2.4.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes assess the most appropriate service infrastructure, and develop the tactical/solution architecture and design specifications to be used to build or source the necessary service infrastructure components, to meet the identified service capability requirements.

Enable Resource Support & Operations

Process Identifier: 1.5.2.5

Hierarchy Level: 3

Brief Description:

Manage the design of any improvements or changes required to the resource operational support processes to support the investment proposals and new resource capabilities and infrastructure

Extended Description:

The Enable Resource Support & Operations processes manage the design of any improvements or changes required to the resource operational support processes to support the investment proposals and new resource capabilities and infrastructure. The processes ensure the identification of operational support groups, required skill sets, and availability of appropriate training programs. These processes ensure the identification, collation and coordination of support requirements from all approved investment proposals, and from any operational support shortfalls identified in the Capture Resource Capability Shortfalls processes.

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Design Resource Operational Support Process Improvements

Process Identifier: 1.5.2.5.1

Hierarchy Level: 4

Brief Description:

These processes manage the design of any improvements or changes required to the resource operational support processes to support the investment proposals and new resource capabilities and infrastructure.

Extended Description:

Not used for this process element

Identify Resource Support Groups, Skills & Training

Process Identifier: 1.5.2.5.2

Hierarchy Level: 4

Brief Description:

The processes ensure the identification of operational support groups, required skill sets, and availability of appropriate training programs.

Extended Description:

Not used for this process element

Identify Resource Support Requirements

Process Identifier: 1.5.2.5.3

Hierarchy Level: 4

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Brief Description:

Not used for this process element.

Extended Description:

These processes ensure the identification, collation and coordination of support requirements from all approved investment proposals, and from any operational support shortfalls identified in the Capture Resource Capability Shortfalls processes.

Manage Resource Capability Delivery

Process Identifier: 1.5.2.6

Hierarchy Level: 3

Brief Description:

Manage the provision, implementation, commissioning and roll-out of the new or enhanced resource capability, and associated operational support processes.

Extended Description:

The Manage Resource Capability Delivery processes manage the provision, implementation, commissioning and roll-out of the new or enhanced resource capability and associated operational support processes. These processes are predominantly program/project management process functions, and require the detailed management and co-ordination of the delivery of individual resource infrastructure components to achieve the delivery of the overall resource capability. Within the Manage Resource Capability Delivery processes separate other parties may be responsible for the delivery of the resource capability, and other parties for the installation and construction. The Manage Resource Capability Delivery processes ensure that the roles and responsibilities of all parties are identified, managed and coordinated.

These processes are responsible to ensure that the quality of the implemented resource capability meets the design specifications. These processes manage the commissioning of the new resource infrastructure by ensuring the availability of test programs and specifications against which to test the new resource infrastructure meets the design requirements.

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These processes leverage the Party Offering Development & Retirement processes as necessary to establish any new sourcing arrangements for the delivery of resource components.

Co-ordinate Resource Capability Delivery

Process Identifier: 1.5.2.6.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes ensure the provision, implementation and roll-out of the new or enhanced resource capability and associated operational support processes. These processes are predominantly program/project management process functions, and require the detailed management and co-ordination of the delivery of individual resource infrastructure components, and any underlying resource infrastructure capability delivery, to achieve the delivery of the overall resource capability. Within these processes, separate other parties may be responsible for the delivery of the resource capability, and other parties for the installation and construction. These processes ensure that the roles and responsibilities of all parties are identified, managed and coordinated.

Ensure Resource Capability Quality

Process Identifier: 1.5.2.6.2

Hierarchy Level: 4

Brief Description:

These processes are responsible to ensure that the quality of the implemented resource capability meets the design specifications.

Extended Description:

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Not used for this process element

Manage Commissioning of New Resource Infrastructure

Process Identifier: 1.5.2.6.3

Hierarchy Level: 4

Brief Description:

These processes manage the commissioning of the new resource infrastructure by ensuring the availability of test programs and specifications against which to test the new resource infrastructure meets the design requirements.

Extended Description:

Not used for this process element

Establish Resource Capability Sourcing

Process Identifier: 1.5.2.6.4

Hierarchy Level: 4

Brief Description:

These processes leverage the Supply Chain Development & Management processes as necessary to establish any new sourcing arrangements for the delivery of resource components.

Extended Description:

Not used for this process element

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Manage Handover to Resource Operations

Process Identifier: 1.5.2.7

Hierarchy Level: 3

Brief Description:

Manage the processes involved in handover of deployed resource infrastructure to operational control

Extended Description:

The Manage Handover to Resource Operations processes manage the processes involved in handover of deployed resource infrastructure to operational control. These processes ensure that all operational and performance design requirements have been met by the installed resource infrastructure, and that all tools, test equipment, operational procedures, support groups, and training is in place to allow for successful operation. These processes include the management and coordination of all stakeholders required to gain approval and acceptance of the handover to operational control.

Co-ordinate Resource Operational Handover

Process Identifier: 1.5.2.7.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes co-ordinate the processes involved in handover of deployed resource infrastructure to operational control, including the management and coordination of all stakeholders required to gain approval and acceptance of the handover.

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Ensure Resource Handover Support

Process Identifier: 1.5.2.7.3

Hierarchy Level: 4

Brief Description:

These processes ensure that all tools, test equipment, operational procedures, support groups, and training is in place to allow for successful operation.

Extended Description:

Not used for this process element

Validate Resource Infrastructure Design

Process Identifier: 1.5.2.7.2

Hierarchy Level: 4

Brief Description:

These processes ensure that all operational and process performance design requirements have been met by the installed resource infrastructure.

Extended Description:

Not used for this process element

Resource Specification Lifecycle Management

Process Identifier: 1.5.3

Hierarchy Level: 2

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Brief Description:

Develop new, or enhance existing technologies and associated resource types, so that new Services can be developed.

Extended Description:

Resource Specification Lifecycle Management processes develop new, or enhance existing technologies and associated resource types, so that new Products are available to be sold to customers. They use the capability definition or requirements defined by Resource Strategy & Planning They also decide whether to acquire resources from outside, taking into account the overall business policy in that respect. These processes also retire or remove technology and associated resource types, which are no longer required by the enterprise.

Resource types may be built, or in some cases leased from other parties. To ensure the most efficient and effective solution can be used, negotiations on network level agreements with other parties are paramount for both building and leasing.

These processes interact strongly with Product and Engaged Party Development processes.

This process was renamed in 24.0 old name was Resource Specification Development & Retirement

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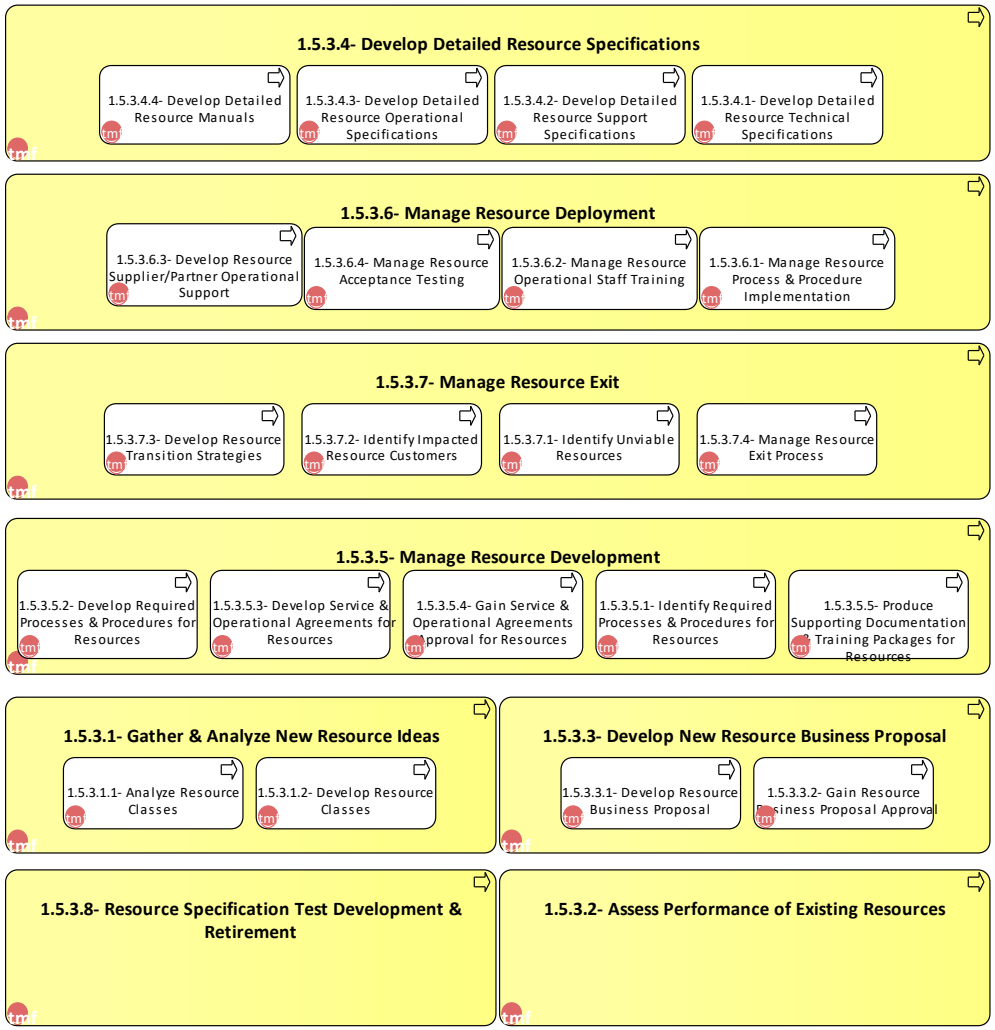

Figure 3: Resource Specification Lifecycle Management Children

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Gather & Analyze New Resource Ideas

Process Identifier: 1.5.3.1

Hierarchy Level: 3

Brief Description:

Combine specific product & service class requirements with demographic, customer, technology and marketing information to identify specific new resource classes/components, or enhancements to existing resource classes/components

Extended Description:

The Gather & Analyze New Resource Ideas processes combine specific product & service class requirements with demographic, customer, technology and marketing information to identify specific new resource classes/components, or enhancements to existing resource classes/components. These processes undertake the necessary analysis to identify potential resource classes, compare current resource classes with the identified required resource classes, and as a result of the analysis develop new resource class ideas.

Analyze Resource Classes

Process Identifier: 1.5.3.1.1

Hierarchy Level: 4

Brief Description:

These processes undertake the necessary analysis to identify potential resource classes, and compare current resource classes with the identified required service classes.

Extended Description:

Not used for this process element

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Develop Resource Classes

Process Identifier: 1.5.3.1.2

Hierarchy Level: 4

Brief Description:

These processes develop new resource class ideas. The new resource class ideas include an analysis of the customer value proposition.

Extended Description:

Not used for this process element

Assess Performance of Existing Resources

Process Identifier: 1.5.3.2

Hierarchy Level: 3

Brief Description:

Analyze the performance of existing resources to identify inadequacies and required improvements

Extended Description:

The Assess Performance of Existing Resources processes analyze the performance of existing resources to identify inadequacies and required improvements. These processes use information from customers and from operational activities to identify required improvements.

Develop New Resource Business Proposal

Process Identifier: 1.5.3.3

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Hierarchy Level: 3

Brief Description:

Develop and document business proposals for the identified new or enhanced Resource ideas

Extended Description:

The Develop New Resource Business Proposal processes develop and document business proposals for the identified new or enhanced Resource ideas (including if necessary a business case). The business proposal (or business case) identifies the resource development (e.g.. network and/or IT resources), management and operations costs and anticipated benefits, including forecast demand, performance gains, productivity gains and/or operational cost improvements specifically associated with the resource business proposal. The business proposal also includes an assessment of the risks and the competitive positioning of the proposal. As a part of the business proposal development a feasibility assessment can be produced. Potential other parties who can assist in the development of the resource classes are also identified (note that commercial arrangements may already be in place with these potential related parties. As a part of the process, the business proposal is appropriately approved, and as a result of the approval, necessary staff and other resources are made available.

Develop Resource Business Proposal

Process Identifier: 1.5.3.3.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes develop and document business proposals for the identified new or enhanced Resource ideas (including if necessary a business case). The business proposal (or business case) identifies the resource development (e.g.. network and/or IT resources), management and operations costs and anticipated benefits, including forecast demand, performance gains, productivity gains and/or operational cost improvements specifically associated with the resource business proposal. The business proposal also includes an assessment of the risks and the competitive positioning of the proposal. As a part

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of the business proposal development a feasibility assessment can be produced. Potential related parties who can assist in the development of the resource classes are also identified (note that commercial arrangements may already be in place with these potential related parties).

Gain Resource Business Proposal Approval

Process Identifier: 1.5.3.3.2

Hierarchy Level: 4

Brief Description:

These processes manage approval of the business proposal, and as a result of the approval, necessary staff and other resources are made available.

Extended Description:

Not used for this process element

Develop Detailed Resource Specifications

Process Identifier: 1.5.3.4

Hierarchy Level: 3

Brief Description:

Develop and document the detailed resource-related technical, performance and operational specifications, and manuals.

Extended Description:

The Develop Detailed Resource Specifications processes develop and document the detailed resource-related technical, performance and operational specifications, and manuals. These processes develop and document the required resource features, the specific technology requirements and selections, the specific operational, performance and quality requirements and support activities, any resource specific data required for the systems and network infrastructure. The Develop Detailed Resource Specifications processes provide input to these specifications. The processes ensure that all detailed

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specifications are produced and appropriately documented. Additionally the processes ensure that the documentation is captured in an appropriate enterprise repository.

Develop Detailed Resource Technical Specifications

Process Identifier: 1.5.3.4.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes develop and document the required resource features for the systems and network infrastructure. The processes ensure that all detailed specifications are produced and appropriately documented. Additionally the processes ensure that the documentation is captured in an appropriate enterprise repository.

Develop Detailed Resource Support Specifications

Process Identifier: 1.5.3.4.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

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These processes develop and document the specific technology requirements and selections required for the systems and network infrastructure. The processes ensure that all detailed specifications are produced and appropriately documented. Additionally the processes ensure that the documentation is captured in an appropriate enterprise repository.

Develop Detailed Resource Operational Specifications

Process Identifier: 1.5.3.4.3

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes develop and document the specific operational, and quality requirements and support activities, and any resource specific data required for the systems and network infrastructure required for the systems and network infrastructure. The processes ensure that all detailed specifications are produced and appropriately documented. Additionally the processes ensure that the documentation is captured in an appropriate enterprise repository.

Develop Detailed Resource Manuals

Process Identifier: 1.5.3.4.4

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

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These processes develop and document the manuals. The processes ensure that all detailed specifications are produced and appropriately documented. Additionally the processes ensure that the documentation is captured in an appropriate enterprise repository.

Manage Resource Development

Process Identifier: 1.5.3.5

Hierarchy Level: 3

Brief Description:

Ensure the co-coordinated delivery in line with the approved business case of all required resource classes/components capabilities for that business case across the enterprise

Extended Description:

The Manage Resource Development processes ensure the co-coordinated development in line with the approved business case of all required resource classes/components for that business case across the enterprise. These processes ensure that all operational processes and procedures, resource changes (e.g. network and/or IT resources), operational procedures, testing tools and procedures, etc. required to support the new resource class/component are identified and developed. These processes ensure that the necessary documentation and training packages are produced to support the operation of the new resource class. These processes also ensure that the required service level agreements and operational level agreements are developed and agreed for each resource class deployed, and that any other parties operational support has been identified and agreed. These processes have both program/project management aspects and technical/operational specification aspects, with the detailed management of individual resource class deployment managed by the Manage Resource Deployment processes.

Note that management of major new or enhanced infrastructure delivery to support product and offer development is managed within the Resource Capability Delivery process.

Note that delivery of resource classes/components within the context of existing commercial arrangements is managed through the Party Offering Development & Retirement process. If new related parties are required, the Party Tender Management and Party Agreement Management processes are used to deliver the necessary commercial arrangements

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Identify Required Processes & Procedures for Resources

Process Identifier: 1.5.3.5.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes ensure that all operational processes and procedures, resource changes (e.g. network and/or IT resources), operational procedures, testing tools and procedures, etc. required to support the new resource class/component are identified.

Develop Required Processes & Procedures for Resources

Process Identifier: 1.5.3.5.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

These processes ensure that all operational processes and procedures, resource changes (e.g. network and/or IT resources), operational procedures, testing tools and procedures, etc. required to support the new resource class/component are developed.

Develop Service & Operational Agreements for Resources

Process Identifier: 1.5.3.5.3

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Hierarchy Level: 4

Brief Description:

These processes ensure that the required service level agreements and operational level agreements are developed for each resource class deployed, and that any other parties operational support has been identified.

Extended Description:

Not used for this process element

Gain Service & Operational Agreements Approval for Resources

Process Identifier: 1.5.3.5.4

Hierarchy Level: 4

Brief Description:

These processes ensure that the required service level agreements and operational level agreements are agreed for each resource class deployed, and that any other parties operational support has been agreed.

Extended Description:

Not used for this process element

Produce Supporting Documentation & Training Packages for Resources

Process Identifier: 1.5.3.5.5

Hierarchy Level: 4

Brief Description:

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These processes ensure that the necessary documentation and training packages are produced to support the operation of the new resource class.

Extended Description:

Not used for this process element

Manage Resource Deployment

Process Identifier: 1.5.3.6

Hierarchy Level: 3

Brief Description:

Ensure the co-coordinated deployment in line with the approved business proposal of all required resource classes/components for that business proposal across the enterprise.

Extended Description:

The Manage Resource Deployment processes ensure the co-coordinated deployment in line with the approved business case of all required resource classes/components for that business case across the enterprise. These processes ensure that all operational processes and procedures, resource changes (e.g. network and/or IT resources), operational procedures, testing tools and procedures, etc. required to support the new resource class/component have been implemented. These processes ensure that appropriate operational staff are identified and have received the necessary training. These processes ensure that the agreed other parties operational support has been implemented. These processes also ensure that acceptance testing is successfully performed to assure that the new or enhanced resources comply with the specifications. These processes have both program/project and management aspects.

Manage Resource Process & Procedure Implementation

Process Identifier: 1.5.3.6.1

Hierarchy Level: 4

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Brief Description:

Not used for this process element.

Extended Description:

These processes ensure that all operational processes and procedures, resource changes (e.g. network and/or IT resources), operational procedures, testing tools and procedures, etc. required to support the new resource class/component have been implemented.

Manage Resource Operational Staff Training

Process Identifier: 1.5.3.6.2

Hierarchy Level: 4

Brief Description:

These processes ensure that appropriate operational staff are identified and have received the necessary training.

Extended Description:

Not used for this process element

Develop Resource Supplier/Partner Operational Support

Process Identifier: 1.5.3.6.3

Hierarchy Level: 4

Brief Description:

These processes ensure that the agreed other parties operational support has been implemented.

Extended Description:

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Not used for this process element

Manage Resource Acceptance Testing

Process Identifier: 1.5.3.6.4

Hierarchy Level: 4

Brief Description:

These processes ensure that acceptance testing is successfully performed to assure that the new or enhanced resources comply with the specifications.

Extended Description:

Not used for this process element

Manage Resource Exit

Process Identifier: 1.5.3.7

Hierarchy Level: 3

Brief Description:

Identify existing resource classes which are unviable and manage the processes to exit the Resource from the market.

Extended Description:

The Manage Resource Exit processes identify existing resource classes which are unviable and manage the process to exit the Resource from the services they support. The processes analyze existing resource classes to identify economically or strategically unviable classes, identify products, services classes & customers impacted by any exit, develop specific exit or migration strategies, develop resource infrastructure transition and/or replacement strategies, and manage the operational aspects of the exit process. A business proposal identifying the competitive threats, risks and costs may be required as a part

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of developing the exit strategy. These processes include any interaction with cross-enterprise co-ordination and management functions to ensure that the needs of all stakeholders are identified and managed.

Identify Unviable Resources

Process Identifier: 1.5.3.7.1

Hierarchy Level: 4

Brief Description:

These processes analyze existing resource classes to identify economically or strategically unviable classes.

Extended Description:

Not used for this process element

Identify Impacted Resource Customers

Process Identifier: 1.5.3.7.2

Hierarchy Level: 4

Brief Description:

These processes identify products, services classes & customers impacted by any exit.

Extended Description:

Not used for this process element

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Develop Resource Transition Strategies

Process Identifier: 1.5.3.7.3

Hierarchy Level: 4

Brief Description:

These processes develop specific exit or migration strategies and develop resource infrastructure transition and/or replacement strategies.

Extended Description:

Not used for this process element

Manage Resource Exit Process

Process Identifier: 1.5.3.7.4

Hierarchy Level: 4

Brief Description:

These processes manage the operational aspects of the exit process.

Extended Description:

Not used for this process element

Resource Specification Test Development & Retirement

Process Identifier: 1.5.3.8

Hierarchy Level: 3

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Brief Description:

Resource Test Development & Retirement is in charge of the Resource Test catalogue.

Extended Description:

Resource Test Development & Retirement is in charge of the Resource Test catalogue.

A type of Resource Test aims at measuring proper functioning and capacities of a Resource.

Resource Test Development & Retirement includes:

- Specifying in detail each Resource Test according to the different context. It includes specifying:
  - the roles authorized to use the Test and quotas for each type of role
  - the method to conduct the Test
  - the rules that define the strategies for conducting the test (including the test plan)
  - the thresholds and related actions
  - the report of test results with rules for enrichment of Resource Tests results according to role asking for it
- Specifying test scenarios defining sequence of Tests with rules about context and planning to trigger it. It includes roles allowed for asking test scenario and corresponding quotas.

Resource Support Management

Process Identifier: 1.5.4

Hierarchy Level: 2

Brief Description:

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Manage resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfillment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on the capabilities and costs of the individual FAB processes.

Extended Description:

Resource Support Management processes are responsible for managing resource infrastructure to ensure that appropriate application, computing and network resources are available and ready to support the Fulfillment, Assurance and Billing processes in instantiating and managing resource instances, and for monitoring and reporting on the capabilities and costs of the individual FAB processes.

Responsibilities of these processes include but are not limited to:

• Supporting the operational introduction of new and/or modified resource infrastructure and conducting operations readiness testing and acceptance;

• Managing planned outages;

• Managing and ensuring the ongoing quality of the Resource Inventory;

• Analyzing availability and performance over time on resources or groups of resources, including trend analysis and forecasting;

• Demand balancing in order to maintain resource capacity and performance

• Performing pro-active maintenance and repair activities;

• Establishing and managing the workforce to support the eTOM processes

• Managing spares, repairs, warehousing, transport and distribution of resources and consumable goods.

• Conducting Vulnerability Management;

• Conducting Threat Assessments;

• Conducting Risk Assessments;

• Conducting Risk Mitigation;

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- Conducting Secure Configuration Activities

This process was renamed in 23.5. old name was Resource Readiness & Support

This process was renamed in 24.0. old name was Resource Support Readiness

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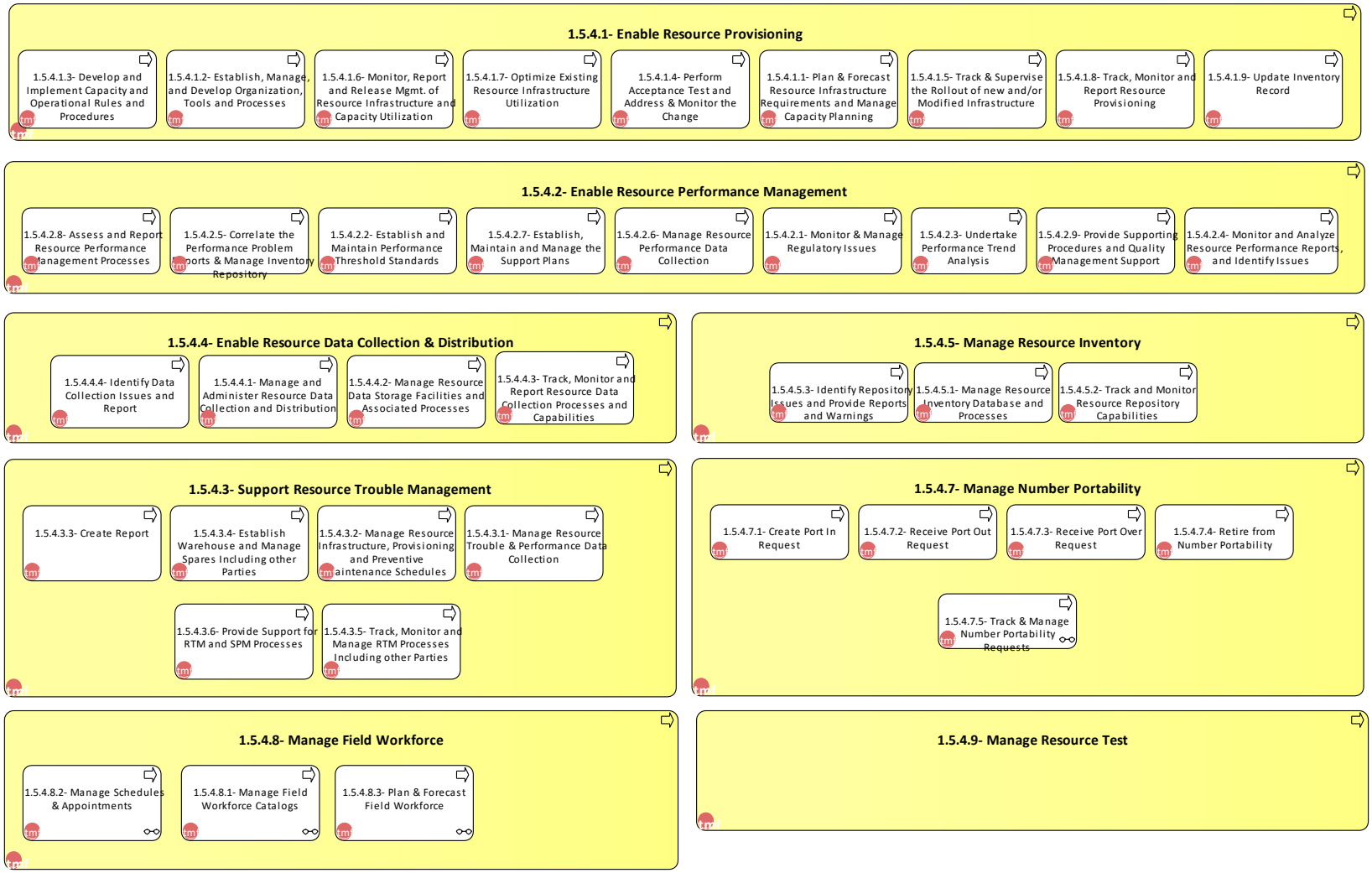

Figure 4: Resource Support Management Children

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Enable Resource Provisioning

Process Identifier: 1.5.4.1

Hierarchy Level: 3

Brief Description:

Planning and deployment of new and/or modified resource infrastructure to ensure availability of sufficient resource infrastructure to support the Resource Provisioning processes, and monitoring, managing and reporting on the capability of the Resource Provisioning processes.

Extended Description:

The responsibilities of the Enable Resource Provisioning processes are twofold - planning and deployment of new and/or modified resource infrastructure to ensure availability of sufficient resource infrastructure to support the Resource Provisioning processes, and monitoring, managing and reporting on the capability of the Resource Provisioning processes.

The responsibilities of these processes include, but are not limited to:

• Forecasting at an operational level resource infrastructure volume requirements and run-out timeframes;

• Managing capacity planning associated with the deployment of new and/or modified resource infrastructure;

• Establishing and monitoring of organizational arrangements to support deployment and operation of new and/or modified resource infrastructure;

• Creating, deploying, modifying and/or upgrading of resource infrastructure deployment support tools (including Resource Inventory) and processes for new and/or modified resource infrastructure;

• Developing and promulgating resource infrastructure capacity deployment rules and controls;

• Authoring, reviewing and approving operational procedures developed by Resource Development & Management processes prior to resource infrastructure deployment;

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• Testing and acceptance of new and/or modified resource infrastructure as part of the handover procedure from the Resource Development & Management processes to Operations;

• Detecting resource infrastructure operational limitations and/or deployment incompatibilities and providing requirements to address these aspects to Resource Development & Management processes;

• Scheduling, managing, tracking and monitoring of the roll-out, in accordance with approved plans, of the approved new and/or modified resource infrastructure; • Monitoring capacity utilization of deployed resource infrastructure to provide early detection of potential resource infrastructure shortfalls;

• Monitoring of, and reporting on, resource infrastructure and resource instance currency and version management;

• Reconfiguring and re-arranging under-utilized deployed resource infrastructure

;• Managing recovery and/or removal of obsolete or unviable resource infrastructure;

• Reporting on deployed resource infrastructure capacity;

• Tracking and monitoring of the Resource Provisioning Management processes and associated costs (including where resource infrastructure is deployed and managed by third parties), and reporting on the capability of the Resource Provisioning Management processes;

• Establishing and managing resource provisioning notification facilities and lists to support the Resource Provisioning notification and reporting processes

• Updating the Resource Inventory of any changes to the available resource infrastructure capacity.

Plan & Forecast Resource Infrastructure Requirements and Manage Capacity Planning

Process Identifier: 1.5.4.1.1

Hierarchy Level: 4

Brief Description:

Planning, managing, deploying, monitoring and reporting of new and/or modified resource infrastructure

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Extended Description:

Planning, managing, deploying, monitoring and reporting of new and/or modified resource infrastructure to ensure availability of sufficient resource infrastructure to support the Resource Provisioning processes. This includes:

• Forecasting at an operational level resource infrastructure volume requirements and run-out timeframes; Volume forecasts may be based on near real-time information and use AI forecasting techniques.

• Managing capacity planning associated with the deployment of new and/or modified resource infrastructure. Managing Capacity Planning shall perform proactively to make sure relevant resources are aligned and planned in-Parallel. It may use of AI based techniques to increase accuracy, agility and shorten TTM.

Establish, Manage, and Develop Organization, Tools and Processes

Process Identifier: 1.5.4.1.2

Hierarchy Level: 4

Brief Description:

Establishing, Creating, Managing, and developing organization, Tools and processes for operations of new/or modified resource infrastructure.

Extended Description:

• Establishing and monitoring of organizational arrangements to support deployment and operation of new and/or modified resource infrastructure;

• Creating, deploying, modifying and/or upgrading of resource infrastructure deployment support tools (including Resource Inventory) and processes for new and/or modified resource infrastructure;

Develop and Implement Capacity and Operational Rules and Procedures

Process Identifier: 1.5.4.1.3

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Hierarchy Level: 4

Brief Description:

Developing and implementing the capacity deployment rules and administrating the infrastructure operational procedures.

Extended Description:

• Developing and promulgating resource infrastructure capacity deployment rules and controls;

• Authoring, reviewing and approving operational procedures developed by Resource Development & Management processes prior to resource infrastructure deployment;

Perform Acceptance Test and Address & Monitor the Change

Process Identifier: 1.5.4.1.4

Hierarchy Level: 4

Brief Description:

Performing acceptance test during the hand over process of new and/or modified resource infrastructure. Also monitoring the new and/or modified infrastructure and addressing the operational limitations.

Extended Description:

• Testing and acceptance of new and/or modified resource infrastructure as part of the handover procedure from the Resource Development & Management processes to Operations;

• Detecting resource infrastructure operational limitations and/or deployment incompatibilities and providing requirements to address these aspects to Resource Development & Management processes;

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Track & Supervise the Rollout of new and/or Modified Infrastructure

Process Identifier: 1.5.4.1.5

Hierarchy Level: 4

Brief Description:

Supervising the rollout activities of new and/or modified resource infrastructure.

Extended Description:

• Scheduling, managing, tracking and monitoring of the roll-out, in accordance with approved plans, of the approved new and/or modified resource infrastructure;

Monitor, Report and Release Mgmt. of Resource Infrastructure and Capacity Utilization

Process Identifier: 1.5.4.1.6

Hierarchy Level: 4

Brief Description:

Monitoring and reporting the capacity utilization of existing infrastructure.

Extended Description:

- Monitoring capacity utilization of deployed resource infrastructure to provide early detection of potential resource infrastructure shortfalls;
- Monitoring of, and reporting on, resource infrastructure and resource instance currency and version management;

In order to improve efficiency and accuracy of monitoring process, different domains' capacity utilization may be monitored at the same time in a single monitoring platform. A prediction model based on different factors may be applied to master capacity monitoring.

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Optimize Existing Resource Infrastructure Utilization

Process Identifier: 1.5.4.1.7

Hierarchy Level: 4

Brief Description:

Optimizing the infrastructure capacity utilization.

Extended Description:

• Reconfiguring and re-arranging under-utilized deployed resource infrastructure

Track, Monitor and Report Resource Provisioning

Process Identifier: 1.5.4.1.8

Hierarchy Level: 4

Brief Description:

Tracking and monitoring the resource provisioning processes and provide the reporting support.

Extended Description:

• Tracking and monitoring of the Resource Provisioning Management processes and associated costs (including where resource infrastructure is deployed and managed by third parties), and reporting on the capability of the Resource Provisioning Management processes;

• Establishing and managing resource provisioning notification facilities and lists to support the Resource Provisioning notification and reporting processes;

Update Inventory Record

Process Identifier: 1.5.4.1.9

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Hierarchy Level: 4

Brief Description:

Updating the inventory record.

Extended Description:

• Updating the Resource Inventory of any changes to the available resource infrastructure capacity.

Enable Resource Performance Management

Process Identifier: 1.5.4.2

Hierarchy Level: 3

Brief Description:

Proactively monitoring and maintaining resource infrastructure, and monitoring, managing and reporting on the capability of the Resource Performance Management processes.

Extended Description:

The responsibilities of the Enable Resource Performance Management processes are twofold - support Resource Performance Management processes by proactively monitoring and assessing resource infrastructure performance, and monitoring, managing and reporting on the capability of the Resource Performance Management processes.

Proactive management is undertaken using a range of performance parameters, whether technical, time, economic or process related.

The responsibilities of the processes include, but are not limited to:

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• Undertaking proactive monitoring regimes of resource infrastructure as required to ensure ongoing performance within agreed parameters over time;

• Developing and maintaining a repository of acceptable performance threshold standards for resource instances to support the Resource Performance Management processes;

• Undertaking trend analysis, and producing reports, of the performance of resource infrastructure to identify any longer term deterioration;

• Monitoring and analyzing the resource instance analyses produced by the Resource Performance Management processes to identify problems that may be applicable to the resource infrastructure as a whole;

• Sourcing details relating to resource instance performance and analysis from the resource inventory to assist in the development of trend analyses;

• Logging the results of the analysis into the resource inventory repository;

• Establishing and managing resource performance data collection schedules, including managing the collection of the necessary information from the Resource Data Collection & Distribution processes, to support proactive monitoring and analysis activity, and requests from Resource Performance

Management processes for additional data to support resource instance performance analysis;

• Establishing and managing facilities to support management of planned resource infrastructure and resource instance outages;

• Establishing, maintaining and managing the testing of resource performance control plans to cater for anticipated resource performance disruptions;

• Proactively triggering the instantiation of control plans to manage performance through programmed and/or foreseen potentially disruptive events, i.e. anticipated traffic loads on Xmas day, planned outages, etc.;

• Tracking and monitoring of the Resource Performance Management processes and associated costs (including where resource infrastructure is deployed and managed by third parties), and reporting on the capability of the Resource Performance Management processes;

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• Establishing and managing resource performance notification facilities and lists to support the Resource Performance Management notification and reporting processes

• Supporting the Support Service Quality Management process.

Monitor & Manage Regulatory Issues

Process Identifier: 1.5.4.2.1

Hierarchy Level: 4

Brief Description:

Proactively monitoring and dealing regulatory issues to ensure the ongoing performance of resource infrastructure.

Extended Description:

• Undertaking proactive monitoring regimes of resource infrastructure as required to ensure ongoing performance within agreed parameters over time;

Establish and Maintain Performance Threshold Standards

Process Identifier: 1.5.4.2.2

Hierarchy Level: 4

Brief Description:

Establishing and maintaining the rules and standards for performance threshold

Extended Description:

• Developing and maintaining a repository of acceptable performance threshold standards for resource instances to support the Resource Performance Management processes;

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Undertake Performance Trend Analysis

Process Identifier: 1.5.4.2.3

Hierarchy Level: 4

Brief Description:

Analyzing the performance trends to identify the performance degradation.

Extended Description:

• Undertaking trend analysis, and producing reports, of the performance of resource infrastructure to identify any longer term deterioration;

Such trend analysis may be at "macro" level or more detailed granular level such as cell level or any level in between for mobile networks.

Monitor and Analyze Resource Performance Reports, and Identify Issues

Process Identifier: 1.5.4.2.4

Hierarchy Level: 4

Brief Description:

Review the resource performance management operational report prepared by resource performance management processes.

Extended Description:

• Monitoring and analyzing the resource instance analyses produced by the Resource Performance Management processes to identify problems that may be applicable to the resource infrastructure as a whole; In order to increase enterprise agility this may be done in near real-time.

Correlate the Performance Problem Reports & Manage Inventory Repository

Process Identifier: 1.5.4.2.5

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Hierarchy Level: 4

Brief Description:

Correlating the Performance issues identified based on the performance report review and logging the results in inventory repository.

Extended Description:

• Sourcing details relating to resource instance performance and analysis from the resource inventory to assist in the development of trend analyses;

• Logging the results of the analysis into the resource inventory repository;

Manage Resource Performance Data Collection

Process Identifier: 1.5.4.2.6

Hierarchy Level: 4

Brief Description:

Developing and managing the resource performance data collection schedules and supporting the monitoring activities to gather the resource performance data.

Extended Description:

• Establishing and managing resource performance data collection schedules, including managing the collection of the necessary information from the Resource Data Collection & Distribution processes, to support proactive monitoring and analysis activity, and requests from Resource Performance;

• Management processes for additional data to support resource instance performance analysis;

Managing resource performance data collection may be supported through automation and whereby information can be shared through a common platform. Information may be collected (near) real-time.

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Establish, Maintain and Manage the Support Plans

Process Identifier: 1.5.4.2.7

Hierarchy Level: 4

Brief Description:

Establishing and managing the support facilities and maintaining the testing activities of resource performance control plans.

Extended Description:

• Establishing and managing facilities to support management of planned resource infrastructure and resource instance outages;

• Establishing, maintaining and managing the testing of resource performance control plans to cater for anticipated resource performance disruptions;

Assess and Report Resource Performance Management Processes

Process Identifier: 1.5.4.2.8

Hierarchy Level: 4

Brief Description:

Track, monitor and assess the resource performance management processes and associated costs and report.

Extended Description:

• Tracking and monitoring of the Resource Performance Management processes and associated costs (including where resource infrastructure is deployed and managed by third parties), and reporting on the capability of the Resource Performance Management processes;

In order to improve efficiency and accuracy of tracking & monitoring process, this process may monitor and report the resource performance in a automated way through a single monitoring platform.

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Provide Supporting Procedures and Quality Management Support

Process Identifier: 1.5.4.2.9

Hierarchy Level: 4

Brief Description:

Establishing, managing and maintaining the supporting facilities and quality management support for resource performance management processes.

Extended Description:

- Establishing and managing resource performance notification facilities and lists to support the Resource Performance Management notification and reporting processes
- Supporting the Support Service Quality Management process.

In order to improve efficiency of notification process, this process may be done in a automated way and provide proactive notifications to the facilities.

Support Resource Trouble Management

Process Identifier: 1.5.4.3

Hierarchy Level: 3

Brief Description:

Proactively undertaking statistically driven preventative and scheduled resource infrastructure maintenance activities, and repair activities, and monitoring, managing and reporting on the capability of the Resource Trouble Management processes.

Extended Description:

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The responsibilities of the Support Resource Trouble Management processes are twofold - support Resource Trouble Management processes by proactively undertaking statistically driven preventative and scheduled resource infrastructure maintenance activities, and repair activities, and monitoring, managing and reporting on the capability of the Resource Trouble Management processes.

These processes are responsible for ensuring that the resource infrastructure is working effectively and efficiently.

Responsibilities of these processes include, but are not limited to:

• Extracting and analyzing, including undertaking trend analysis, historical and current resource instance trouble reports and performance reports to identify potential resource infrastructure or resource instances requiring proactive maintenance and/or replacement;

• Requesting scheduling of additional resource instance data collection to assist in the analysis activity;

• Requesting scheduling of resource instance performance testing to assist in analysis activity;

• Developing and managing resource infrastructure and resource instance proactive maintenance programs;

• Requesting resource provisioning activity to prevent anticipated resource troubles associated with capacity limitations identified in the analysis activities;

• Reporting outcomes of trend analysis to Resource Development & Management processes to influence new and/or modified resource infrastructure development;

• Establishing and managing resource instance spares holding facilities, including arrangements with other parties for vendor managed spares (the actual commercial agreement is negotiated using Party Offering Development & Retirement processes);

• Management of issuing and re-stocking of spares;

• Establishing and managing resource instance return and repair programs and associated processes, including both service provider and other parties repair activities;

• Tracking and monitoring of the Resource Trouble Management processes and associated costs (including where resource infrastructure is deployed and managed by third parties), and reporting on the capability of the Resource Trouble Management processes;

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• Establishing and managing resource trouble notification facilities and lists to support the Resource Trouble Management notification and reporting processes

• Supporting the Support Service Problem Management process

Manage Resource Trouble & Performance Data Collection

Process Identifier: 1.5.4.3.1

Hierarchy Level: 4

Brief Description:

Reviewing the trend analysis and undertaking the analysis of trouble and performance report to identify the necessary preventative activities.

Extended Description:

• Extracting and analyzing, including undertaking trend analysis, historical and current resource instance trouble reports and performance reports to identify potential resource infrastructure or resource instances requiring proactive maintenance and/or replacement;

Manage Resource Infrastructure, Provisioning and Preventive Maintenance Schedules

Process Identifier: 1.5.4.3.2

Hierarchy Level: 4

Brief Description:

Proactively undertaking statistically driven preventative and scheduled resource infrastructure maintenance activities, and repair activities, and monitoring, managing and reporting on the capability of the Resource Trouble Management processes.

Extended Description:

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• Requesting scheduling of additional resource instance data collection to assist in the analysis activity;

• Requesting scheduling of resource instance performance testing to assist in analysis activity;

• Developing and managing resource infrastructure and resource instance proactive maintenance programs;

• Requesting resource provisioning activity to prevent anticipated resource troubles associated with capacity limitations identified in the analysis activities;

Create Report

Process Identifier: 1.5.4.3.3

Hierarchy Level: 4

Brief Description:

Proactively generating reports based on the trend analysis.

Extended Description:

• Reporting outcomes of trend analysis to Resource Development & Management processes to influence new and/or modified resource infrastructure development;

Establish Warehouse and Manage Spares Including other Parties

Process Identifier: 1.5.4.3.4

Hierarchy Level: 4

Brief Description:

Establishing and managing the spares houses and managing the spares including other parties spares.

Extended Description:

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• Establishing and managing resource instance spares holding facilities, including arrangements with other parties for vendor managed spares (the actual commercial agreement is negotiated using Supply Chain Development & Management processes);

• Management of issuing and re-stocking of spares;

• Establishing and managing resource instance return and repair programs and associated processes, including both service provider and other parties repair activities;

Track, Monitor and Manage RTM Processes Including other Parties

Process Identifier: 1.5.4.3.5

Hierarchy Level: 4

Brief Description:

Track, monitor and assess the resource performance management processes and associated costs and report.

Extended Description:

• Tracking and monitoring of the Resource Trouble Management processes and associated costs (including where resource infrastructure is deployed and managed by third parties), and reporting on the capability of the Resource Trouble Management processes;

Provide Support for RTM and SPM Processes

Process Identifier: 1.5.4.3.6

Hierarchy Level: 4

Brief Description:

Establishing, managing and maintaining the supporting facilities and quality management support for resource trouble management processes.

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Extended Description:

• Establishing and managing resource trouble notification facilities and lists to support the Resource Trouble Management notification and reporting processes

• Supporting the Support Service Problem Management process

Enable Resource Data Collection & Distribution

Process Identifier: 1.5.4.4

Hierarchy Level: 3

Brief Description:

Administering and management of the processes which enable the effective operation of the resource data collection and data distribution network, and monitoring, managing and reporting on the capability of the Resource Data Collection & Distribution processes

Extended Description:

The responsibilities of the Enable Resource Data Collection & Distribution processes are twofold - administering and management of the processes which enable the effective operation of the resource data collection and data distribution infrastructure, and monitoring, managing and reporting on the capability of the Resource Data Collection & Distribution processes.

The Resource Data Collection & Distribution processes may be either scheduled activities, or may be triggered as a result of ad-hoc events.

Responsibilities of these processes include, but are not limited to:

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• Administering and managing the scheduling of resource data collection and resource data distribution;

• Managing the registration and access control processes used by other processes to gain access to the collected resource data;

• Managing the registration and access control processes that enable processes to download resource data to be distributed to identified resource instances;

• Establishing and managing resource data storage facilities, and associated management processes, within the resource data collection and resource data distribution infrastructure, to be used as temporary data holding facilities as required;

• Tracking and monitoring of the Resource Data Collection and Distribution processes and associated costs, and reporting on the capability of the Resource Data Collection and Distribution processes

• Identifying any technical driven shortcomings of the resource data collection and resource data distribution infrastructures, and providing input to Resource Development & Management processes to rectify these issues.

Note that the underlying resource infrastructure used for the actual transport of resource data are managed as appropriate by other processes within the Resource-Ops and Resource-SIP horizontal process groupings

Manage and Administer Resource Data Collection and Distribution

Process Identifier: 1.5.4.4.1

Hierarchy Level: 4

Brief Description:

Administering and managing the processes which enable the effective operation of the resource data collection and data distribution network.

Extended Description:

• Administering and managing the scheduling of resource data collection and resource data distribution;

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- Managing the registration and access control processes used by other processes to gain access to the collected resource data;
- Managing the registration and access control processes that enable processes to download resource data to be distributed to identified resource instances;

In order to improve efficiency and accuracy, this process may be supported through automation and whereby information can be shared through a common platform. Information may be collected (near) real-time.

In addition, Managing the registration and access control processes may be supported through automation and grant different level of resource data accesses to other processes.

Manage Resource Data Storage Facilities and Associated Processes

Process Identifier: 1.5.4.4.2

Hierarchy Level: 4

Brief Description:

Establishing, managing and maintaining the supporting facilities.

Extended Description:

- Establishing and managing resource data storage facilities, and associated management processes, within the resource data collection and resource data distribution infrastructure, to be used as temporary data holding facilities as required;

Track, Monitor and Report Resource Data Collection Processes and Capabilities

Process Identifier: 1.5.4.4.3

Hierarchy Level: 4

Brief Description:

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Monitoring, managing and reporting on the capability of the Resource Data Collection & Distribution processes.

Extended Description:

• Tracking and monitoring of the Resource Data Collection and Distribution processes and associated costs, and reporting on the capability of the Resource Data Collection and Distribution processes;

In order to improve efficiency and accuracy of tracking, monitoring and reporting, this process may be supported through automation through a common platform.

Identify Data Collection Issues and Report

Process Identifier: 1.5.4.4.4

Hierarchy Level: 4

Brief Description:

Managing and Identifying any Data Collection capability issues and providing warnings.

Extended Description:

• Identifying any technical driven shortcomings of the resource data collection and resource data distribution infrastructures, and providing input to Resource Development & Management processes to rectify these issues.

Manage Resource Inventory

Process Identifier: 1.5.4.5

Hierarchy Level: 3

Brief Description:

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Establish, manage and administer the enterprise's resource inventory, as embodied in the Resource Inventory Database, and monitor and report on the usage and access to the resource inventory, and the quality of the data maintained in it

Extended Description:

The responsibilities of the Manage Resource Inventory processes are twofold - establish, manage and administer the enterprise's resource inventory, as embodied in the Resource Inventory Database, and monitor and report on the usage and access to the resource inventory, and the quality of the data maintained in it.

The resource inventory maintains records of all resource infrastructure and resource instance configuration, version, and status details. It also records test and performance results and any other resource related- information, required to support Resource-Ops and other processes.

The resource inventory is also responsible for maintaining the association between service instances and resource instances, created as a result of the Resource Provisioning Management processes.

Responsibilities of these processes include, but are not limited to:

• Identifying the inventory-relevant information requirements to be captured for resource infrastructure and resource instances;

• Identifying, establishing and maintaining resource inventory repository facilities;

• Establishing and managing the resource inventory management and information capture processes;

• Managing the registration and access control processes that enable processes to create, modify, update, delete and/or download resource data to and from the resource inventory;

• Ensuring the resource inventory repository accurately captures and records all identified resource infrastructure and resource instance details, through use of automated or manual audits;

• Tracking and monitoring of the usage of, and access to, the resource inventory repository and associated costs, and reporting on the findings

• Identifying any technical driven shortcomings of the resource inventory repository, and providing input to Resource Development & Management processes to rectify these issues.

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Manage Resource Inventory Database and Processes

Process Identifier: 1.5.4.5.1

Hierarchy Level: 4

Brief Description:

Establishing, managing and administering the enterprise's resource inventory

Extended Description:

• Identifying the inventory-relevant information requirements to be captured for resource infrastructure and resource instances;

• Identifying, establishing and maintaining resource inventory repository facilities;

• Establishing and managing the resource inventory management and information capture processes;

• Managing the registration and access control processes that enable processes to create, modify, update, delete and/or download resource data to and from the resource inventory;

Track and Monitor Resource Repository Capabilities

Process Identifier: 1.5.4.5.2

Hierarchy Level: 4

Brief Description:

Monitoring and reporting on the usage and access to the resource inventory, and the quality of the data maintained in it.

Extended Description:

• Tracking and monitoring of the usage of, and access to, the resource inventory repository and associated costs, and reporting on the findings;

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Identify Repository Issues and Provide Reports and Warnings

Process Identifier: 1.5.4.5.3

Hierarchy Level: 4

Brief Description:

Managing and Identifying any Inventory Repository issues and providing warnings.

Extended Description:

· Identifying any technical driven shortcomings of the resource inventory repository, and providing input to Resource Development & Management processes to rectify these issues.

Manage Number Portability

Process Identifier: 1.5.4.7

Hierarchy Level: 3

Brief Description:

The Manage Number Portability process is responsible for receiving, managing and tracking number portability requests issued by customer orders or a third CSP.

Extended Description:

The Manage Number Portability process is responsible for receiving, managing and tracking number portability requests issued by customer orders or a third CSP. The number portability process allows the execution of different types of portability such as Port In, Port Out, Port Over and Retirement from Number Portability.

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Responsibilities of these processes include, but are not limited to:

- Issuing and reception of number portability requests
- Modifying number portability request status
- Validating number portability requests
- Modifying or canceling open number portability requests
- Undertaking necessary tracking of the execution process
- Adding additional information to number portability request
- Reporting the completion status of number portability request
- Receiving portability requests from a third CSP
- Scheduling number portability request
- Managing the communication flow with Number Portability Clearinghouse
- Implementing business rules
- Implementing regulatory policies
- Determining the number portability feasibility

Create Port In Request

Process Identifier: 1.5.4.7.1

Hierarchy Level: 4

Brief Description:

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The Create Port In Request is responsible for receiving and process Port In requests issued by customer orders.

Extended Description:

The Create Port In Request is responsible for receiving and process Port In requests issued by customer orders. The process allows a customer to come up with his/her number from a third CSP.

Responsibilities of these processes include, but are not limited to:

- Issuing and reception of Port In request
- Validating Port In request
- Adding additional information to Port In request
- Managing Port In communication flow with Number Portability Clearinghouse

Receive Port Out Request

Process Identifier: 1.5.4.7.2

Hierarchy Level: 4

Brief Description:

The Receive Port Out Request is responsible for receiving and process Port Out requests issued by a third CSP.

Extended Description:

The Receive Port Out Request is responsible for receiving and process Port Out requests issued by a third CSP through the Number Portability Clearinghouse. The process allows a customer to port his/her number to a third CSP.

Responsibilities of these processes include, but are not limited to:

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- Receiving Port Out request
- Accepting or Rejecting Port Out request
- Adding additional information to Port Out request
- Managing Port Out communication flow with Number Portability Clearinghouse

Receive Port Over Request

Process Identifier: 1.5.4.7.3

Hierarchy Level: 4

Brief Description:

The Receive Port Over Request process is responsible for receiving notifications of number portability between third CSPs.

Extended Description:

The Receive Port Over Request process is responsible for receiving notifications of number portability between third CSPs. The process allows notify other systems or applications about number portability processes between third CSPs for different.

Responsibilities of these processes include, but are not limited to:

- Receiving Port Over notifications
- Reporting Port Over notifications to other systems or applications

Retire from Number Portability

Process Identifier: 1.5.4.7.4

Hierarchy Level: 4

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Brief Description:

The Retire from Number Portability process is responsible for issuing and receiving requests for Number Portability Retirement.

Extended Description:

The Retire from Number Portability process is responsible for issuing and receiving requests for Number Portability Retirement. This process allows to send back telephone numbers that were previously ported in, and receive telephone numbers that were previously ported out.

Responsibilities of these processes include, but are not limited to:

- Issuing Retirement request to send back previously ported in telephone numbers
- Receiving Retirement request of previously ported out telephone numbers
- Validating Retirement request

Track & Manage Number Portability Requests

Process Identifier: 1.5.4.7.5

Hierarchy Level: 4

Brief Description:

This process ensures that the portability activities are assigned, managed and tracked efficiently in order to meet the business rules and regulatory policies.

Extended Description:

The purpose of the Track & Manage Number Portability Requests is to ensure that the portability activities are assigned, managed and tracked efficiently in order to meet the business rules and regulatory policies.

Responsibilities of these processes include, but are not limited to:

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- Modifying the number portability request status
- Validating number portability request
- Modifying or canceling open number portability request
- Undertaking necessary tracking of the execution progress
- Adding additional information to number portability request
- Reporting the completion status of number portability request
- Generating number portability reports
- Implementing business rules
- Implementing regulatory policies

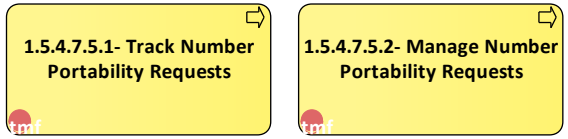

Figure 5: Track & Manage Number Portability Requests Children

Track Number Portability Requests

Process Identifier: 1.5.4.7.5.1

Hierarchy Level: 5

Brief Description:

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Undertake necessary tracking of the execution process, modify the number portability request status and monitor the jeopardy status of open portability requests to ensure that any open number portability request in jeopardy are identified and reported. This process is necessary to meet the business rules and regulatory policies.

Extended Description:

Undertake necessary tracking of the execution process, modify the number portability request status and monitor the jeopardy status of open portability requests to ensure that any open number portability request in jeopardy are identified and reported.

Responsibilities of these processes include, but are not limited to:

- Reporting the completion status of number portability request
- Generating number portability reports

Manage Number Portability Requests

Process Identifier: 1.5.4.7.5.2

Hierarchy Level: 5

Brief Description:

The Manage Number Portability Requests process ensures that the portability activities are assigned and managed efficiently and effectively.

Extended Description:

The Manage Number Portability Requests process ensures that the portability activities are assigned and managed efficiently and effectively.

Responsibilities of these processes include, but are not limited to:

- Modifying or canceling open number portability requests
- Modifying the number portability request status

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- Scheduling number portability request
- Adding additional information to number portability request
- Reporting the completion status of number portability request
- Managing communication flow with Number Portability Clearinghouse

Manage Field Workforce

Process Identifier: 1.5.4.8

Hierarchy Level: 3

Brief Description:

"Managing the staff performing manual activities along with managing the actual activity being performed.

- Note: The current focus of the Manage Field Workforce processes is field Staff and others managed through work orders, etc.

Extended Description:

The responsibilities of the Manage Field Workforce processes is to manage the lifecycle of work assignments or work orders carried out by the workforce. This will include managing the workforce staff (directly or indirectly) employed by, or operating as part of, the enterprise (i.e. technicians, clerks, managers, etc.) that are assigned to, and perform the work specified. The staff directly managed by these processes include all employees, contractors and who are paid by the enterprise. The staff indirectly managed by these processes includes all employees, consultants and contractors paid by third parties who have commercial arrangements with the enterprise. In the cases where the third parties own and manage the service and/or resource infrastructure the Manage Field Workforce processes are responsible for requesting activities to be performed rather than directly assigning specific staff. The Manage Workforce processes also enable reporting and monitoring of assigned activities. Responsibilities of these processes include, but are not limited to:

- Establishing and managing work assignment queues through which requests for work activities are received from eTOM processes;

- Establishing and managing staff lists, containing details about assignable staff such as location, skills, availability for assignment etc.;

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- Establishing, managing and distributing individuals work assignments to staff outlining the daily, or other time breadth, work assignments;

- Establishing and managing fast-track and jeopardy re-assignment capabilities to allow for modification of work assignments as required to meet jeopardy or other high priority conditions;

- Managing appointment schedules, including viewing and scheduling of appointments;

- Managing work orders, including issuance and closure;

- Forecasting assignable staffing requirements on a daily, weekly and longer period basis, based on historic work volume records, and forecast activity volumes;

- Determining work activity time estimates for all known work activities, based on actual historic results or on forward estimates, to be used as a parameter for scheduling work rosters;

- Establishing and managing recall capabilities to allow for out-of-hours staff recall in the event of unforeseen circumstances;

- Managing the registration and access control processes that enable processes to create, modify, update, delete and/or download scheduling and work assignment data to and from the workforce management system(s);

- Establishing and managing information transfer between the enterprise workforce management system(s) and those of external third parties (when the infrastructure is owned and operated by third parties);

- Ensuring MFW system(s) accurately captures and records all assignment and work scheduling details, through use of automated or manual audits;

- Tracking and monitoring of the usage of, and access to, the MFW system(s) and associated costs of the Manage Field Workforce processes, and reporting on the findings

- Identifying any technical driven shortcomings of the workforce management system(s), and providing input to Resource Development & Management processes to rectify these issues.-

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Manage Field Workforce Catalogs

Process Identifier: 1.5.4.8.1

Hierarchy Level: 4

Brief Description:

Performing the activities necessary to configure a variety of workforce management catalogs and settings required to assure that the assignable workforce is properly and efficiently utilized.

Extended Description:

The Administer and Configure Workforce Management processes perform the activities required to configure a variety of workforce management catalogs and settings required to assure that the assignable workforce is properly and efficiently utilized, and is positioned to get the required work assigned and completed within the required time frame. Administer and Configure Workforce Management scope includes the following:

•{T}Classifying different types of manual activities, including association with required skill sets and job code descriptions

•{T}Managing work order specifications and their structure filled with sequences of dependent work items. A work order can be decomposed into elementary parts that can be completed by different people in different places and at different times. These elementary pieces of work are called work items.

•{T}Managing individual technicians, allocating them to various workforce groups (workforce pools, crews, etc.) and associating them with aggregations of skills they possess, locations they cover, etc. This also includes potential "borrowing" of staff from one group to another.

•{T}Establishing and managing organization lists, containing organizational charts, serving area covered, etc.

•{T}Consolidating employee and/or third-party working hours on a specific timeframe basis (day, week, month) including management of various types of events; providing this information to the Financial Management process for generating payroll and/or to the other parties Settlement & Payments Management process for paying third parties.

•{T}Enabling and managing recall capabilities to allow for out-of-hours staff recall in the event of unforeseen circumstances

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•{T}Administering the appointment schedule, including the hours of operation of the schedule, number of appointments allowed within a certain time block, etc.

•{T}Setting up and configuring the information transfer and integration with third-party systems

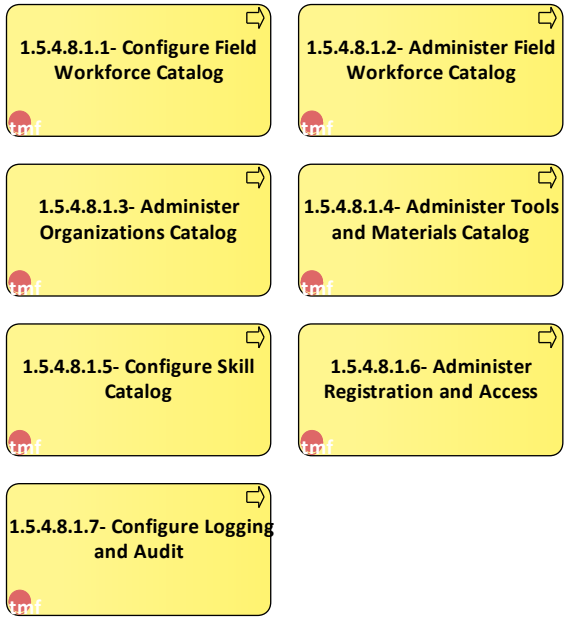

Figure 6: Manage Field Workforce Catalogs Children

Configure Field Workforce Catalog

Process Identifier: 1.5.4.8.1.1

Hierarchy Level: 5

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Brief Description:

This process configures the specification of Work and Work Order, including description, relationships, and rules.

Extended Description:

•Classifying different types of manual activities (or Work Items), including association with required skill sets and job code descriptions

•Managing work order specifications and their dependent work items

•Establishing the conditions for work items to be instantiated while decomposing work orders

•Managing standard durations of work items to be performed, depending on a variety of parameters

Administer Field Workforce Catalog

Process Identifier: 1.5.4.8.1.2

Hierarchy Level: 5

Brief Description:

This process configures the catalog of field technicians, including description, grouping, hours, relationships, and rules.

Extended Description:

• Managing individual technicians, allocating them to various workforce groups, (workforce pools, crews, etc.), forming holistic assignable units

• Associating human resources with aggregations of skills they posses, locations they cover, tools and devices they can operate, etc.

• Establishing basic working and non-working hours of human resources, including the allocation of technicians to shifts

• Managing a list of technician activities divided into two types: extending available working hours (because of overtime, for example) and reducing working hours (to allow for activities such as training and meetings)

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• Establishing the rules by which some types of activities can be ignored

• Establishing interaction with HR systems for data transfer and synchronization

Administer Organizations Catalog

Process Identifier: 1.5.4.8.1.3

Hierarchy Level: 5

Brief Description:

This process manages the description of the organizational structure of the CSP or field service partners.

Extended Description:

A CSP may have their own employees who fulfill all assurance and fulfillment jobs. The company may also delegate some of its work (or all of its work) to third-party companies that are called other parties. Both the CSP company and other parties are called organizations.

•Establishing and managing an organization list, containing organizational chart, covered serving areas, etc.

Administer Tools and Materials Catalog

Process Identifier: 1.5.4.8.1.4

Hierarchy Level: 5

Brief Description:

This process manages the catalog of tools and materials for field service.

Extended Description:

This process is responsible for managing the catalog of tools and materials that are required to fulfill various types of work.

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Configure Skill Catalog

Process Identifier: 1.5.4.8.1.5

Hierarchy Level: 5

Brief Description:

This process manages the catalog of technician's skills, including description and combination.

Extended Description:

A skill represents an ability of a person or a group of people to fulfill a particular activity.

•Managing the list of single skills and their combinations into whole sets and their descriptions

•Synchronizing the skill catalog with HR systems

Administer Registration and Access

Process Identifier: 1.5.4.8.1.6

Hierarchy Level: 5

Brief Description:

This process manages the registration and access rights for various users, including third-party systems, of the workforce management solution.

Extended Description:

•Defining user access rights, permissions, and rules depending on operations or actions

•Establishing rules and permissions for third-party system integrations and data exchange

•Tracking and monitoring the use of, and access to, the workforce management system(s) and associated costs of the Workforce Management processes

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•Establishing and managing information transfer between the enterprise workforce management system(s) and those of external third parties

•Identifying any technically driven shortcomings of the workforce management system(s), and providing input to Resource Development & Management processes to rectify these issues

Configure Logging and Audit

Process Identifier: 1.5.4.8.1.7

Hierarchy Level: 5

Brief Description:

This process manages the logging and audit functions of the workforce management solution.

Extended Description:

•Establishing system event logging, rules, and events that should be handled

•Configuring logging details and storage

•Establishing audit functions and rules

•Configuring notifications and post-functions (action to be performed when a task is complete)

Manage Schedules & Appointments

Process Identifier: 1.5.4.8.2

Hierarchy Level: 4

Brief Description:

Manages the appointment schedule of assignable staff.

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Extended Description:

The responsibilities of the Manage Appointment Schedule processes include the ability to view the availability of appropriate assignable staff as well as being able to schedule an appointment. Both the viewing and the scheduling may be based on availability of a collection of staff and not a specific person. Similarly, times could be windows of time (i.e. - AM or PM) or a specific time.

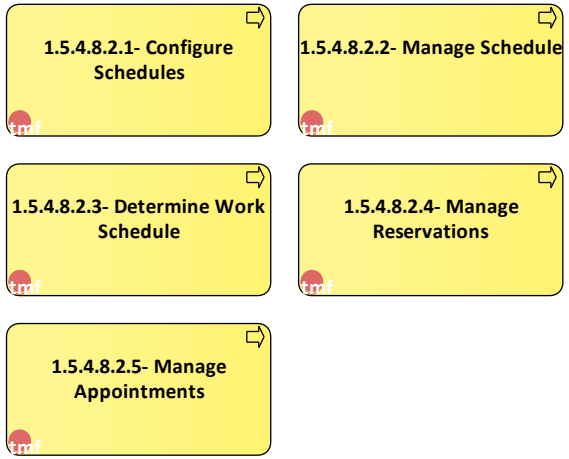

Figure 7: Manage Schedules & Appointments Children

Configure Schedules

Process Identifier: 1.5.4.8.2.1

Hierarchy Level: 5

Brief Description:

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This process creates the schedules for the organization, which show work requests and employee capabilities and availabilities. The process also manages rules for appointment, reservation, and assignment.

Extended Description:

•{T}Establishing and managing the information transfer through which other parties report on their workforce parameters and schedules

•{T}Establishing a workforce schedule which provides a single view of their capabilities and availabilities, taking into account both their own company's workforces and workforces provided by other parties

•{T}Identifying a resource plan, which is the time frame-based association of workforce with types of work. This means that associated workforce units are supposed to perform particular types of work in a specified timeframe. This schedule provides a view of the workforce in terms of work availability. It may be used for limiting and balancing work and appointments between several workforce sources and for optimizing the use of the total workforce.

•{T}Managing assignment/re-assignment rules, prioritization criteria, based on the organizational settings, type of location, type of jobs, and minimizing the cost of reassignments

•{T}Defining restriction rules for appointment scheduling, such as the number of appointments allowed within a certain block of time, etc.

•{T}Defining restriction rules for workforce reservations such as number of concurrent reservations for a particular workforce source, etc.

Manage Schedule

Process Identifier: 1.5.4.8.2.2

Hierarchy Level: 5

Brief Description:

This process is responsible for keeping the schedule of available workforce actualized, recalculating the availability schedule, and synchronizing actual workforce schedules with third parties based on information about reserved and available units.

Extended Description:

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This process is responsible for keeping the schedule of the available workforce actualized, recalculating the availability schedule and synchronizing actual workforce schedules with third parties based on information about reserved and available units.

The Workforce Management Schedule scope includes the following:

•{T}Tracking and managing workforce schedules, reflecting the actual information on workforce availability. Exchanging this information with involved third parties

•{T}Gathering and providing actual information on working hours to the Financial Management process for generating the payroll and/or to the Party Settlement & Payments Management process for paying third parties

•{T}Enabling and managing recall capabilities to allow for out-of-hours staff recall in the event of unforeseen circumstances

•{T}Managing the registration and access control processes that enable processes to create, modify, update, delete, and/or download scheduling and work assignment data to and from the workforce management system(s)

•{T}Ensuring that the workforce management system(s) accurately captures and records all assignment and work scheduling details, through use of automated or manual audits

Determine Work Schedule

Process Identifier: 1.5.4.8.2.3

Hierarchy Level: 5

Brief Description:

This process answers inquiries about available work capacity for a given time slot.

Extended Description:

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This process is responsible for answering inquiries from other processes (like Customer-Ops), given a particular work task, about what timeframes the work can be performed in, considering dependencies between work tasks. This process does not look for a particular executor, but checks available work capacity in a particular time interval.

Manage Reservations

Process Identifier: 1.5.4.8.2.4

Hierarchy Level: 5

Brief Description:

This process is responsible for managing workforce reservations.

Extended Description:

This process is responsible for managing workforce reservations. Once the reservation is made for a work item, this work item is scheduled (not necessarily assigned yet).

The scope of Manage Reservations includes the following:

•{T}Creating, modifying, and deleting workforce reservations for particular work items, taking into account dependencies between them

•{T}Confirming, canceling, or expiring workforce reservations, including defining the reservation expiration period

Manage Appointments

Process Identifier: 1.5.4.8.2.5

Hierarchy Level: 5

Brief Description:

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Manages the appointment schedule of assignable staff.

Extended Description:

This process is responsible for managing the schedule of appointments which are formal arrangements with customers about planned site visits. Appointments are associated with particular work items to be done. The responsibilities of the Manage Appointment Schedule process include the ability to view the availability of appropriate assignable staff as well as the ability to schedule an appointment. Both the viewing and the scheduling may be based on workforce availability and skill and not a specific person. Similarly, times can be windows of time (i.e. AM or PM) or a specific time.

Manage Appointment Schedule scope includes the following:

•{T}Creating and canceling appointments, including required coordination with the customer or request to Customer-Ops systems, for such coordination and updates

•{T}In-flight modification, rescheduling, and cancellation of appointments

Plan & Forecast Field Workforce

Process Identifier: 1.5.4.8.3

Hierarchy Level: 4

Brief Description:

This process is responsible for planning and forecasting future workload and workforce availability demands and for making adjustments based on reports and forecasts.

Extended Description:

Plan and Forecast Workforce processes are responsible for planning and forecasting the workload of manual activities to be assigned to technicians within the enterprise and/or third parties.

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Responsibilities of these processes include, but are not limited to:

•{T}Forecasting the future demand for workforces based on work order statistics

•{T}Forecasting the workload and respective assignable staffing requirements on a daily, weekly, and longer period basis, based on workload history, considering special events (extraordinary activities), staff vacation considerations, and distributing the available workforce in order to best meet the estimated load

•{T}Making adjustments, whenever necessary, to the estimated load:

•{T}Adjust duration estimates for all known work activities, based on actual historic results or on forward estimates

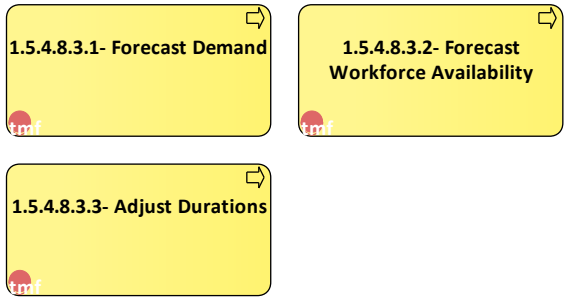

Figure 8: Plan & Forecast Field Workforce Children

Forecast Demand

Process Identifier: 1.5.4.8.3.1

Hierarchy Level: 5

Brief Description:

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This process is responsible for forecasting the future demand of workforces based on work order statistics.

Extended Description:

This process is responsible for forecasting the future demand for workforces based on work order statistics.

Forecast Workforce Availability

Process Identifier: 1.5.4.8.3.2

Hierarchy Level: 5

Brief Description:

Forecasting workforce availability on a daily, weekly, and longer period basis.

Extended Description:

Forecasting workforce availability on a daily, weekly, and longer period basis, based on workload history, considering special events (extraordinary activities), staff vacation considerations, and distributing the available workforce to best meet the estimated load

Adjust Durations

Process Identifier: 1.5.4.8.3.3

Hierarchy Level: 5

Brief Description:

This process is responsible for periodically recalculating the average time for performing a work task.

Extended Description:

This process is responsible for periodically recalculating the average time for performing a work task. This duration is recorded in Work Specification.

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Manage Resource Test

Process Identifier: 1.5.4.9

Hierarchy Level: 3

Brief Description:

Manage Resource Test Resource processes manage the end-to-end execution of a test or test scenario for resources not specific to a customer's subscribed product(s).

Extended Description:

processes manage the end-to-end execution of a test or test scenario for resources not specific to a customer. Tests can be manual or automated.

Resource Test Management processes rely on Product and Service Manage Resource Test process due to dependencies between product tests, service tests and resource tests.

Resource Test Management includes:

- Identification of resource tests needed according to fulfillment and assurance issues

- triggering of resource tests in manual or automated mode

- Execution of resource tests

- Verification of test authorization (role and context) and quota management

- Identification and prioritization of tests

- Setting up the test context and configuration

- Triggering appropriate product and service tests

- Tests can be on-demand or planned according to specific needs

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- Enrichment with product and service tests results based on applicable resource test rules

- Reporting of test results

Resource Order Management

Process Identifier: 1.5.5

Hierarchy Level: 2

Brief Description:

Resource Order Management business process directs and controls ordering, scheduling, and allocation of resources (such as materials, equipment, and personnel) within the business.

Extended Description:

Resource Order Management business process directs and controls ordering, scheduling, and allocation of resources (such as materials, equipment, and personnel) within the business.

Resource Order Management includes managing the capture of resource orders, scheduling works to support the resource order, managing the fullfillment of resource orders, picking/packing, shipping, tracking and closing orders.

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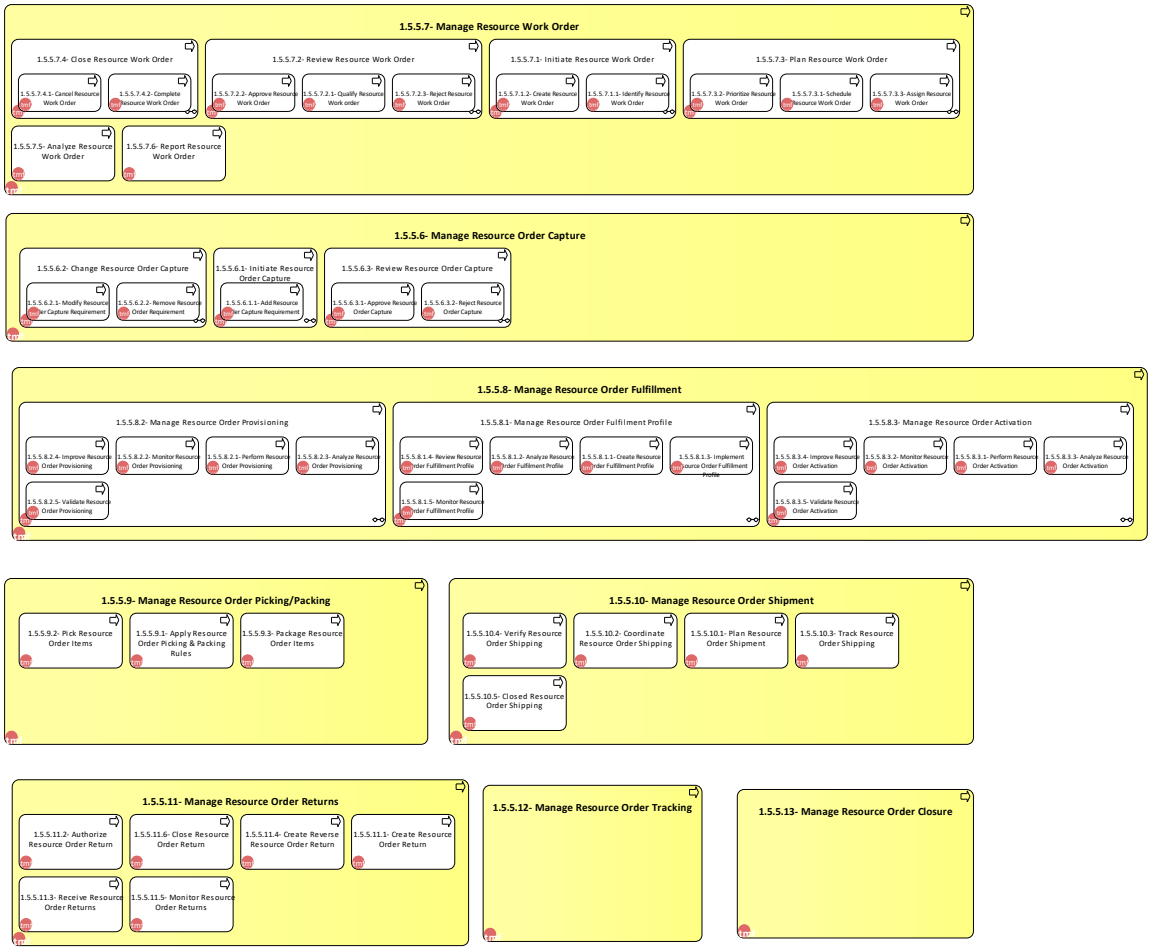

Figure 9: Resource Order Management Children

Manage Resource Order Capture

Process Identifier: 1.5.5.6

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Hierarchy Level: 3

Brief Description:

Manage Resource Order Capture is responsible for directing and controlling the capture and collection of resource orders from internal and external customers.

Extended Description:

Manage Resource Order Capture is responsible for directing and controlling the capture and collection of resource orders from internal and external customers.

The business activity begins with the receipt of an order for resource(s), checks orders for completeness and accuracy, and ensures missing or incorrect information is requested from the customer.

Initiate Resource Order Capture

Process Identifier: 1.5.5.6.1

Hierarchy Level: 4

Brief Description:

Initiate Resource Order Capture business activity is responsible for the initial activity of capturing and collecting resource orders from internal and external customers.

Extended Description:

Initiate Resource Order Capture business activity is responsible for the initial activity of capturing and collecting resource orders from internal and external customers.

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This business activity begins with the identification of the needed resources, either by a "customer" and facilitating creating the request for the resources. This business activity will gather the necessary information to complete the request order, such as the type and quantity of resources needed, delivery location, and any special instructions.

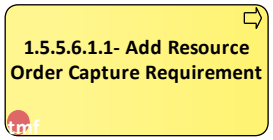

Figure 10: Initiate Resource Order Capture Children

Add Resource Order Capture Requirement

Process Identifier: 1.5.5.6.1.1

Hierarchy Level: 5

Brief Description:

Add Resource Order Capture Requirement business activity specifies the requirements of the resource order, such as the type and quantity of resources needed, delivery location, and any special instructions.

Extended Description:

Add Resource Order Capture Requirement business activity specifies the requirements of the resource order, such as the type and quantity of resources needed, delivery location, and any special instructions.

The business activity process ensures gathering any special instructions or requirements related to the order, such as certifications, specific packaging, or special handling requirements.

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Change Resource Order Capture

Process Identifier: 1.5.5.6.2

Hierarchy Level: 4

Brief Description:

Change Resource Order Capture business activity manages changes to resource orders that have been previously captured and processed to ensure that any changes to resource orders are approved and modified in a timely and accurate manner, and that the necessary resources are available.

Extended Description:

Change Resource Order Capture business activity manages changes to resource orders that have been previously captured and processed to ensure that any changes to resource orders are approved and modified in a timely and accurate manner, and that the necessary resources are available.

This business activity process ensures that any change to an order are tracked, managed and updated in a timely and accurate manner after receipt of a change request. Changes may pertain to type and quantity of resources needed, delivery location, and any special instructions.

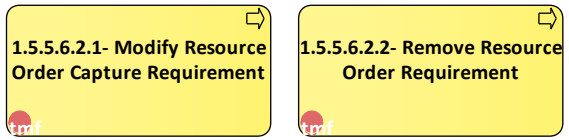

Figure 11: Change Resource Order Capture Children

Modify Resource Order Capture Requirement

Process Identifier: 1.5.5.6.2.1

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Hierarchy Level: 5

Brief Description:

Modify Resource Order Capture business activity make changes to the details of a resource order that has been previously captured and processed.

Extended Description:

Modify Resource Order Capture business activity make changes to the details of a resource order that has been previously captured and processed.

This business activity supports Change Resource Order Capture business activity by ensuring modifications to an order are updated in a timely and accurate manner.

Remove Resource Order Requirement

Process Identifier: 1.5.5.6.2.2

Hierarchy Level: 5

Brief Description:

Remove Resource Order Capture business activity cancels or removes an order that has been previously captured to be processed.

Extended Description:

Remove Resource Order Capture business activity cancels or removes an order that has been previously captured to be processed.

This business activity ensures that any cancellation to an order are done in a timely and accurate manner without any impact.

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Review Resource Order Capture

Process Identifier: 1.5.5.6.3

Hierarchy Level: 4

Brief Description:

Review Resource Order Capture business activity ensures that resource orders captured can be approved for further processing, or rejected in a timely manner to enable any changes to be done.

Extended Description:

Review Resource Order Capture business activity ensures that resource orders captured can be approved for further processing, or rejected in a timely manner to enable any changes to be done.

This business activity ensures that resource orders meet the organization's policies and standards. It supports checking the order for completeness and accuracy, and supporting change to resource order captured, including verifying any change requests that have been made to the order.

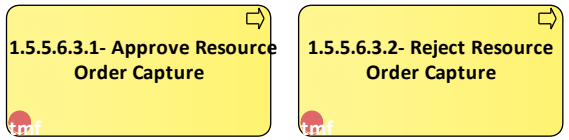

Figure 12: Review Resource Order Capture Children

Approve Resource Order Capture

Process Identifier: 1.5.5.6.3.1

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Hierarchy Level: 5

Brief Description:

Approve Resource Order Capture business activity authorizes and confirm that a resource order capture is accurate, complete and meets the organization's policies and standards.

Extended Description:

Approve Resource Order Capture business activity authorizes and confirm that a resource order capture is accurate, complete and meets the organization's policies and standards.

Approve Resource Order Capture business activity check the order for completeness and accuracy, verify that the necessary resources are available, and ensuring that the order meets the organization's policies and standards and that it can be sent for further processing, such as payment and inventory reservation.

Reject Resource Order Capture

Process Identifier: 1.5.5.6.3.2

Hierarchy Level: 5

Brief Description:

Reject Resource Order Capture business activity ensures that resource orders that are captured are denied when they do not meet the organization's policies and standards.

Extended Description:

Reject Resource Order Capture business activity ensures that resource orders that are captured are denied when they do not meet the organization's policies and standards.

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This business activity supports review of resource order capture by providing the information necessary to support the rejection and rejection reason to the customer.

Manage Resource Work Order

Process Identifier: 1.5.5.7

Hierarchy Level: 3

Brief Description:

Manage Resource Order Work business activity directs and controls all work that are required to fulfill an approved resource order by ensuring the work related to the order is planned, executed and closed in a timely and efficient manner.

Extended Description:

Manage Resource Order Work business activity directs and controls all work that are required to fulfill an approved resource order by ensuring the work related to the order is planned, executed and closed in a timely and efficient manner.

Manage Resource Order Work business activity includes activities to "Initiate Resource Work Order", "Create Resource Work Order", "Review Resource Work Order", "Plan Resource Work Order", "Close Resource Work Order", "Analyze Resource Work Order" and "Report Resource Work Order".

Initiate Resource Work Order

Process Identifier: 1.5.5.7.1

Hierarchy Level: 4

Brief Description:

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Initiate Resource Work Order business activity starts a new work order for a specific resource along with all work orders tasks, roles and supporting resources that are need.

Extended Description:

Initiate Resource Work Order business activity starts a new work order for a specific resource along with all work orders tasks, roles and supporting resources that are need.

Initiate Resource Work Order establishes the need for a new resource work order and makes sure that work order is created sufficiently in a timely and accurate manner. It includes identifying the need for a resource, managing tasks necessary, creating the work order and communicating with relevant processes and parties.

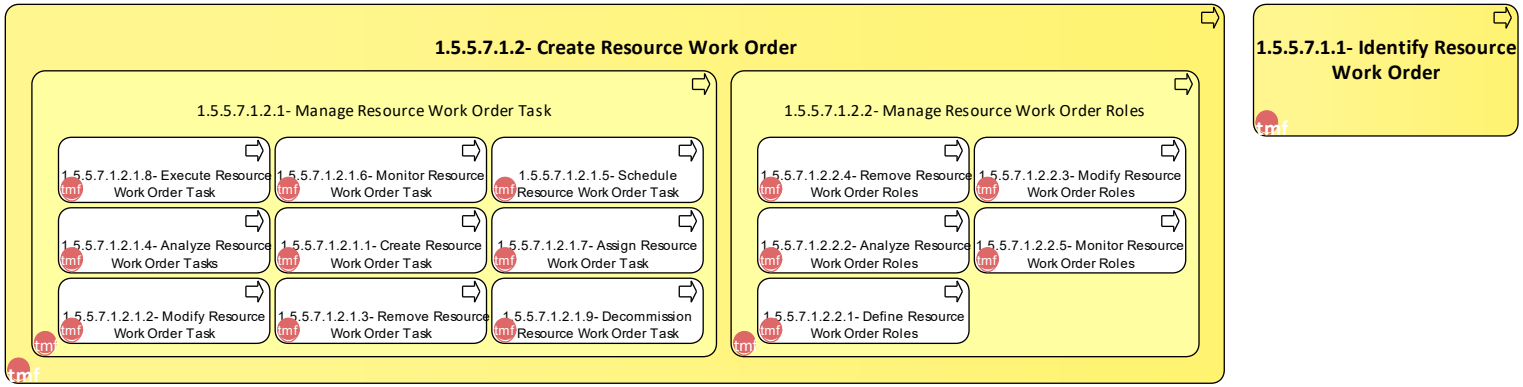

Figure 13: Initiate Resource Work Order Children

Identify Resource Work Order

Process Identifier: 1.5.5.7.1.1

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Hierarchy Level: 5

Brief Description:

Identify Resource Order Work Tasks business activity identifies the tasks that needs to be done in the work order.

Extended Description:

Identify Resource Order Work business activity identifies the tasks that needs to be done in the work order.

Identify Resource Order Work Tasks business activity sets the foundation for the work order, including identifying the tasks required, and any specific task based on the work order instructions or requirements.

Create Resource Work Order

Process Identifier: 1.5.5.7.1.2

Hierarchy Level: 5

Brief Description:

Create Resource Work Order business activity establishes all the details of the work order based on tasks that support type specifications and quantities of resources needed, delivery location/address, and any specific instructions related to scheduling of the work order.

Extended Description:

Create Resource Work Order business activity establishes all the details of the work order based on tasks that support type specifications and quantities of resources needed, delivery location/address, and any specific instructions related to scheduling of the work order.

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Create Resource Work Order business activity ensures that all the tasks relevant to the work order are in scope of the work order to enable further processing of the order.

Manage Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1

Hierarchy Level: 6

Brief Description:

Manage Resource Work Order Tasks business activity directs ad controls the execution of the tasks that have been identified for a resource work order.

Extended Description:

Manage Resource Work Order Tasks business activity directs ad controls the execution of the tasks that have been identified for a resource work order.

Manage Resource Work Order Tasks helps to ensure that the tasks are completed on time, that tasks are allocated to the appropriate parties, scheduling the tasks, monitoring progress, and making any necessary adjustments to keep the work on track and ultimately ensuring that the tasks are completed to meet the requirement of the work order.

Create Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.1

Hierarchy Level: 7

Brief Description:

Create Resource Work Order Task business activity sets up and defines tasks that need to be completed.

Extended Description:

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Create Resource Work Order Task business activity sets up and defines tasks that need to be completed.

Create Resource Work Order Task business activity establishes clear instructions and expectations for the work order task. It includes defining the task actions with deadlines.

Modify Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.2

Hierarchy Level: 7

Brief Description:

Modify Resource Work Order Task business activity makes changes or updates to an existing resource work order task.

Extended Description:

Modify Resource Work Order Task business activity makes changes or updates to an existing resource work order task.

Modify Resource Work Order Task business activity includes making changes to the task's description, changing the tasks specification, and other characteristics that define that scope of the task. This includes communicating changes to the relevant parties.

Remove Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.3

Hierarchy Level: 7

Brief Description:

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Remove Resource Work Order Task business activity is to remove a task from a resource work order.

Extended Description:

Remove Resource Work Order Task business activity is to remove a task from a resource work order.

Remove Resource Work Order Task business activity includes identifying the task to be removed, removing the task from all relevant references, and communicating the removal to relevant parties.

Analyze Resource Work Order Tasks

Process Identifier: 1.5.5.7.1.2.1.4

Hierarchy Level: 7

Brief Description:

Analyze Resource Work Order Tasks business activity reviews a resource work order task and its progress.

Extended Description:

Analyze Resource Work Order Tasks business activity reviews a resource work order task and its progress.

Analyze Resource Work Order Tasks business activity ensures that a task is on track and that any issues or risks are identified and addressed. This includes actions that review the task in progress, identify any issues or risks and providing recommendations on mitigatory actions.

Schedule Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.5

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Hierarchy Level: 7

Brief Description:

Schedule Resource Work Order Task business activity arranges the plan, including establishing the assignment of resources, and determining the time frame for completion.

Extended Description:

Schedule Resource Work Order Task business activity arranges the plan, including establishing the assignment of resources, and determining the time frame for completion.

Schedule Resource Work Order Task business activity will develop a plan for the task, assign required resources for the task, and provide details on intended events and times to carry out the task.

Monitor Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.6

Hierarchy Level: 7

Brief Description:

Monitor Resource Work Order Task business activity tracks the progress of the task.

Extended Description:

Monitor Resource Work Order Task business activity tracks the progress of the task.

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Monitor Resource Work Order Task business activity ensures that the task is tracked to be completed on time and to the required or established standards. It includes tracking progress, communicating progress and supports the making of needed adjustments to keep the task on track.

Assign Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.7

Hierarchy Level: 7

Brief Description:

Assign Resource Work Order Task business activity allocate specific related tasks needed to be completed to the resource work order along with the role required to complete the task.

Extended Description:

Assign Resource Work Order Task business activity allocate specific related tasks needed to be completed to the resource work order along with the role required to complete the task.

Assign Resource Work Order Task business activity will assign resource work order tasks to the appropriate roles. This business activity ensures that all tasks have been clearly defined and that the right roles are in place to execute and complete them. This activity involves reviewing the resource work order, identifying the necessary tasks and determining the roles to complete them.

Execute Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.8

Hierarchy Level: 7

Brief Description:

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Execute Resource Work Order Task business activity carries out the work order task that have been assigned as part of a resource work order.

Extended Description:

Execute Resource Work Order Task business activity carries out the work order task that have been assigned as part of a resource work order.

Execute Resource Work Order Task business activity ensures that tasks are completed in a timely and efficient manner, and that the outcome meets the requirements of the resource work order. This includes activities such as coordination, providing guidance and support as needed, monitoring progress to ensure task completion to schedule according to expected levels of quality.

Decommission Resource Work Order Task

Process Identifier: 1.5.5.7.1.2.1.9

Hierarchy Level: 7

Brief Description:

Decommission Resource Work Order Task business activity withdraws or removes a task safely and effectively from a resource work order.

Extended Description:

Decommission Resource Work Order Task business activity withdraws or removes a task safely and effectively from a resource work order.

Decommission Resource Work Order Task business activity ensures that resources work order tasks are properly taken offline or made inoperable/dismantles/withdrawn and removed in a way that avoids any risk. This includes preparing sub-tasks to remove the resource work order, preparing the resource to be removed, documenting the decommissioning, and inspecting resource has been safely and properly decommissioned.

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Manage Resource Work Order Roles

Process Identifier: 1.5.5.7.1.2.2

Hierarchy Level: 6

Brief Description:

Manage Resource Work Order Roles business activity direct and controls identification and assignment of the roles and responsibilities of the parties involved in fulfilling the resource order.

Extended Description:

Manage Resource Work Order Roles business activity direct and controls identification and assignment of the roles and responsibilities of the parties involved in fulfilling the resource order.

Manage Resource Work Order Roles business activity helps to ensure that the resource work orders are completed by the right parties. The underlying activities include identifying the roles and responsibilities, assigning them to the appropriate parties, and communicating the roles and responsibilities to all involved parties. This business activity is also responsible for ensuring that the roles and responsibilities are clearly defined and that there is no confusion about responsibilities.

Define Resource Work Order Roles

Process Identifier: 1.5.5.7.1.2.2.1

Hierarchy Level: 7

Brief Description:

Define Resource Work Order Roles business activity identifies and sets the roles and responsibilities of the parties involved in fulfilling a resource order.

Extended Description:

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Define Resource Work Order Roles business activity identifies and sets the roles and responsibilities of the parties involved in fulfilling a resource order.

Define Resource Work Order Roles helps ensure that there are right roles/skills and responsibilities available to meet the requirement of honoring a resource work order.

Analyze Resource Work Order Roles

Process Identifier: 1.5.5.7.1.2.2.2

Hierarchy Level: 7

Brief Description:

Analyze Resource Work Order Roles business activity is responsible for evaluating and examining the roles/skills and responsibilities of parties involved in fulfilling the resource order.

Extended Description:

Analyze Resource Work Order Roles business activity is responsible for evaluating and examining the roles/skills and responsibilities of parties involved in fulfilling the resource order.

Analyze Resource Work Order Roles business activity helps to ensure that the organization can evaluate roles that are responsible for work orders needed to meet the organization's resource order needs.

Modify Resource Work Order Roles

Process Identifier: 1.5.5.7.1.2.2.3

Hierarchy Level: 7

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Brief Description:

Modify Resource Work Order Roles business activity responsible for making changes to roles and responsibilities of the parties that are involved in fulfilling resource orders.

Extended Description:

Modify Resource Work Order Roles business activity responsible for making changes to roles and responsibilities of the parties that are involved in fulfilling resource orders.

Modify Resource Work Order Roles business activity helps to ensure that the changes needed to facilitate efficient and effective resource work orders are made.

This includes analyzing resource orders for the roles and responsibilities needed and making changes that pair the right roles to parties involved in the work order.

Remove Resource Work Order Roles

Process Identifier: 1.5.5.7.1.2.2.4

Hierarchy Level: 7

Brief Description:

Remove Resource Work Order Roles business activity is responsible for removing roles and responsibilities involved in fulfilling the resource order.

Extended Description:

Remove Resource Work Order Roles business activity is responsible for removing roles and responsibilities involved in fulfilling the resource order.

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Remove Resource Work Order Roles business activity helps to ensure that work order roles that are not necessary can be removed according to the organizations internal process. This process is also responsible for ensuring that the role removal is validated, and that the removal is communicated to all involved parties.

Monitor Resource Work Order Roles

Process Identifier: 1.5.5.7.1.2.2.5

Hierarchy Level: 7

Brief Description:

Monitor Resource Work Order Roles business activity is responsible for monitoring the roles and responsibilities involved in fulfilling the resource order.

Extended Description:

Monitor Resource Work Order Roles business activity is responsible for monitoring the roles and responsibilities involved in fulfilling the resource order.

Monitor Resource Work Order Roles business activity ensures that the work order roles are monitored to identify any issues or problems that occur, and enable make any necessary adjustments to keep the work orders on track.

Review Resource Work Order

Process Identifier: 1.5.5.7.2

Hierarchy Level: 4

Brief Description:

Review Resource Work Order business activity is responsible for ensuring that the work order meets the organization's standards and policies, and that all the necessary information is included.

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Extended Description:

Review Resource Work Order business activity is responsible for ensuring that the work order meets the organization's standards and policies, and that all the necessary information is included.

Review Resource Work Order business activity includes checking for any potential issues, such as lead times, delivery schedule possibility, or any underlying capacity constraints. It is important that a work order is accurate, qualified and complete before it is sent for further processing.

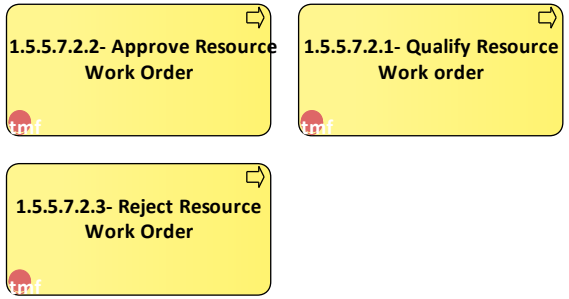

Figure 14: Review Resource Work Order Children

Qualify Resource Work order

Process Identifier: 1.5.5.7.2.1

Hierarchy Level: 5

Brief Description:

Qualify Resource Work Order business activity determine the validity and accuracy of work order requests.

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Extended Description:

Qualify Resource Work Order business activity determine the validity and accuracy of work order requests.

Qualify Resource Work Order business activity ensures that the work order is complete, accurate, and meets all necessary requirements before it is approved. This business activity includes reviewing the work order for completeness, accuracy and compliance policies and procedures.

Approve Resource Work Order

Process Identifier: 1.5.5.7.2.2

Hierarchy Level: 5

Brief Description:

Approve Resource Work Order business activity grants the consent with acceptance of the resource work order.

Extended Description:

Approve Resource Work Order business activity grants the consent with acceptance of the resource work order.

Approve Resource Work Order business activity ensures that the work order has been properly reviewed, and that it meets all necessary requirements before it is approved for onward execution. It includes verifying the qualification of the roles assigned to the work order, and signing off any works in the work order.

Reject Resource Work Order

Process Identifier: 1.5.5.7.2.3

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Hierarchy Level: 5

Brief Description:

Reject Resource Work Order business activity identifies work orders that fail to meet necessary requirements and denies their onward processing.

Extended Description:

Reject Resource Work Order business activity identifies work orders that fail to meet necessary requirements and denies their onward processing.

Reject Resource Work Order business activity ensures that only properly qualified and accurate work orders are approved for execution. This business activity includes supports the review of the work order, identifies any discrepancies, and communicates reasons for rejection.

Plan Resource Work Order

Process Identifier: 1.5.5.7.3

Hierarchy Level: 4

Brief Description:

Plan Resource Work Order business activity is responsible for planning and managing the scheduling of appointments for the work order and associated tasks that need to be done to fulfill the resource work order.

Extended Description:

Plan Resource Work Order business activity is responsible for planning and managing the scheduling of appointments for the work order and associated tasks that need to be done to fulfill the resource work order.

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Plan Resource Work Order business activity includes determining the availability of the necessary resources, such as materials, equipment, and personnel, and creating a fulfillment schedule for the work to be completed on time, and that there are any disruptions are effectively managed to support the resource work order.

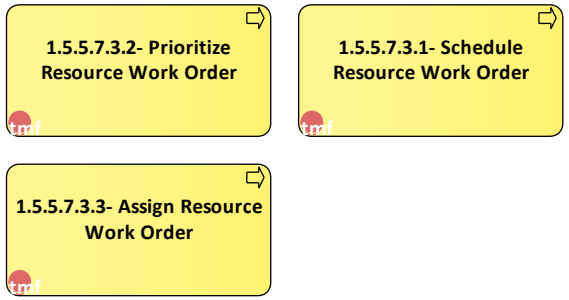

Figure 15: Plan Resource Work Order Children

Schedule Resource Work Order

Process Identifier: 1.5.5.7.3.1

Hierarchy Level: 5

Brief Description:

Schedule Resource Work Order business activity is responsible for determining the appropriate time for the completion of a work order, including all milestones, activities, and deliverables.

Extended Description:

Schedule Resource Work Order business activity is responsible for determining the appropriate time for the completion of a work order, including all milestones, activities, and deliverables.

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Schedule Resource Work Order business activity ensures that the work order can be completed within the required timeframe, and that there is an appointment in place in order not to conflict with other work orders or work order tasks. This includes liaising with relevant stakeholders to identify appropriate times, determining the duration of the work order and work order tasks, identifying any dependencies or constraints, and creating a schedule that takes into account all relevant factors.

Prioritize Resource Work Order

Process Identifier: 1.5.5.7.3.2

Hierarchy Level: 5

Brief Description:

Prioritize Resource Work Order business activity determines the relative importance of a work order.

Extended Description:

Prioritize Resource Work Order business activity determines the relative importance of a work order.

Prioritize Resource Work Order business activity ensures that the most important work order is completed ahead of less important ones, and that resources are allocated accordingly. It includes evaluating impact of different work orders, determining their urgency and importance, and creating a prioritization scheme that takes into account all relevant factors.

Assign Resource Work Order

Process Identifier: 1.5.5.7.3.3

Hierarchy Level: 5

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Brief Description:

Assign Resource Work Order business activity determines the resources required and dispatches work order tasks to facilitate completion of the work order.

Extended Description:

Assign Resource Work Order business activity dispatches work order tasks to responsible parties (resources) to execute the work order tasks.

Assign Resource Work Order business activity ensures that work order is completed with efficient allocation and use of supporting resources. Underlying activities includes identifying all supporting resources (e.g. agents, task crew, field agents etc.) for the work order, determining their availability and suitability, and allocating them to the work order task according to the schedule and priorities established.

Close Resource Work Order

Process Identifier: 1.5.5.7.4

Hierarchy Level: 4

Brief Description:

Close Resource Work Order business activity is responsible for closing out the work order when all information on the work order has been documented with an end state.

Extended Description:

Close Resource Work Order business activity is responsible for closing out the work order when all information on the work order has been documented with an end state.

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Close Resource Work Order business activity includes updating inventory levels, updating associated purchase orders, and updating the relevant accounts (e.g. customer, product, service etc). Closed Resource Work Order business activity also check that the work was cancelled or completed to the satisfaction of the requesting party.

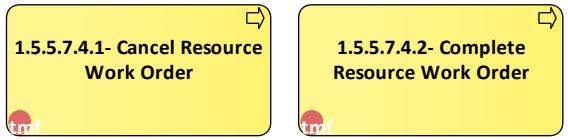

Figure 16: Close Resource Work Order Children

Cancel Resource Work Order

Process Identifier: 1.5.5.7.4.1

Hierarchy Level: 5

Brief Description:

Cancel Resource Work Order business activity terminates a resource work order that has been deemed unnecessary or invalid.

Extended Description:

Cancel Resource Work Order business activity terminates a resource work order that has been deemed unnecessary or invalid.

Cancel Resource Work Order business activity includes actions to identifying all underpinning tasks and terminating/canceling them, communicating the cancellation or termination to the relevant parties, and updating all resources and processes (intruding updates to stock and inventory reservations).

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Complete Resource Work Order

Process Identifier: 1.5.5.7.4.2

Hierarchy Level: 5

Brief Description:

Complete Resource Work Order business activity finalizes a resource work order that has been successfully completed.

Extended Description:

Complete Resource Work Order business activity finalizes a resource work order that has been successfully completed.

Complete Resource Work Order business activity ensures that there is no further work to be done. Activities include performing a final review of the work order with all task in completed status, communicating completion of the work order to the relevant parties, and updating information on the order to their respective processes.

Analyze Resource Work Order

Process Identifier: 1.5.5.7.5

Hierarchy Level: 4

Brief Description:

Analyze Resource Work Order business activity is responsible for examining work orders in a way to improve efficiency and identify any issues or problems that occurred.

Extended Description:

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Analyze Resource Work Order business activity is responsible for examining work orders in a way to improve efficiency and identify any issues or problems that occurred.

Analyze Resource Work Order business activity collects data, analyzes it, and creates reports to be shared with management in relation to the resource order management process.

Report Resource Work Order

Process Identifier: 1.5.5.7.6

Hierarchy Level: 4

Brief Description:

Report Resource Work Order business activity gives an account of resource work order activities.

Extended Description:

Report Resource Work Order business activity gives an account of resource work order activities.

Report Resource Work Order business activity including an account of issue or problems that occurred, and any work-arounds or improvements that were made to enable realize the resource order. This process is critical for ensuring that management is informed and can make decisions to improve the work order process.

Manage Resource Order Fulfillment

Process Identifier: 1.5.5.8

Hierarchy Level: 3

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Brief Description:

Manage Resource Order Fulfillment business activity directs and controls activities that ensure that resource orders are described, internally satisfied and delivered accordingly.

Extended Description:

Manage Resource Order Fulfillment business activity directs and controls activities that ensure that resource orders are described, internally satisfied and delivered accordingly.

Manage Resource Order Fulfillment business activity will coordinate with various business processes, such as inventory management, purchasing and logistics to ensure that resources are readily available and can be shipped to the customer in a timely manner.

Manage Resource Order Fulfilment Profile

Process Identifier: 1.5.5.8.1

Hierarchy Level: 4

Brief Description:

Manage Resource Order Fulfillment Profile business activity directs and controls the fulfillment information (e.g. set of predefined rules and procedures) that dictate how resource orders are processed.

Extended Description:

Manage Resource Order Fulfillment Profile business activity directs and controls the fulfillment information (e.g. set of pre-defined rules and procedures) that dictate how resource orders are processed.

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Manage Resource Order Fulfillment Profile business activity includes tasks that detail any preferred shipping methods, handling instructions, and special requirements or considerations that must be taken into account when fulfilling a particular type of resource order. This business activity helps to streamline resource order fulfillment processing, helping to ensure that resource orders are processed and delivered with minimal or no risk.

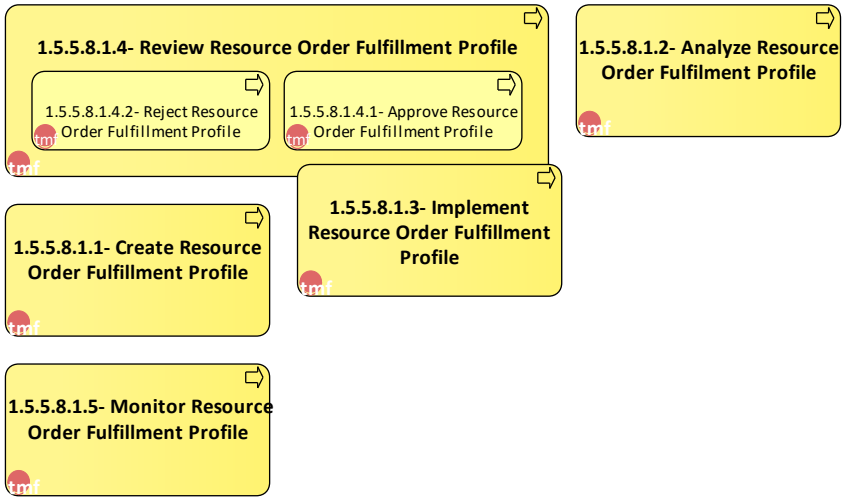

Figure 17: Manage Resource Order Fulfilment Profile Children

Create Resource Order Fulfillment Profile

Process Identifier: 1.5.5.8.1.1

Hierarchy Level: 5

Brief Description:

Create Resource Order Fulfillment Profile business activity generates the resource order fulfillment profile.

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Extended Description:

Create Resource Order Fulfillment Profile business activity generates the resource order fulfillment profile.

Create Resource Order Fulfillment Profile business activity identifies the specific information requirement of the resource order and creates the fulfillment plan that meets the resource order.

Analyze Resource Order Fulfilment Profile

Process Identifier: 1.5.5.8.1.2

Hierarchy Level: 5

Brief Description:

Analyze Resource Order Fulfillment Profile business activity parses existing fulfillment profiles to determine their suitability for the resource order.

Extended Description:

Analyze Resource Order Fulfillment Profile business activity parses existing fulfillment profiles to determine their suitability for the resource order.

Analyze Resource Order Fulfillment Profile business activity reviews and learns from historical fulfillment profiles to identify improvements. This includes reviewing historical data to identify patterns and trends.

Implement Resource Order Fulfillment Profile

Process Identifier: 1.5.5.8.1.3

Hierarchy Level: 5

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Brief Description:

Implement Resource Order Fulfillment Profile business activity puts into action the fulfillment profile created or improved through the analysis.

Extended Description:

Implement Resource Order Fulfillment Profile business activity puts into action the fulfillment profile created or improved through the analysis.

Implement Resource Order Fulfillment Profile business activity ensures there is coordination with all parties/agents in the fulfillment stage for successful execution of the fulfillment profile.

Review Resource Order Fulfillment Profile

Process Identifier: 1.5.5.8.1.4

Hierarchy Level: 5

Brief Description:

Review Resource Order Fulfillment Profile business activity approves or rejects resource order fulfillment profiles applied to resource orders.

Extended Description:

Review Resource Order Fulfillment Profile business activity approves or rejects resource order fulfillment profiles applied to resource orders.

Review Resource Order Fulfillment Profile business activity will validate resource order data with performance requirements (capacity and capability constraints) for resource orders.

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Approve Resource Order Fulfillment Profile

Process Identifier: 1.5.5.8.1.4.1

Hierarchy Level: 6

Brief Description:

Approve Resource Order Fulfillment Profile business activity supports Review Resource Order Fulfillment Profile by rendering a resource order fulfillment profile as valid for the resource order.

Extended Description:

Approve Resource Order Fulfillment Profile business activity supports Review Resource Order Fulfillment Profile by rendering a resource order fulfillment profile as valid for the resource order.

Approve Resource Order Fulfillment Profile business activity assures resource order fulfillment profiles applied to resource orders follow the organization's established business rules.

Reject Resource Order Fulfillment Profile

Process Identifier: 1.5.5.8.1.4.2

Hierarchy Level: 6

Brief Description:

Reject Resource Order Fulfillment Profile business activity supports Review Resource Order Fulfillment Profile by rendering a resource order fulfillment profile as invalid or ineffective for the resource order.

Extended Description:

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Reject Resource Order Fulfillment Profile business activity supports Review Resource Order Fulfillment Profile by rendering a resource order fulfillment profile as invalid or ineffective for the resource order.

Reject Resource Order Fulfillment Profile business activity assures resource order fulfillmet profiles matched to resource orders are best match based on the organization's established business rules.

Monitor Resource Order Fulfillment Profile

Process Identifier: 1.5.5.8.1.5

Hierarchy Level: 5

Brief Description:

Monitor Resource Order Fulfillment Profile business activity observes and tracks resource order fulfillment profile to ensure they meet the needs of the resource order.

Extended Description:

Monitor Resource Order Fulfillment Profile business activity observes and tracks resource order fulfillment profile to ensure they meet the needs of the resource order.

Monitor Resource Order Fulfillment Profile business activity tracks performance and identifies deviations from established resource order fulfillment profile. It supports the taking of actions, if necessary, that ensure resource order fulfillment profiles remain effective.

Manage Resource Order Provisioning

Process Identifier: 1.5.5.8.2

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Hierarchy Level: 4

Brief Description:

Manage Resource Order Provisioning business activity directs and controls the arrangement/composition/organization and setup of underlying resources that enable the organization to fulfill a resource order.

Extended Description:

Manage Resource Order Provisioning business activity directs and controls the arrangement/composition/organization and setup of underlying resources that enable the organization to fulfill a resource order.

Manage Resource Order Provisioning business activity ensures the allocation/assignment of all constraints (capabilities and capacities), the installation of necessary supporting resources, and testing of provisioned resources to meet the requirement readiness of the resource order.

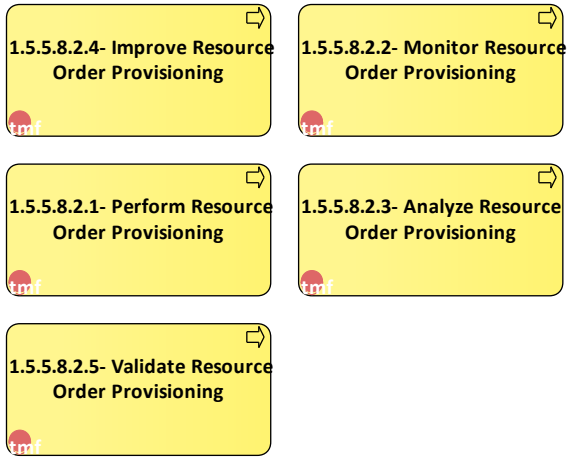

Figure 18: Manage Resource Order Provisioning Children

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Perform Resource Order Provisioning

Process Identifier: 1.5.5.8.2.1

Hierarchy Level: 5

Brief Description:

Perform Resource Order Provisioning business activity executes the tasks that prepare and allocate resources to requesting parties.

Extended Description:

Perform Resource Order Provisioning business activity executes the tasks that prepare and allocate resources to requesting parties.

Perform Resource Order Provisioning business activity acquires and allocates resources, and makes them available when they are needed. This business activity ensures that the right resources are available when they are needed in an efficient and effective way.

Monitor Resource Order Provisioning

Process Identifier: 1.5.5.8.2.2

Hierarchy Level: 5

Brief Description:

Monitor Resource Order Provisioning business activity tracks the progress of resource order provisioning and accounts for all activities in support of the resource order provisioning.

Extended Description:

Monitor Resource Order Provisioning business activity tracks the progress of resource order provisioning and accounts for all activities in support of the resource order provisioning.

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Monitor Resource Order Provisioning business activity ensure that resource order provisioning activities are done without any issues, and that resource provisioning is proceeding as planned (by work order). This business activity supports making appropriate resource order escalations.

Analyze Resource Order Provisioning

Process Identifier: 1.5.5.8.2.3

Hierarchy Level: 5

Brief Description:

Analyze Resource Order Provisioning business activity examine and review resource order provisioning tasks to identify improvements.

Extended Description:

Analyze Resource Order Provisioning business activity examine and review resource order provisioning tasks to identify improvements.

Analyze Resource Order Provisioning business activity helps to identify inefficiencies in the provisioning process before committing them, and to support improvement changes that will enhance the overall performance of a resource order provisioning process.

Improve Resource Order Provisioning

Process Identifier: 1.5.5.8.2.4

Hierarchy Level: 5

Brief Description:

Improve Resource Order Provisioning business activity implements changes or optimizations to enhance the resource order provisioning.

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Extended Description:

Improve Resource Order Provisioning business activity implements changes or optimizations to enhance the resource order provisioning.

Improve Resource Order Provisioning business activity helps to ensure that the resource order provisioning requests are done in an efficient and effective way.

Validate Resource Order Provisioning

Process Identifier: 1.5.5.8.2.5

Hierarchy Level: 5

Brief Description:

Validate Resource Order Provisioning business activity checks and authenticates resource order provisioning requests have been done.

Extended Description:

Validate Resource Order Provisioning business activity checks and authenticates resource order provisioning requests have been done.

Validate Resource Order Provisioning business activity ensures that all steps in the resource order provisioning process have been completed correctly, and that the results match requirements in the resource order provisioning request.

Manage Resource Order Activation

Process Identifier: 1.5.5.8.3

Hierarchy Level: 4

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Brief Description:

Manage Resource Order Activation business activity directs and controls the validation of the provisioned resources, the completion of all underlying work orders. and communication of the resource order activation status.

Extended Description:

Manage Resource Order Activation business activity directs and controls the validation of the provisioned resources, the completion of all underlying work orders. and communication of the resource order activation status.

Manage Resource Order Activation business activity ensures that the resource order owner/agent can be advised of capability to use the ordered resource(s).

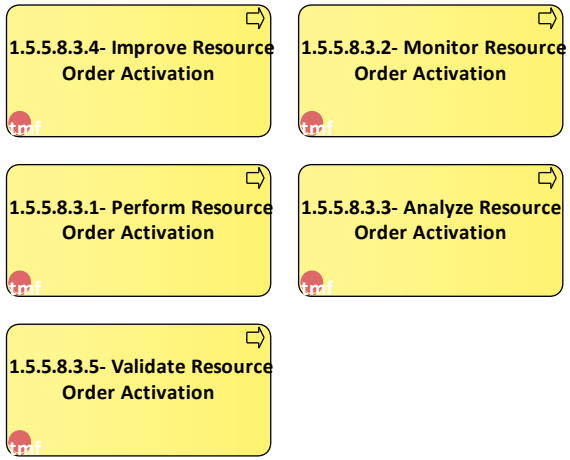

Figure 19: Manage Resource Order Activation Children

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Perform Resource Order Activation

Process Identifier: 1.5.5.8.3.1

Hierarchy Level: 5

Brief Description:

Perform Resource Order Activation business activity executes tasks that make the resource available and usable by the end user.

Extended Description:

Perform Resource Order Activation business activity executes tasks that make the resource available and usable by the end user.

Perform Resource Order Activation business activity will includes tasks that configure the readiness requirement for the resource, set final availability permissions, and ensure the readiness of underlying capability-enablers (accounts, monitors, alerts etc.) for the resource.

Monitor Resource Order Activation

Process Identifier: 1.5.5.8.3.2

Hierarchy Level: 5

Brief Description:

Monitor Resource Order Activation business activity tracks the progress and accounts of all activations in a resource order.

Extended Description:

Monitor Resource Order Activation business activity tracks the progress and accounts of all activations in a resource order.

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Monitor Resource Order Activation business activity ensures that a resource order is being activated according to implied requirements. This business activity supports identifying any issues to escalate in a timely manner in order to meet the resource order request.

Analyze Resource Order Activation

Process Identifier: 1.5.5.8.3.3

Hierarchy Level: 5

Brief Description:

Anallze Resource Order Activation business activity examine and review resource order activation tasks to identify improvements.

Extended Description:

Analyze Resource Order Activation business activity examine and review resource order activation tasks to identify improvements.

Analyze Resource Order Activation business activity helps to identify opportunities to optimize the activation process, and to improve its efficiency. This includes reviewing any data and information related to the activation, identifying patterns and trends, and making recommendations for improvements.

Improve Resource Order Activation

Process Identifier: 1.5.5.8.3.4

Hierarchy Level: 5

Brief Description:

Improve Resource Order Activation business activity implements improvements identified from analyzing a resource order activation request.

Extended Description:

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Improve Resource Order Activation business activity implements improvements identified from analyzing a resource order activation request.

Improve Resource Order Activation business activity ensures that a resource order activation request is competed effectively and efficiently by selecting the best recommendation from analyzing the resource order activation request, testing the recommendation and enable support monitoring impact.

Validate Resource Order Activation

Process Identifier: 1.5.5.8.3.5

Hierarchy Level: 5

Brief Description:

Validate Resource Order Activation business activity verifies the resource order activation request has been done and that the resource order activation is working as intended.

Extended Description:

Validate Resource Order Activation business activity verifies the resource order activation request has been done and that the resource order activation is working as intended.

Validate Resource Order Activation business activity will test the activation status of the resource order, support the evaluation of test results, and enable the necessary adjustments.

Manage Resource Order Picking/Packing

Process Identifier: 1.5.5.9

Hierarchy Level: 3

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Brief Description:

Manage Resource Order Picking/Packing business activity directs and controls the preparation of resources for delivery to the customer.

Extended Description:

Manage Resource Order Picking/Packing business activity directs and controls the preparation of resources for delivery to the customer.

Manage Resource Order Picking/Packing business activity will select the resources from inventory, package them accordingly for delivery (based on the shipment method of the resource order), applying the right mark/label/designation.

Apply Resource Order Picking & Packing Rules

Process Identifier: 1.5.5.9.1

Hierarchy Level: 4

Brief Description:

Apply Resource Order Picking & Packing Rules business activity matches established guidelines and regulations for picking and packing resource order items.

Extended Description:

Apply Resource Order Picking & Packing Rules business activity matches established guidelines and regulations for picking and packing resource order items.

Apply Resource Order Picking & Packing Rules business activity ensure that general and specific handling requirements such as compliance with safety, quality and regularity mandates, are in effect when picking and packing resource order items.

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Pick Resource Order Items

Process Identifier: 1.5.5.9.2

Hierarchy Level: 4

Brief Description:

Pick Resource Order Items gathers the appropriate item/s from inventory to fulfill the resource order.

Extended Description:

Pick Resource Order Items gathers the appropriate item/s from inventory to fulfill the resource order.

Pick Resource Order Items includes activities to locate the items from the right inventory/inventories, managing the physical of virtual picking of the items in readiness for onward packaging. This activity coordinates and is supported by warehousing, quality control, logistics and accurate and timely fulfillment of the resource order.

Package Resource Order Items

Process Identifier: 1.5.5.9.3

Hierarchy Level: 4

Brief Description:

Package Resource Order Items business activity packs resource order items appropriately and properly.

Extended Description:

Package Resource Order Items business activity packs resource order items appropriately and properly.

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Package Resource Order Items business activity ensures safe and secure delivery of the resource order item/items. This includes selecting the appropriate packaging, preparing the items for delivery with proper labeling and documenting of the package for delivery. This activity coordinates and is supported by logistics and delivery partners.

Manage Resource Order Shipment

Process Identifier: 1.5.5.10

Hierarchy Level: 3

Brief Description:

Manage Resource Order Shipment business activity directs and controls the arrangements for delivery of resources.

Extended Description:

Manage Resource Order Shipment business activity directs and controls the arrangements for delivery of resources.

Manage Resource Order Shipment business activity includes planning, coordinating with logistics, delivery functions to schedule the actual delivery, tracking the delivery and providing the resource order consignee with related information (e.g. tracking number).

Plan Resource Order Shipment

Process Identifier: 1.5.5.10.1

Hierarchy Level: 4

Brief Description:

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Plan Resource Order Shipment business activity determines the shipping method, carrier and route for the resource order.

Extended Description:

Plan Resource Order Shipment business activity determines the shipping method, carrier and route for the resource order.

Plan Resource Order Shipment business activity includes creating the shipping schedule, and determining the needed supporting resources.

Coordinate Resource Order Shipping

Process Identifier: 1.5.5.10.2

Hierarchy Level: 4

Brief Description:

Coordinate Resource Order Shipping liaises and communicates with inter-linked processes and resources to ensure timely and correct shipping.

Extended Description:

Coordinate Resource Order Shipping liaises and communicates with inter-linked processes and resources to ensure timely and correct shipping.

Coordinate Resource Order Shipping will coordinate with the organizations internal functions and external business partners, such as couriers, carriers etc., to ensure the shipment is picked up and delivered on schedule.

Track Resource Order Shipping

Process Identifier: 1.5.5.10.3

Hierarchy Level: 4

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Brief Description:

Track Resource Order Shipping business activity monitors the shipment over the shipment lifecycle.

Extended Description:

Track Resource Order Shipping business activity monitors the shipment over the shipment lifecycle.

Track Resource Order Shipping business activity ensures the shipment is tracked through all transit points, and tracking information updated accordingly. This business activity helps to proactively identify and address any issues that arise during the shipping process.

Verify Resource Order Shipping

Process Identifier: 1.5.5.10.4

Hierarchy Level: 4

Brief Description:

Verify Resource Order Shipping business activity reviews and validates delivery information.

Extended Description:

Verify Resource Order Shipping business activity reviews and validates delivery information.

Verify Resource Order Shipping business activity include checking order and delivery details to ensure accuracy, checking for discrepancies, and rectifying them before they shipment is delivered.

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Closed Resource Order Shipping

Process Identifier: 1.5.5.10.5

Hierarchy Level: 4

Brief Description:

Close Resource Order Shipping business activity finalizes shipment with an update to all relevant information.

Extended Description:

Close Resource Order Shipping business activity finalizes shipment with an update to all relevant information.

Close Resource Order Shipping business activity includes support to close the resource order, and attaining a delivery confirmation.

Manage Resource Order Returns

Process Identifier: 1.5.5.11

Hierarchy Level: 3

Brief Description:

Manage Resource Order Returns business activity directs and controls returns of ordered resources.

Extended Description:

Manage Resource Order Returns business activity directs and controls returns of ordered resources.

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Manage Resource Order Returns business activity coordinates with requesting parties all returns, inspection of returned resources, processing refunds or replacements.

Create Resource Order Return

Process Identifier: 1.5.5.11.1

Hierarchy Level: 4

Brief Description:

Create Resource Order Return business activity initiates the return of a resource order that was previously fulfilled.

Extended Description:

Create Resource Order Return business activity initiates the return of a resource order that was previously fulfilled.

Create Resource Order Return business activity include identifying the resource order that a return request if filed against and gathering needed information about the order, contacting the related parties to formalize a return authorization, preparing necessary documentation to support approvals, providing any return information (such as return address etc.), arranging for the return delivery and updating the order status.

Authorize Resource Order Return

Process Identifier: 1.5.5.11.2

Hierarchy Level: 4

Brief Description:

Authorize Resource Order Return business activity reviews and approves/rejects a resource order return according to the organizations return policy.

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Extended Description:

Authorize Resource Order Return business activity reviews and approves/rejects a resource order return according to the organizations return policy.

Authorize Resource Order Return business activity includes verifying that a return order is eligible for return, checking the conditions for return, confirming all supporting information to warrant a return are provided, assess reasons for return, approving or denying the return, and communicating authorization decision appropriately.

Receive Resource Order Returns

Process Identifier: 1.5.5.11.3

Hierarchy Level: 4

Brief Description:

Receive Resource Order Return business activity accepts returns of resource order and associated resource order items and verify they meet the conditions for return acceptance and processing.

Extended Description:

Receive Resource Order Return business activity accepts returns of resource order and associated resource order items and verify they meet the conditions for return acceptance and processing.

Receive Resource Order Return business activity includes inspecting the returned resource order items, verifying that they are in their original forms and that they meet the organization's return policies.

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Create Reverse Resource Order Return

Process Identifier: 1.5.5.11.4

Hierarchy Level: 4

Brief Description:

Create Reverse Resource Order Return business activity handle tasks to act on returned resource order and resource order items.

Extended Description:

Create Reverse Resource Order Return business activity handle tasks to act on returned resource order and resource order items.

Process Resource Order Returns business activity includes identifying root cause of returns, and efficiently handle returned resource order items.

Monitor Resource Order Returns

Process Identifier: 1.5.5.11.5

Hierarchy Level: 4

Brief Description:

Monitor Resource Order Returns business activity tracks and accounts for resource order returns and return items.

Extended Description:

Monitor Resource Order Returns business activity tracks and accounts for resource order returns and return items.

Monitor Resource Order Returns business activity includes activities to track volume, types, trends with returns.

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Close Resource Order Return

Process Identifier: 1.5.5.11.6

Hierarchy Level: 4

Brief Description:

Close Resource Order Return business activity finalizes and updates the resource order return for no further action.

Extended Description:

Close Resource Order Return business activity finalizes and updates the resource order return for no further action.

Close Resource Order Return business activity includes disposing f or restocking returned items and appropriately updating relevant records. This business activity includes verifying that returns have met the organizations return conditions, updating all party information, generating required reports or documentation, and communicating the outcome of the resource return order to any relevant parties.

Manage Resource Order Tracking

Process Identifier: 1.5.5.12

Hierarchy Level: 3

Brief Description:

Manage Resource Order Tracking business activity directs and controls the monitoring of resource order status from the time the order is placed to the time is confirmed delivered.

Extended Description:

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Manage Resource Order Tracking business activity directs and controls the monitoring of resource order status from the time the order is placed to the time is confirmed delivered.

Manage Resource Order Tracking business activity tracks the status of the resource order, provides updates to all related parties, and ensures that issues are escalated and managed promptly.

Manage Resource Order Closure

Process Identifier: 1.5.5.13

Hierarchy Level: 3

Brief Description:

Manage Resource Order Closure business activity directs and controls the closure of an order and finalizing all supporting business activities.

Extended Description:

Manage Resource Order Closure business activity directs and controls the closure of an order and finalizing all supporting business activities.

Manage Resource Order Closure business activity will support order invoicing, order payment processing, and updating the resource order status based on completion status of the order.

Resource Data Management

Process Identifier: 1.5.7

Hierarchy Level: 2

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Brief Description:

Resource Data Management business process directs and controls the engineering, generation, integration auditing, distributing, analyzing, and visualizing resource data as valuable resource to unlock tactical and strategic potential for an organization.

Extended Description:

Resource Data Management business process directs and controls the engineering, generation, integration auditing, distributing, analyzing, and visualizing resource data as valuable resource to unlock tactical and strategic potential for an organization. Resource Data Management includes all the activities for proactively defining data required, ensuring resource data collection, and organizing the data. It is supported and aligned to the Enterprise's data management process.

this process was renamed in 23.5. it was called Resource Data Collection & Distribution

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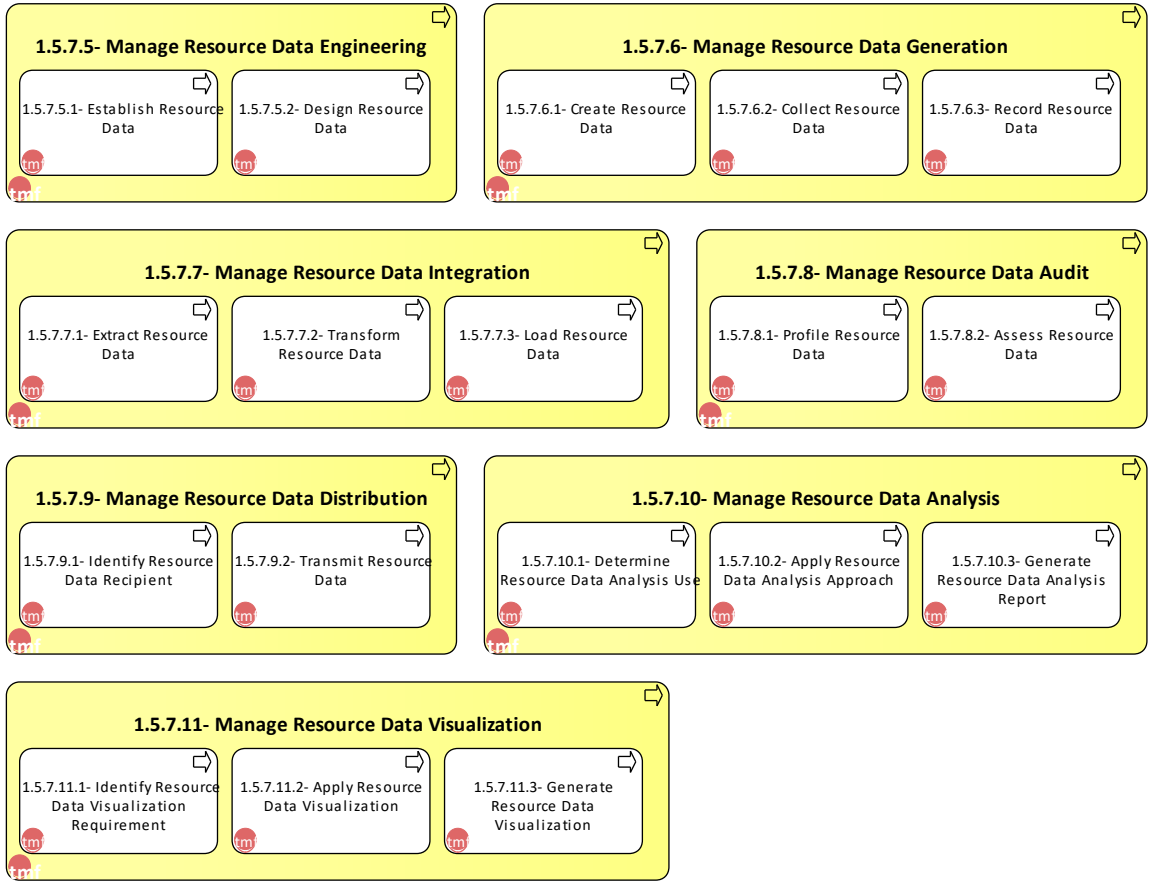

Figure 20: Resource Data Management Children

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Manage Resource Data Engineering

Process Identifier: 1.5.7.5

Hierarchy Level: 3

Brief Description:

Manage Resource Data Engineering business activity controls Resource Data Definition and Resource Data Development activities that realize resource data operations: from generating, collection, storage, integrating, auditing, analysis and visualization.

Extended Description:

Manage Resource Data Engineering business activity controls Resource Data Definition and Resource Data Development activities that realize resource data operations: from generating, collection, storage, integrating, auditing, analyzes and visualization.

Manage Resource Data Engineering supports translation, and validation of resource data for analysis. This business activity builds the underlying resource management systems to make resource data accessible. This enables business intelligence in the organization and supports, evaluation and optimization of resource performance.

Establish Resource Data

Process Identifier: 1.5.7.5.1

Hierarchy Level: 4

Brief Description:

Establish Resource Data business activity defines the data specific to a resource to enable business and technology operations.

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Extended Description:

Establish Resource Data business activity defines the data specific to a resource to enable business and technology operations.

Establish Resource Data involves specifying the characteristics or features of the resource as data objects that can support the management of the resource.

Design Resource Data

Process Identifier: 1.5.7.5.2

Hierarchy Level: 4

Brief Description:

Design Resource Data business activity plans the generation of resource data.

Extended Description:

Design Resource Data business activity plans the generation of resource data.

Design Resource Data ensures that data generated for or on resources support the organizations interests.

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Manage Resource Data Generation

Process Identifier: 1.5.7.6

Hierarchy Level: 3

Brief Description:

Generate Resource Data business activity creates/produces, collections and stores new resource data to support business and technology operations.

Extended Description:

Generate Resource Data business activity creates/produces, collections and stores new resource data to support business and technology operations.

Generate Resource Data business activity may apply different methodologies or approach to support collecting and recording resource data.

Create Resource Data

Process Identifier: 1.5.7.6.1

Hierarchy Level: 4

Brief Description:

Create Resource Data business activity is responsible for the production of new resource data.

Extended Description:

Create Resource Data business activity is responsible for the production of new resource data.

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Create Resource Data generates new data that can be used for various purposes, such as analysis, decision-making, and reporting.

Collect Resource Data

Process Identifier: 1.5.7.6.2

Hierarchy Level: 4

Brief Description:

Collect Resource Data business activity is responsible for gathering all resource data of interest and required.

Extended Description:

Collect Resource Data business activity is responsible for gathering all resource data of interest and required.

Collect Resource Data ensures all resource data required by the organization to answer, test, evaluate and report about a resource are captured.

Record Resource Data

Process Identifier: 1.5.7.6.3

Hierarchy Level: 4

Brief Description:

Record Resource Data business activity stores resource data in an organized and systematic manner.

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Extended Description:

Record Resource Data business activity stores resource data in an organized and systematic manner.

Record Resource Data includes accounting of collected resource data into databases or other forms of documentation (physical or digital) in order to make them permanent for access and use.

Manage Resource Data Integration

Process Identifier: 1.5.7.7

Hierarchy Level: 3

Brief Description:

Manage Resource Data Integration business activity controls the extraction and transformation of resource data to support the organization's use.

Extended Description:

Manage Resource Data Integration business activity controls the extraction and transformation of resource data to support the organization's use.

Manage Resource Data Integration includes combining data from different sources into a single unified view, enriching data through practices like federation and/or transforming resource data through techniques like data virtualization.

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Extract Resource Data

Process Identifier: 1.5.7.7.1

Hierarchy Level: 4

Brief Description:

Extract Resource Data business activity is responsible for getting data from one or more sources where resource data is recorded or stored.

Extended Description:

Extract Resource Data business activity is responsible for getting data from one or more sources where resource data is recorded or stored.

Extract Resource Data ensures resource data can be ingested and aggregated for use.

Transform Resource Data

Process Identifier: 1.5.7.7.2

Hierarchy Level: 4

Brief Description:

Transform Resource Data business activity is responsible for sorting, cleansing (which is data cleaning and data scrubbing), and enriching resource data etc.

Extended Description:

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Transform Resource Data business activity is responsible for sorting, cleansing (which is data cleaning and data scrubbing), and enriching resource data etc.

Transform Resource Data ensures that resource data can support intended needs and use.

Load Resource Data

Process Identifier: 1.5.7.7.3

Hierarchy Level: 4

Brief Description:

Load Resource Data business activity is responsible for making resource data integration available to a destination.

Extended Description:

Load Resource Data business activity is responsible for making resource data integration available to a destination.

Load Resource Data enhances resource data use intelligence, and operational performance by providing timely, quality and consistent resource data.

Manage Resource Data Audit

Process Identifier: 1.5.7.8

Hierarchy Level: 3

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Brief Description:

Manage Resource Data Audit business activity controls inspection of resource data to assess fit-for-purpose.

Extended Description:

Manage Resource Data Audit business activity controls inspection of resource data to assess fit-for-purpose.

Manage Resource Data Audit ensures that fit-for-purpose is compliant with the enterprises' data management policies and data governance procedures.

Profile Resource Data

Process Identifier: 1.5.7.8.1

Hierarchy Level: 4

Brief Description:

Profile Resource Data Use business activity is responsible for learning about resource data use.

Extended Description:

Profile Resource Data Use business activity is responsible for learning about resource data use.

Profile Resource Data Use ensure resource data engineering, resource data generation and resource data distribution meet the needs of resource management.

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Assess Resource Data

Process Identifier: 1.5.7.8.2

Hierarchy Level: 4

Brief Description:

Assess Resource Data Impact business activity is responsible for assessing quality of resource data on business and technology use.

Extended Description:

Assess Resource Data Impact business activity is responsible for assessing quality of resource data on business and technology use.

Assess Resource Data Impact ensures impact of data management goals with business/technology performance.

Manage Resource Data Distribution

Process Identifier: 1.5.7.9

Hierarchy Level: 3

Brief Description:

Manage Resource Data Distribution business activity controls the transfer of resource data to entity/entities that require resource data use.

Extended Description:

Manage Resource Data Distribution business activity

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controls the transfer of resource data to entity/entities that require resource data use.

Manage Resource Data Distribution includes activities that ensure that data can be properly identified and transmitted using the right methodology and technique necessary to meet the organizations' data management practice.

Identify Resource Data Recipient

Process Identifier: 1.5.7.9.1

Hierarchy Level: 4

Brief Description:

Identify Resource Data Recipient business activity establishes the entity/entities to receive resource data.

Extended Description:

Identify Resource Data Recipient business activity establishes the entity/entities to receive resource data.

Identify Resource Data Recipient ensures that resource data can be transmitted to the correct receiving destination.

Transmit Resource Data

Process Identifier: 1.5.7.9.2

Hierarchy Level: 4

Brief Description:

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Transmit Resource Data business activity transfers resource data between entities.

Extended Description:

Transmit Resource Data business activity transfers resource data between entities.

Transmit Resource Data business ensures that resource data can be transferred or exchanged.

Manage Resource Data Analysis

Process Identifier: 1.5.7.10

Hierarchy Level: 3

Brief Description:

Manage Resource Data Analysis business activity controls activities to makes sense of resource data.

Extended Description:

Manage Resource Data Analysis business activity controls activities to makes sense of resource data.

Manage Resource Data Analysis includes making sure resource data are analyzed correctly, including examining the data, synthesizing the data, inspecting the data etc. and to support creation of valuable and/or actionable insights.

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Determine Resource Data Analysis Use

Process Identifier: 1.5.7.10.1

Hierarchy Level: 4

Brief Description:

Determine Resource Data Analysis Use business activity establishes the problem or opportunity for resource data analysis.

Extended Description:

Determine Resource Data Analysis Use business activity establishes the problem or opportunity for resource data analysis.

Determine Resource Data Analysis Use includes identifying the intended use of the resource data analysis, supporting the identification of the right analysis approach and goals underpinning use of the resource data analysis.

Apply Resource Data Analysis Approach

Process Identifier: 1.5.7.10.2

Hierarchy Level: 4

Brief Description:

Apply Resource Data Analysis Approach business activity identifies for use, the right analytics approach for resource data.

Extended Description:

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Apply Resource Data Analysis Approach business activity identifies for use, the right analytics approach for resource data.

Apply Resource Data Analysis Approach ensures use of appropriate methods, tools and techniques in analyzing resource data.

Generate Resource Data Analysis Report

Process Identifier: 1.5.7.10.3

Hierarchy Level: 4

Brief Description:

Generate Resource Data Analysis Report business activity creates the resource data analysis outcome dataset and report objects.

Extended Description:

Generate Resource Data Analysis Report business activity creates the resource data analysis outcome dataset and report objects.

Generate Resource Data Analysis Report ensures formal communication of resource data analysis.

Manage Resource Data Visualization

Process Identifier: 1.5.7.11

Hierarchy Level: 3

Brief Description:

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Manage Resource Data Visualization business activity controls activities to visually present resource data in comprehendible ways.

Extended Description:

Manage Resource Data Visualization business activity controls activities to visually present resource data in comprehendible ways. Manage Resource Data Visualization includes activities to identify resource data visualization requirement, select appropriate resource data visualization to communicate a meaningful insight, through virtualization of resource data.

Identify Resource Data Visualization Requirement

Process Identifier: 1.5.7.11.1

Hierarchy Level: 4

Brief Description:

Identify Resource Data Visualization Requirement is responsible for identifying the needs to satisfy use of visuals for resource data.

Extended Description:

Identify Resource Data Visualization Requirement is responsible for identifying the needs to satisfy use of visuals for resource data.

Identify Resource Data Visualization Requirement ensures that the need for resource data visualization supports the goal of managing resource data visualization.

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Apply Resource Data Visualization

Process Identifier: 1.5.7.11.2

Hierarchy Level: 4

Brief Description:

Apply Resource Data Visualization business activity is responsible for identifying and using the appropriate visualization for resource data.

Extended Description:

Apply Resource Data Visualization business activity is responsible for identifying and using the appropriate visualization for resource data.

Apply Resource Data Visualization ensures that appropriate visuals for resource data based on established need(s) is/are used.

Generate Resource Data Visualization

Process Identifier: 1.5.7.11.3

Hierarchy Level: 4

Brief Description:

Generate Resource Data Visualization business activity is responsible for the creating the Resource Data Visualization.

Extended Description:

Generate Resource Data Visualization business activity is responsible for the creating the Resource Data Visualization.

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Generate Resource Data Visualization ensures that resource data visuals generated convey meaningful information.

Resource Trouble Management

Process Identifier: 1.5.8

Hierarchy Level: 2

Brief Description:

Responsible for the management of troubles with specific resources.

Extended Description:

Resource Trouble Management processes are responsible for the management of troubles, including security events, associated with specific resources. The objectives of these processes are to efficiently and effectively manage reported resource trouble, isolate the root cause and act to resolve the resource trouble.

Responsibilities of the Resource Trouble Management processes include, but are not limited to:

• Detecting, analyzing, managing and reporting on resource alarm event notifications;

• Initiating and managing resource trouble reports;

• Performing resource trouble localization analysis;

• Correcting and resolving resource trouble;

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- Reporting progress on resource trouble reports to other processes;
- Assigning & tracking resource trouble testing and repair activities
- Managing resource trouble jeopardy conditions.

On one hand, resource troubles may relate to Problems in the Service domain and therefore also potentially in the customer domain. On the other hand, they may relate to specific resource failures or performance degradations, which are caused by resource faults.

As such, the Resource Trouble Management processes work with specific resource alarm event notifications received from Resource Data Collection & Distribution, specific resource performance notifications from Resource Performance Management, and potential specific resource trouble notifications from Service Problem Management processes.

Resource Trouble Management processes perform analysis, decide on the appropriate actions/responses and carry them out with the intent of restoring normal operation on specific resources.

However these activities need to interact with the Service Problem Management processes, as the latter have a view on service impact. Resource Trouble Management processes are responsible for informing Service Problem Management of any potential service problems. Where the original report arose as a result of service problems, the Resource Trouble Management processes may be coordinated by Service Problem Management processes.

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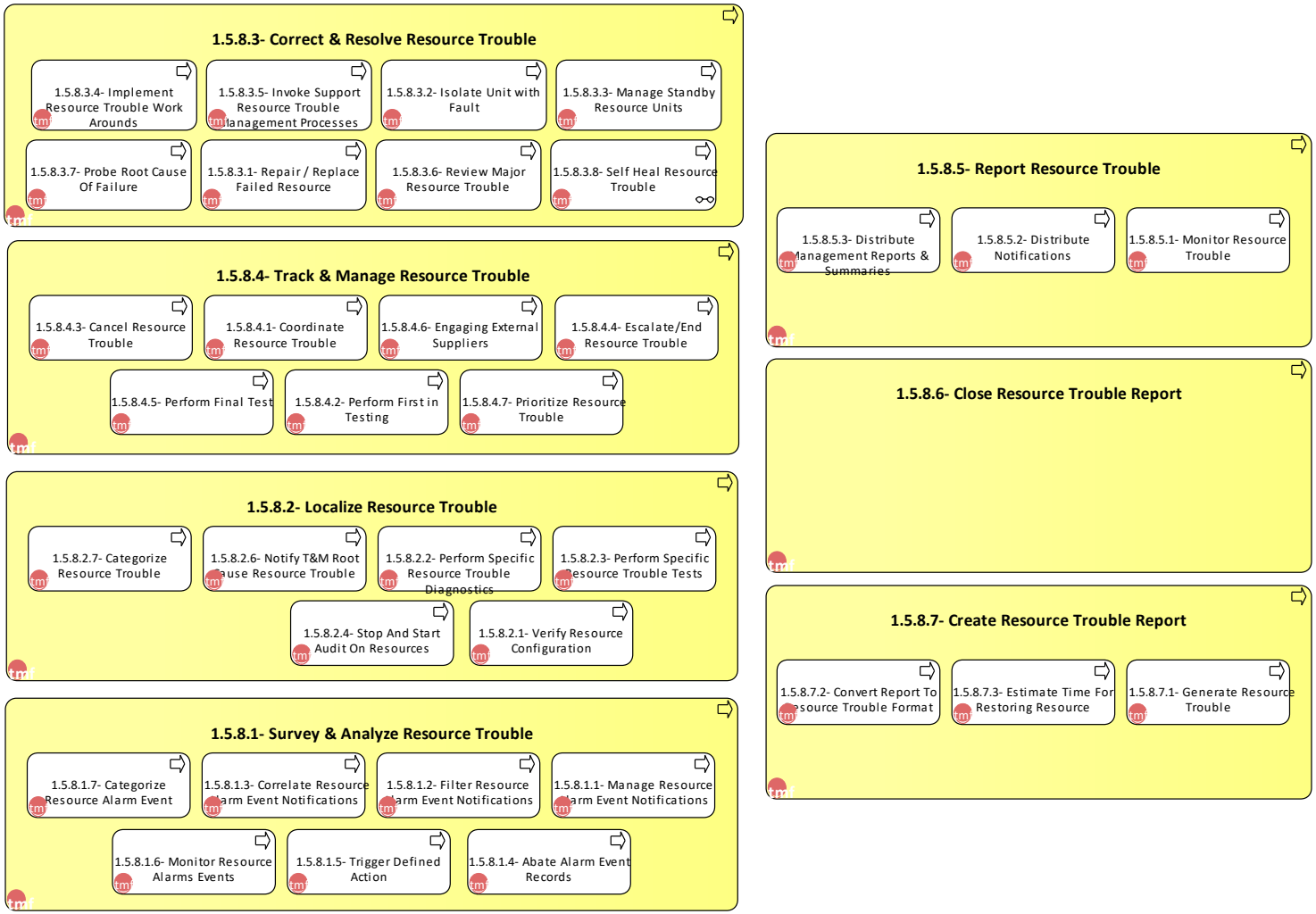

Figure 21: Resource Trouble Management Children

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Survey & Analyze Resource Trouble

Process Identifier: 1.5.8.1

Hierarchy Level: 3

Brief Description:

Monitor resource alarm event notifications and manage resource alarm event records in real-time. Resource alarm event notifications include those alarms related to security events.

Extended Description:

The objective of the Survey & Analyze Resource Trouble processes is to monitor resource alarm event notifications and manage resource alarm event records in real-time.

Responsibilities of the Survey & Analyze Resource Trouble processes include, but are not limited to:

• Detecting and collecting resource alarm event notifications;

• Initiating and managing resource alarm event records;

• Performing resource alarm event notification localization analysis;

• Correlating and filtering resource alarm event records;

• Reporting resource alarm event record status changes to other processes

• Managing resource alarm event record jeopardy conditions.

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Resource alarm event notification analysis encompasses the identification of the resource alarm event in terms of reporting entity and nature of the resource alarm event. It will then analyze the resource alarm events based on a number of criteria and then suppress redundant, transient or implied resource alarm events by means of filtering and correlation. It includes the notification of new resource alarm event records, or status changes of previously reported resource alarm event records, as well as abatement messages when resource alarm event records have been cleared.

The analysis will correlate resource alarm event notifications to planned outage notifications to remove false resource alarm event notifications arising as a result of the planned outage activity.

These processes may determine that a resource alarm event notification may represent a service impacting condition. In these circumstances this process is responsible for indicating a potential service problem to the Service Problem Management processes. As a part of this indication this process is responsible for identifying the impacted service instances associated with the resource instances presenting alarm event notifications and passing this information to the Service Problem Management processes.

Resource alarm event record correlation and filtering encompasses the correlation of redundant, transient or implied resource alarm event notifications with a specific “root cause” resource alarm event notification and associated resource alarm event record.

The Survey & Analyze Resource Trouble processes might trigger a well-defined action based on specific resource alarm event notification information as well as the non-arrival of resource alarm event notification information after a specific time interval has elapsed.

These processes are also responsible for monitoring and triggering the appropriate action when a resource alarm event record is not cleared within a pre-defined period of time.

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Manage Resource Alarm Event Notifications

Process Identifier: 1.5.8.1.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process may determine that a resource alarm event notification may represent a service impacting condition. In these circumstances this process is responsible for indicating a potential service problem to the Service Problem Management processes. As a part of this indication this process is responsible for identifying the impacted service instances associated with the resource instances presenting alarm event notifications and passing this information to the Service Problem Management processes.

Filter Resource Alarm Event Notifications

Process Identifier: 1.5.8.1.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process encompasses the identification of the resource alarm event in terms of reporting entity and nature of the resource alarm event. It then analyzes the resource alarm events based on a number of criteria and then suppresses redundant, transient or implied resource alarm events by means of filtering and correlation

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Correlate Resource Alarm Event Notifications

Process Identifier: 1.5.8.1.3

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

Resource alarm event record correlation and filtering encompasses the correlation of redundant, transient or implied resource alarm event notifications with a specific ⧫root cause⧫ resource alarm event notification and associated resource alarm event record. This process correlates resource alarm event notifications to planned outage notifications to remove false resource alarm event notifications arising as a result of the planned outage activity.

Abate Alarm Event Records

Process Identifier: 1.5.8.1.4

Hierarchy Level: 4

Brief Description:

This process includes the notification of new resource alarm event records, or status changes of previously reported resource alarm event records, as well as abatement messages when resource alarm event records have been cleared.

Extended Description:

Not used for this process element

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Trigger Defined Action

Process Identifier: 1.5.8.1.5

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process triggers a well-defined action based on specific resource alarm event notification information as well as the non-arrival of resource alarm event notification information after a specific time interval has elapsed. This process is also responsible for monitoring and triggering the appropriate action when a resource alarm event record is not cleared within a pre-defined period of time.

Monitor Resource Alarms Events

Process Identifier: 1.5.8.1.6

Hierarchy Level: 4

Brief Description:

This process is responsible for collecting and monitoring events and alarms provided by resources though systems and monitoring tools.

Extended Description:

This process is responsible for collecting and monitoring events and alarms provided by resources though systems and monitoring tools.

Categorize Resource Alarm Event

Process Identifier: 1.5.8.1.7

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Hierarchy Level: 4

Brief Description:

This process is responsible for categorizing the resource alarm events in order to support the management, filtering and correlation of events at Resource-Ops level.

Extended Description:

This process is responsible for categorizing the resource alarm events in order to support the management, filtering and correlation of events at Resource-Ops level.

Localize Resource Trouble

Process Identifier: 1.5.8.2

Hierarchy Level: 3

Brief Description:

Perform analysis to identify the root cause of the specific resource trouble including those resource troubles related to security events.

Extended Description:

The objective of the Localize Resource Trouble processes is to identify the root cause of the specific resource trouble. These processes are invoked by the Track & Manage Resource Trouble processes.

The responsibilities of these processes include, but are not limited to:

•Verifying whether the resource configuration matches the appropriate service features;

•Performing diagnostics against the specific resources;•

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•Running tests against the specific resources;

•Starting and stopping audits against specific resources

•Scheduling routine testing of the specific resources.

The Localize Resource Trouble processes will make the results of the root cause analysis available to other processes. The Localize Resource Trouble processes will update the open resource trouble report, as required during the assessment, and when the root cause has been identified.

When the process is complete the Localize Resource Trouble processes will notify the Track & Manage Resource Trouble processes.

Verify Resource Configuration

Process Identifier: 1.5.8.2.1

Hierarchy Level: 4

Brief Description:

This process verifies whether the resource configuration matches the appropriate service features.

Extended Description:

Not used for this process element

Perform Specific Resource Trouble Diagnostics

Process Identifier: 1.5.8.2.2

Hierarchy Level: 4

Brief Description:

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This process performs diagnostics against the specific resources. This includes application of signatures and other methods to capture anomalous events or malware.

Extended Description:

Not used for this process element

Perform Specific Resource Trouble Tests

Process Identifier: 1.5.8.2.3

Hierarchy Level: 4

Brief Description:

This process runs tests against the specific resources. This can include penetration tests to check for vulnerabilities.

Extended Description:

Not used for this process element

Stop And Start Audit On Resources

Process Identifier: 1.5.8.2.4

Hierarchy Level: 4

Brief Description:

This process starts and stops audits against specific resources.

Extended Description:

Not used for this process element

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Notify T&M Root Cause Resource Trouble

Process Identifier: 1.5.8.2.6

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process makes the results of the root cause analysis available to other processes. It updates the open resource trouble report, as required during the assessment, and when the root cause has been identified. This process notifies the Track & Manage Resource Trouble processes.

Categorize Resource Trouble

Process Identifier: 1.5.8.2.7

Hierarchy Level: 4

Brief Description:

This process is responsible for categorizing the resource trouble according to the type of the trouble, impact, and standardized criteria at Resource-Ops level. Moreover, this activity will be carried out with appropriate diligence, in order to provide inputs to the "Track & Manage Resource Trouble" process.

Extended Description:

This process is responsible for categorizing the resource trouble according to the type of the trouble, impact, and standardized criteria at Resource-Ops level. Moreover, this activity will be carried out with appropriate diligence, in order to provide inputs to the "Track & Manage Resource Trouble" process.

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Correct & Resolve Resource Trouble

Process Identifier: 1.5.8.3

Hierarchy Level: 3

Brief Description:

Restore or replace resources that have failed as efficiently as possible

Extended Description:

The objective of the Correct & Resolve Resource Trouble processes is to restore or replace resources that have failed as efficiently as possible.

Based on the nature of the resource failure leading to the associated resource alarm event notification, automatic restoration procedures might be triggered. Manual restoration activity is assigned to the Correct & Resolve Resource Trouble processes from the Track & Manage Resource Trouble processes.

Depending on the nature of the specific resource failure, these processes may possibly repair or replace the failed unit or specific resource. These processes are also responsible for

- to isolate a unit with a fault
- to manage the redundant resource units (e.g. hot standby)
- to (auto) heal a resource
- to identify the root cause of failure
- to repair failed specific resource (including hardware failure and software failure)

For large resource failures requiring extensive repair and/or replacement activity to restore normal operation, these processes will attempt to implement work-arounds to recover the specific resource operation. In these circumstances, recover of normal operation may require invocation of the Support Resource Trouble Management processes.

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They will also report successful restoration of normal operation, restoration through temporary work-arounds or an unsuccessful attempt at restoration to Track & Manage Resource Trouble through updates to the associated resource trouble report.

Repair / Replace Failed Resource

Process Identifier: 1.5.8.3.1

Hierarchy Level: 4

Brief Description:

This process may possibly repair, reconfigure, or replace the failed unit or specific resource (including hardware failure and software failure).

Extended Description:

Depending on the nature of the specific resource failure, this process may possibly repair, reconfigure, or replace the failed unit or specific resource. It also reports successful restoration of normal operation or an unsuccessful attempt at restoration to Track & Manage Resource Trouble through updates to the associated resource trouble report.

The responsibilities of this process include:

- Check reserve hardware parts for replacement (in case of hardware failure).

- Replace hardware parts (in case of hardware failure).

- Procure hardware if there is no spare parts (in case of hardware failure).

- Perform testing after replacement (in case of hardware failure)

- Upgrade software (in case of software failure).

- Reporting the result of the attempt to restore normal operations

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Isolate Unit with Fault

Process Identifier: 1.5.8.3.2

Hierarchy Level: 4

Brief Description:

This process is responsible for isolating a unit with a fault.

Extended Description:

Not used for this process element

Manage Standby Resource Units

Process Identifier: 1.5.8.3.3

Hierarchy Level: 4

Brief Description:

This process is responsible for managing the redundant resource units (e.g. hot standby).

Extended Description:

Not used for this process element

Implement Resource Trouble Work Arounds

Process Identifier: 1.5.8.3.4

Hierarchy Level: 4

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Brief Description:

Not used for this process element.

Extended Description:

For large service failures requiring extensive re-assignment and/or re-configuration activity to restore normal operation, this process will attempt to implement workarounds to recover the specific service operation. It also reports restoration through temporary workarounds or an unsuccessful attempt at restoration to Track & Manage Service Problem through updates to the associated Service Trouble report.

Invoke Support Resource Trouble Management Processes

Process Identifier: 1.5.8.3.5

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process recovers normal operation by invocation of the Support Resource Trouble Management processes. It also reports successful restoration of normal operation or an unsuccessful attempt at restoration to Track & Manage Resource Trouble through updates to the associated resource trouble report.

Review Major Resource Trouble

Process Identifier: 1.5.8.3.6

Hierarchy Level: 4

Brief Description:

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To review the resolution of resource troubles that have been categorized as major impact, in order to prevent recurrence and learn lessons for the future.

Extended Description:

To review the resolution of resource troubles that have been categorized as major impact, in order to prevent recurrence and learn lessons for the future.

Probe Root Cause Of Failure

Process Identifier: 1.5.8.3.7

Hierarchy Level: 4

Brief Description:

Probe the root cause of failure to identify repair point and prevention of recurrence.

Extended Description:

The objective of Probe Root Cause Of Failure process is to find out a root cause of failure causing resource trouble for dealing with hardware repair and software upgrade. The process is performed after resource healing, and analyzes the problem (e.g., the failure software bugs and/or hardware faults causing failure of network function).

The responsibilities of these processes include:

-Perform testing of resource for isolation of cause of failure .

-Identify a real cause of failure.

-Obtain existing knowledge information related to recovery.

-Determine a optimal way to repair.

-Determine schedule for repair work.

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-Update knowledge information coping with recovery if the event is unknown failure.

Self Heal Resource Trouble

Process Identifier: 1.5.8.3.8

Hierarchy Level: 4

Brief Description:

Self Heal failed resource before identification of root cause of failure.

Extended Description:

The objective of Self heal Resource Trouble process is to promptly recover failed resource, especially software-based resource (e.g., virtual network function: VNF) before identification of the detailed root cause of failure, such as software bugs like memory leak, and hardware failure like fan fault and link down.

The responsibilities of these processes include:

- Isolate failed resource (e.g., failed VNF) from service.

- Check available resource related to infrastructure (e.g., CPU, memory, Disk etc..) to create new network function for recovery.

-Create new network function for recovery.

-Perform testing in order to check normality of the created function.

The objective of Recover Resource Trouble process is to promptly recover failed resource, especially software-based resource (e.g., virtual network function: VNF) before identification of the detailed root cause of failure, such as software bugs like memory leak, and hardware failure like fan fault and link down.

The responsibilities of these processes include:

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- Decide healing option based on a healing policy/rule of operator.

- Isolate failed resource (e.g., failed VNF) from service.

- Check available resource related to infrastructure (e.g., CPU, memory, Disk etc..) to create new network function for recovery.

-Create new network function for recovery.

-Perform testing in order to check normality of the created function.

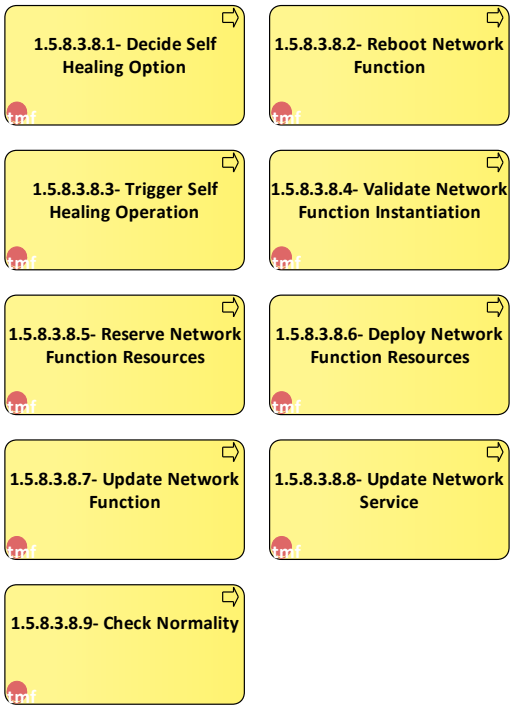

Figure 22: Self Heal Resource Trouble Children

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Decide Self Healing Option

Process Identifier: 1.5.8.3.8.1

Hierarchy Level: 5

Brief Description:

Decide Self Healing Option based on a healing policy/rule of operator.

Extended Description:

The objective of "Decide Self Healing Option" process is a responsibility to decide how recover the failure based on a self healing policy (e.g. total number of times of violating a certain threshold, emergency level of a detected alarm or service level agreement that depends on each service provider). For example, the patterns of recover are divided into following patterns:

- Physical network function self healing

- Virtual network self function healing

- Network service self healing

Reboot Network Function

Process Identifier: 1.5.8.3.8.2

Hierarchy Level: 5

Brief Description:

Execute reboot operation for network function resources.

Extended Description:

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The objective of "Reboot Network Function" process is to execute reboot operation. If the failure status still remains after the process execution, it may trigger other healing process or hardware exchange process.

Trigger Self Healing Operation

Process Identifier: 1.5.8.3.8.3

Hierarchy Level: 5

Brief Description:

Trigger lifecycle operation processes for executing network function self healing.

Extended Description:

The objective of "Trigger Self Healing Operation" process is to call healing operation processes with lifecycle management tasks. In case of success, complete or partial healing has been executed. In case of failure, appropriate error information is returned.

Validate Network Function Instantiation

Process Identifier: 1.5.8.3.8.4

Hierarchy Level: 5

Brief Description:

Validate to deploy network function.

Extended Description:

The objective of "Validate Network Function Instantiation" process is to check a feasibility for deploying network function. This process grants an authorization of a lifecycle operation. When applicable, this process give a permission for reserving network function resources (e.g., virtualized infrastructure with location information) based on network function catalog.

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Reserve Network Function Resources

Process Identifier: 1.5.8.3.8.5

Hierarchy Level: 5

Brief Description:

Reserve required network function, infrastructure and network resources for self healing.

Extended Description:

Reserve Network Function Resources process is responsible to reserve the required infrastructure resources for self healing which may corresponding to following elements, but not limited to:

- Number of CPU core, memory size

- Resource type, minimum bandwidth, QoS

- Storage type, size

Deploy Network Function Resources

Process Identifier: 1.5.8.3.8.6

Hierarchy Level: 5

Brief Description:

Deploy reserved infrastructure and network resources for self healing.

Extended Description:

The objective of "Deploy Network Function Resources." process is to deploy the needed infrastructure resources and network connectivity.

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Update Network Function

Process Identifier: 1.5.8.3.8.7

Hierarchy Level: 5

Brief Description:

Update configuration of network function.

Extended Description:

The objective of "Update Network Function" process is to activate deployed network function instances by modifying a configuration. The result of configuration may corresponds to failed VNF configuration or template.

Update Network Service

Process Identifier: 1.5.8.3.8.8

Hierarchy Level: 5

Brief Description:

Update configuration of a constituent instances of network service.

Extended Description:

The objective of "Update Network Service" process is to activate deployed network connectivity which are constituents of healed network service.

Check Normality

Process Identifier: 1.5.8.3.8.9

Hierarchy Level: 5

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Brief Description:

Check normality of network function instance/network connectivity.

Extended Description:

The objective of "Check Normality" process is to perform testing of virtual network function instance/connection statement (e.g. to check a current status of alarms regarding failed resources).

Track & Manage Resource Trouble

Process Identifier: 1.5.8.4

Hierarchy Level: 3

Brief Description:

Ensure testing, repair and restoration activities are assigned, coordinated and tracked efficiently, and that escalation is invoked as required for any open resource trouble reports in jeopardy

Extended Description:

The objective of the Track & Manage Resource Trouble is to ensure testing, repair and restoration activities are assigned, coordinated and tracked efficiently, and that escalation is invoked as required for any open resource trouble reports in jeopardy. Responsibilities of these processes include, but are not limited to:

•Initiating first-in testing using automated remote testing capabilities;

•Adding additional information to an open resource trouble report based on the first-in testing;

•Scheduling, assigning and coordinating repair and restoration activities;

•Initiate any final testing to confirm clearance of the service problem;

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•Undertake necessary tracking of the execution progress;

•Modifying information in an existing resource trouble report based on assignments;

•Modifying the resource trouble report status;

•Canceling a resource trouble report when the specific trouble was related to a false alarm event

•Monitoring the jeopardy status of open resource trouble reports, and escalating resource trouble reports as necessary.

These processes will co-ordinate all the actions necessary in order to guarantee that all tasks are finished at the appropriate time and in the appropriate sequence.

The Track & Manage Resource Trouble processes are responsible for engaging external suppliers in correction and recovery activities when higher level expertise and/or higher level support is required to resolve the resource trouble. This engagement can be linked to the priority of the resource trouble report, and could occur automatically for highest priority resource trouble reports.

The Track & Manage Resource Trouble processes will also inform the Close Resource Trouble processes by modifying the resource trouble report status to cleared when the resource trouble has been resolved.

Coordinate Resource Trouble

Process Identifier: 1.5.8.4.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

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This process coordinates all the actions necessary in order to guarantee that all tasks are finished at the appropriate time and in the appropriate sequence. It schedules, assigns and coordinates repair and restoration activities, undertakes necessary tracking of the execution progress, modifies information in an existing resource trouble report based on assignments, and modifies the resource trouble report status;

Perform First in Testing

Process Identifier: 1.5.8.4.2

Hierarchy Level: 4

Brief Description:

This process initiates first-in testing using automated remote testing capabilities, and adds additional information to an open resource trouble report based on the first-in testing.

Extended Description:

Not used for this process element

Cancel Resource Trouble

Process Identifier: 1.5.8.4.3

Hierarchy Level: 4

Brief Description:

This process cancels a resource trouble report when the specific trouble was related to a false alarm event.

Extended Description:

Not used for this process element

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Escalate/End Resource Trouble

Process Identifier: 1.5.8.4.4

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process provides the escalations between support levels and actors when a resource trouble cannot be solved by the current assigned group. Therefore, the escalation can be carried out between supports levels at Resource-Ops level, or, it can be escalated to others actors, such as, supplier/partners. It is essential the ability to rapidly escalate resource troubles according to clear and agreed rules.

Perform Final Test

Process Identifier: 1.5.8.4.5

Hierarchy Level: 4

Brief Description:

This process initiates any final testing to confirm clearance of the service problem.

Extended Description:

Not used for this process element

Engaging External Suppliers

Process Identifier: 1.5.8.4.6

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Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process is responsible for engaging external suppliers in correction and recovery activities when higher level expertise and/or higher level support is required to resolve the resource trouble. This engagement can be linked to the priority of the resource trouble report, and could occur automatically for highest priority resource trouble reports.

Prioritize Resource Trouble

Process Identifier: 1.5.8.4.7

Hierarchy Level: 4

Brief Description:

This process is responsible for assigning the prioritization to resource trouble in order to establish an order for managing it. The prioritization is assigned (usually automatically) according to criteria, such as, categorization and resources affected.

Extended Description:

This process is responsible for assigning the prioritization to resource trouble in order to establish an order for managing it. The prioritization is assigned (usually automatically) according to criteria, such as, categorization and resources affected.

Report Resource Trouble

Process Identifier: 1.5.8.5

Hierarchy Level: 3

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Brief Description:

Monitor the status of resource trouble reports provide notifications of any changes and provide management reports. This includes resource trouble caused by security events.

Extended Description:

The objective of the Report Resource Trouble processes is to monitor the status of resource trouble reports, provide notifications of any changes and provide management reports.

These processes are responsible for continuously monitoring the status of resource trouble reports and managing notifications to processes and other parties registered to receive notifications of any status changes, for example, Resource Performance Management and Service Quality Management. Notification lists are managed and maintained by the Support Resource Trouble Management processes.

These processes record, analyze and assess the resource trouble report status changes to provide management reports and any specialized summaries of the efficiency and effectiveness of the overall Resource Trouble Management process. These specialized summaries could be specific reports required by specific audiences.

These processes will make the necessary reports about the resource trouble that occurred, the root cause and the activities carried out for restoration.

Monitor Resource Trouble

Process Identifier: 1.5.8.5.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

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This process is responsible for continuously monitoring the status of resource trouble reports and managing notifications to processes and other parties registered to receive notifications of any status changes, for example, Resource Performance Management and Service Quality Management. Notification lists are managed and maintained by the Support Resource Trouble Management processes.

Distribute Notifications

Process Identifier: 1.5.8.5.2

Hierarchy Level: 4

Brief Description:

This process makes the necessary reports about the Service Problem that occurred, the root cause and the activities carried out for restoration.

Extended Description:

Not used for this process element

Distribute Management Reports & Summaries

Process Identifier: 1.5.8.5.3

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process records, analyzes and assesses the Service Performance Degradation Report status changes to provide management reports and any specialized summaries of the efficiency and effectiveness of the overall Service Quality Management process. These specialized summaries could be specific reports required by specific audiences.

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Close Resource Trouble Report

Process Identifier: 1.5.8.6

Hierarchy Level: 3

Brief Description:

Close a resource trouble report when the resource problem has been resolved

Extended Description:

The objective of the Close Service Trouble Report processes is to close a service trouble report when the service problem has been resolved.

These processes monitor the status of all open service trouble reports, and recognize that a service trouble report is ready to be closed when the status is changed to cleared.

Create Resource Trouble Report

Process Identifier: 1.5.8.7

Hierarchy Level: 3

Brief Description:

Create a new resource trouble report

Extended Description:

The objective of the Create Resource Trouble Report process is to create a new resource trouble report.

A new resource trouble report may be created as a result of resource alarm event notification analysis, and subsequent creation of new resource alarm event records, undertaken by the Survey & Analyze Resource Trouble processes, or at the request of analysis undertaken by other processes in the

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Service-Ops or Resource-Ops (in particular a Service Trouble Report can generate one or more Resource Trouble Reports) or Engaged Party-Ops and Resource-Ops layers which detect that some form of failure has occurred for which resource restoration activity is required to restore normal operation.

If the resource trouble report is created as a result of a notification or request from processes other than the Survey & Analyze Resource Trouble processes, the Create Resource Trouble Report processes are responsible for converting the received information into a form suitable for the Resource Trouble Management processes, and for requesting additional information if required.

These processes will make estimates of the time to restore resource which will be included in the new resource trouble report so that other processes can gain access to this information.

Generate Resource Trouble

Process Identifier: 1.5.8.7.1

Hierarchy Level: 4

Brief Description:

This process creates a new resource trouble report.

Extended Description:

Not used for this process element

Convert Report To Resource Trouble Format

Process Identifier: 1.5.8.7.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

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Extended Description:

If the resource trouble report is created as a result of a notification or request from processes other than the Survey & Analyze Resource Trouble processes, this process is responsible for converting the received information into a form suitable for the Resource Trouble Management processes, and for requesting additional information if required.

Estimate Time For Restoring Resource

Process Identifier: 1.5.8.7.3

Hierarchy Level: 4

Brief Description:

This process estimates the time to restore service which is included in the new Service Trouble report so that other processes can gain access to this information.

Extended Description:

Not used for this process element

Resource Performance Management

Process Identifier: 1.5.9

Hierarchy Level: 2

Brief Description:

Managing, tracking, monitoring, analyzing, controlling and reporting on the performance of specific resources

Extended Description:

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Resource Performance Management processes encompass managing, tracking, monitoring, analyzing, controlling and reporting on the performance of specific resources. They work with basic information received from the Resource Data Collection & Distribution processes.

If the analysis identifies a resource performance violation or a potential service performance violation, information will be passed to Resource Trouble Management and/or Service Quality Management as appropriate. The latter processes are responsible for deciding on and carrying out the appropriate action/response. This may include requests to the Resource Performance Management processes to install controls to optimize the specific resource performance.

The Resource Performance Management processes will continue to track the resource performance problem, ensuring that resource performance is restored to a level required to support services.

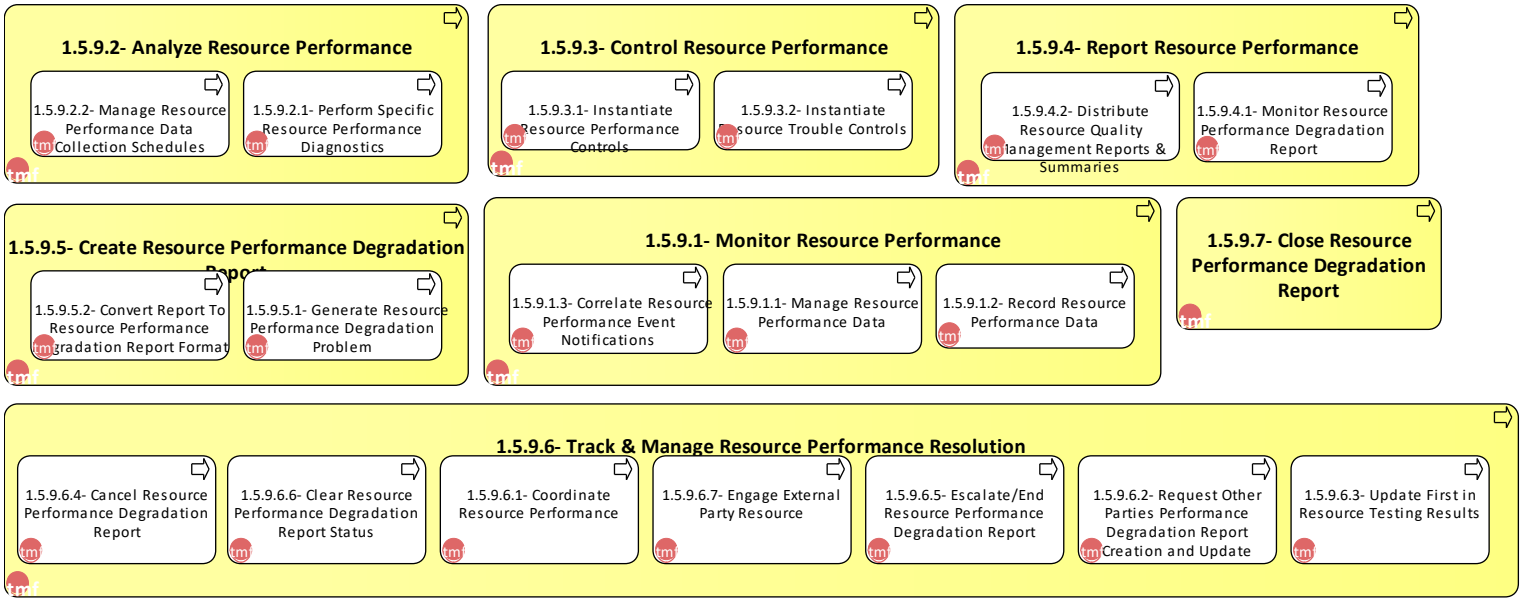

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Figure 23: Resource Performance Management Children

Monitor Resource Performance

Process Identifier: 1.5.9.1

Hierarchy Level: 3

Brief Description:

Monitor received resource performance information and undertake first-in detection.

Extended Description:

The objective of the Monitor Resource Performance processes is to monitor received resource performance information and undertake first-in detection.

The responsibilities of the processes include, but are not limited to:

•Undertaking the role of first in detection by monitoring the received specific resource performance data;

•Comparing the received specific resource performance data to performance standards set for each specific resource (available from the Resource Inventory);

•Assessing and recording received specific resource performance data which is within tolerance limits for performance standards, and for which continuous monitoring and measuring of specific resource performance is required;

•Recording the results of the continuous monitoring for reporting through the Report Resource Performance processes;

•Detecting performance threshold violations which represent specific resource failures due to abnormal performance;

•Passing information about resource failures due to performance threshold violations to Resource Trouble Management to manage any necessary restoration activity as determined by that process;

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•Passing information about potential specific service performance degradations arising from specific resource degradations to Service Quality Management to manage any necessary restoration activity as determined by that process;

•Detecting performance degradation for specific resources which provide early warning of potential issues;

•Forwarding resource performance degradation notifications to other Resource Performance Management processes, which manage activities to restore normal specific resource performance

•Logging specific resource performance degradation and violation details within the repository in the Manage Resource Inventory processes to ensure historical records are available to support the needs of other processes.

Manage Resource Performance Data

Process Identifier: 1.5.9.1.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process monitors and logs the received specific resource performance quality data, compares the received specific resource performance data to performance standards set for each specific resource (available from the Resource Inventory), detects performance threshold violations which represent specific resource failures due to abnormal performance, and detects performance degradation for specific resources which provide early warning of potential issues. This process undertakes the role of first in detection by monitoring the received specific resource performance data;

Record Resource Performance Data

Process Identifier: 1.5.9.1.2

Hierarchy Level: 4

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Brief Description:

Not used for this process element.

Extended Description:

This process assesses and records received specific resource performance data which is within tolerance limits for performance standards, and for which continuous monitoring and measuring of specific resource performance is required. It records the results of the continuous monitoring for reporting through the Report Resource Performance processes, and logs specific resource performance degradation and violation details within the repository in the Manage Resource Inventory processes to ensure historical records are available to support the needs of other processes.

Correlate Resource Performance Event Notifications

Process Identifier: 1.5.9.1.3

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process passes information about potential specific service performance degradations arising from specific resource degradations to Service Quality Management to manage any necessary restoration activity as determined by that process. It passes information about resource failures due to performance threshold violations to Resource Trouble Management to manage any necessary restoration activity as determined by that process. It forwards resource performance degradation notifications to other Resource Performance Management processes, which manage activities to restore normal specific resource performance

Analyze Resource Performance

Process Identifier: 1.5.9.2

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Hierarchy Level: 3

Brief Description:

Analyze and evaluate the performance of specific resources

Extended Description:

The objective of the Analyze Resource Performance processes is to analyze the information received from the Monitor Resource Performance process to evaluate the performance of a specific resource.

The responsibilities of the processes include, but are not limited to:

•Undertaking analysis as required on specific resource performance information received from the Monitor Resource Performance processes;

•Initiating, modifying and cancelling continuous performance data collection schedules for specific resources required to analyze specific resource performance. These schedules are established through requests sent to the Enable Resource Data Collection & Distribution processes;

•Determining the root causes of specific resource performance degradations and violations;

•Recording the results of the analysis and intermediate updates in the Resource Inventory for historical analysis and for use as required by other processes

•Undertaking specific detailed analysis (if the original requested came from Service Quality Management processes) to discover the root cause of service performance degradations that may be arising due to interactions between resource instances, without any specific resource instance having an unacceptable performance in its own right.

Perform Specific Resource Performance Diagnostics

Process Identifier: 1.5.9.2.1

Hierarchy Level: 4

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Brief Description:

Not used for this process element.

Extended Description:

This process performs analysis as required on specific resource performance information received from the Monitor Service Quality processes. It determines the root causes of specific resource performance degradations and violations, records the results of the analysis and intermediate updates in the Resource Inventory for historical analysis and for use as required by other processes, and undertakes specific detailed analysis (if the original requested came from Service Quality Management processes) to discover the root cause of customer QoS performance degradations that may be arising due to interactions between resource instances, without any specific resource instance having an unacceptable performance in its own right.

Manage Resource Performance Data Collection Schedules

Process Identifier: 1.5.9.2.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process initiates, modifies and cancels continuous performance data collection schedules for specific resources required to analyze specific resource performance. These schedules are established through requests sent to the Enable Resource Data Collection & Distribution processes.

Control Resource Performance

Process Identifier: 1.5.9.3

Hierarchy Level: 3

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Brief Description:

Apply controls to resources in order to optimize the resource performance

Extended Description:

The objective of the Control Resource Performance processes is to apply controls to resource instances in order to optimize the resource performance.

The responsibilities of the processes include, but are not limited to:

•Instantiating controls to attempt to restore resource instances to normal operation, at the request of Analyze Resource Performance processes. These controls may be based on established control plans, or the controls may be developed within the Control Resource Performance processes depending on circumstances.

•Instantiating controls to attempt to restore failed resource instances to normal operation, at the request of Resource Trouble Management or Service Quality Management processes. These controls may be based on established control plans, or the controls may be developed within the Control Resource Performance process depending on circumstances.

Instantiate Resource Performance Controls

Process Identifier: 1.5.9.3.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

Instantiate controls to attempt to restore resource instances to normal operation, at the request of Analyze Resource Performance processes. These controls may be based on established control plans, or the controls may be developed within the Control Resource Performance processes depending on circumstances.

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Instantiate Resource Trouble Controls

Process Identifier: 1.5.9.3.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

Instantiate controls to attempt to restore failed resource instances to normal operation, at the request of Resource Trouble Management or Service Quality Management processes. These controls may be based on established control plans, or the controls may be developed within the Control Resource Performance process depending on circumstances.

Report Resource Performance

Process Identifier: 1.5.9.4

Hierarchy Level: 3

Brief Description:

Monitor the status of resource performance degradation reports, provide notifications of any changes and provide management reports

Extended Description:

The objective of the Report Resource Performance processes is to monitor the status of resource performance degradation reports, provide notifications of any changes and provide management reports.

These processes are responsible for continuously monitoring the status of resource performance degradation reports and managing notifications to other processes in the Resource-Ops and other layers, and to other parties registered to receive notifications of any status changes. Notification lists are managed and maintained by the Enable Resource Performance Management processes.

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These processes record, analyze and assess the resource performance degradation report status changes to provide management reports and any specialized summaries of the efficiency and effectiveness of the overall Resource Performance Management process. These specialized summaries could be specific reports required by specific audiences.

Monitor Resource Performance Degradation Report

Process Identifier: 1.5.9.4.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process is responsible for continuously monitoring the status of Resource Performance Degradation Reports and managing notifications to processes in the Resource-Ops and other layers, and to other parties registered to receive notifications of any status changes. Notification lists are managed and maintained by the Enable Resource Performance Management processes.

Distribute Resource Quality Management Reports & Summaries

Process Identifier: 1.5.9.4.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

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This process records, analyzes and assesses the Resource Performance Degradation Report status changes to provide management reports and any specialized summaries of the efficiency and effectiveness of the overall Resource Performance Management process. These specialized summaries could be specific reports required by specific audiences.

Create Resource Performance Degradation Report

Process Identifier: 1.5.9.5

Hierarchy Level: 3

Brief Description:

Create a new resource performance degradation report

Extended Description:

The objective of the Create Resource Performance Degradation Report process is to create a new resource performance degradation report, modify existing resource performance degradation reports, and request cancellation of existing resource performance degradation reports.

A new resource performance degradation report may be created as a result of specific resource performance notifications undertaken by the Monitor Resource Performance processes, or at the request of analysis undertaken by other Service-Ops or Resource-Ops or party management processes which detect that some form of deterioration or failure has occurred requires an assessment of the specific resource performance.

If the resource performance degradation report is created as a result of a notification or request from processes other than Monitor Resource Performance processes, the Create Resource Performance Degradation Report processes are responsible for converting the received information into a form suitable for the Resource Performance Management processes, and for requesting additional information if required.

Generate Resource Performance Degradation Problem

Process Identifier: 1.5.9.5.1

Hierarchy Level: 4

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Brief Description:

Not used for this process element.

Extended Description:

This process creates a new Resource Performance Degradation Report as a result of specific resource performance notifications undertaken by the Monitor Resource Performance processes, or at the request of analysis undertaken by other Service-Ops or Resource-Ops or Resource-Ops processes which detect that some form of deterioration or failure has occurred requires an assessment of the specific resource performance.

Convert Report To Resource Performance Degradation Report Format

Process Identifier: 1.5.9.5.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

If the resource performance degradation report is created as a result of a notification or request from processes other than Monitor Resource Performance processes, this process responsible for converting the received information into a form suitable for the Resource Performance Management processes, and for requesting additional information if required.

Track & Manage Resource Performance Resolution

Process Identifier: 1.5.9.6

Hierarchy Level: 3

Brief Description:

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Ensure testing, repair and restoration activities are assigned, coordinated and tracked efficiently, and that escalation is invoked as required for any open resource performance degradation reports in jeopardy

Extended Description:

The objective of the Track & Manage Resource Performance Resolution processes is to efficiently assign, coordinate and track specific resource performance analysis and control activities, and escalate any open resource performance degradation reports in jeopardy.

Responsibilities of these processes include, but are not limited to:

• Adding additional information to an open resource performance degradation report based on the first-in and on-going analysis;

• Scheduling, assigning and coordinating analysis and specific resource performance restoration activities and/or repair activities delegated to other processes;

• Generating the respective other parties problem report creation request(s) to Initiate other parties Problem Report processes based on specific resource performance degradation reports where analysis the root cause is related to other parties products;

• Modifying information in an existing resource performance degradation report based on assignments;

• Modifying the resource performance degradation report status;

• Canceling a resource performance degradation report when the specific request was related to a false resource failure event

• Monitoring the jeopardy status of open resource performance degradation reports, and escalating resource performance degradation reports as necessary.

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These processes will co-ordinate all the actions necessary in order to guarantee that all tasks are finished at the appropriate time and in the appropriate sequence.

The Track & Manage Resource Performance Resolution processes will also inform the Close Resource Performance Degradation Report processes by modifying the resource performance degradation report status to cleared when the specific resource performance issues have been resolved.

Coordinate Resource Performance

Process Identifier: 1.5.9.6.1

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

This process schedules, assigns and coordinates analysis and specific service performance restoration activities and/or repair activities delegated to other processes, undertakes necessary tracking of the execution progress, modifies information in an existing Resource Performance Degradation Report based on assignments, and modifies the Resource Performance Degradation Report status

Request Other Parties Performance Degradation Report Creation and Update

Process Identifier: 1.5.9.6.2

Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

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This process generates the respective other parties problem report creation request(s) to Initiate other parties Problem Report processes based on specific resource performance degradation reports where analysis the root cause is related to other parties products.

Update First in Resource Testing Results

Process Identifier: 1.5.9.6.3

Hierarchy Level: 4

Brief Description:

This process adds additional information to an open Resource Performance Degradation Report based on the first-in testing.

Extended Description:

Not used for this process element

Cancel Resource Performance Degradation Report

Process Identifier: 1.5.9.6.4

Hierarchy Level: 4

Brief Description:

This process cancels a resource Performance Degradation Report when the specific trouble was related to a false resource failure event.

Extended Description:

Not used for this process element

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Escalate/End Resource Performance Degradation Report

Process Identifier: 1.5.9.6.5

Hierarchy Level: 4

Brief Description:

This process monitors the jeopardy status of open Resource Performance Degradation Reports, and escalates Resource Performance Degradation Reports as necessary.

Extended Description:

Not used for this process element

Clear Resource Performance Degradation Report Status

Process Identifier: 1.5.9.6.6

Hierarchy Level: 4

Brief Description:

This process informs the Close Resource Performance Degradation Report process by modifying the Resource Performance Degradation Report status to cleared when the specific resource performance quality issues have been resolved.

Extended Description:

Not used for this process element

Engage External Party Resource

Process Identifier: 1.5.9.6.7

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Hierarchy Level: 4

Brief Description:

Not used for this process element.

Extended Description:

If some specific resource components are owned and managed by other parties, this process is responsible for initiating requests, through other parties Performance Management, for resolution by the other parties of the specific resource components. This process will co-ordinate all the actions necessary in order to guarantee that all tasks are finished at the appropriate time and in the appropriate sequence.

Close Resource Performance Degradation Report

Process Identifier: 1.5.9.7

Hierarchy Level: 3

Brief Description:

Close a resource performance degradation report when the resource performance has been resolved

Extended Description:

The objective of the Close Resource Performance Degradation Report processes is to close a resource performance degradation report when the resource performance has been resolved.

These processes monitor the status of all open resource performance degradation reports, and recognize that a resource performance degradation report is ready to be closed when the status is changed to cleared.

Resource Mediation & Reporting

Process Identifier: 1.5.10

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Hierarchy Level: 2

Brief Description:

Manage resource events by correlating and formatting them into a useful format.

Extended Description:

Resource Mediation & Reporting processes manage resource events by correlating and formatting them into a useful format. These processes include the mediation and reporting of resource records. Investigation of resource related billing event problems is also part of these processes.

These processes are often handled by appropriate network elements.

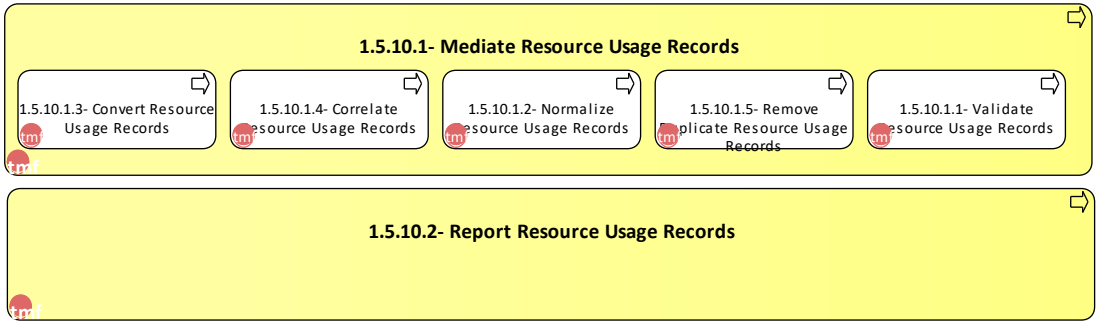

Figure 24: Resource Mediation & Reporting Children

Mediate Resource Usage Records

Process Identifier: 1.5.10.1

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Hierarchy Level: 3

Brief Description:

Validate, normalize, convert and correlate usage records collected from the network.

Extended Description:

The purpose of the Mediate Usage Records processes is to validate, normalize, convert and correlate usage records collected from various pieces of equipment in the network. It also removes any duplicate usage records that have already been processed.

Validate Resource Usage Records

Process Identifier: 1.5.10.1.1

Hierarchy Level: 4

Brief Description:

Validate resource usage record collected from the network.

Extended Description:

Not used for this process element

Normalize Resource Usage Records

Process Identifier: 1.5.10.1.2

Hierarchy Level: 4

Brief Description:

Normalize resource usage records to specific expression format.

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Extended Description:

Not used for this process element

Convert Resource Usage Records

Process Identifier: 1.5.10.1.3

Hierarchy Level: 4

Brief Description:

Convert resource usage records to specific data format.

Extended Description:

Not used for this process element

Correlate Resource Usage Records

Process Identifier: 1.5.10.1.4

Hierarchy Level: 4

Brief Description:

Correlate collected resource usage records.

Extended Description:

Not used for this process element

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Remove Duplicate Resource Usage Records

Process Identifier: 1.5.10.1.5

Hierarchy Level: 4

Brief Description:

Remove duplicate resource usage records.

Extended Description:

Not used for this process element

Report Resource Usage Records

Process Identifier: 1.5.10.2

Hierarchy Level: 3

Brief Description:

Generate reports on resource usage records based on requests from other processes.

Extended Description:

The purpose of the Report Resource Usage Records is to generate reports on usage records based on requests from other processes. These processes produce reports that may identify abnormalities, which may be caused by fraudulent activity or related to customer complaints or network problems.

Resource Capacity Management

Process Identifier: 1.5.14

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Hierarchy Level: 2

Brief Description:

Resource Capacity Management business process manage the activities required to ensure a business is able to plan, analyze, optimize, monitor and report on capacity and constraints associated to resources in response to business objectives in a timely fashion

Extended Description:

Resource Capacity Management business process manage the activities required to ensure a business is able to plan, analyze, optimize, monitor and report on capacity and constraints associated to resources in response to business objectives in a timely fashion.

The underlying business activities supporting Resource Capacity Management ensure that there is an effective understanding and management of business objectives, based on underlying Resource constraints.

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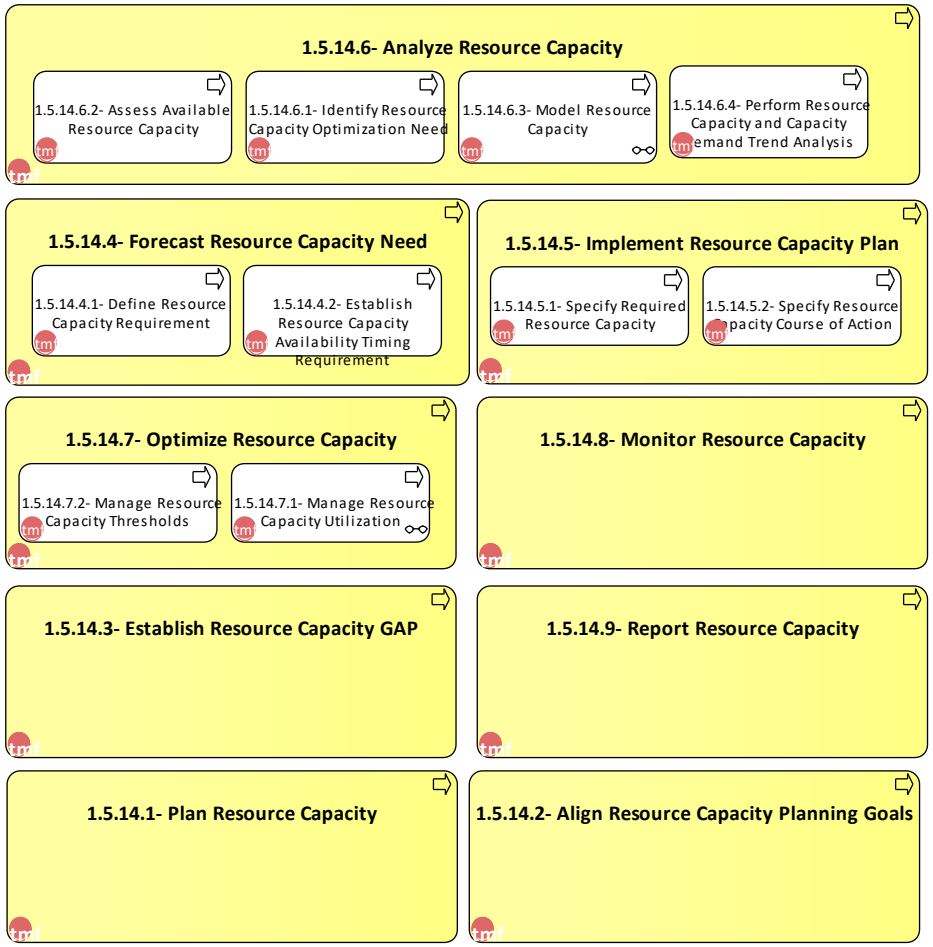

Figure 25: Resource Capacity Management Children

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Plan Resource Capacity

Process Identifier: 1.5.14.1

Hierarchy Level: 3

Brief Description:

Plan Resource Capacity business activity is in charge of determining the production capacity needed to meet changing demand for resources over a period of time, and implements the plan.

Extended Description:

Plan Resource Capacity business activity is in charge of determining the production capacity needed to meet changing demand for resources over a period of time, and implements the plan.

Plan Resource Capacity ensures Resource Management aligns capacity planning strategies with business goals; while ensuring that there are sufficient resources or supporting resource capabilities to support planned and forecasted business objectives; identifying gaps in capacity allocation early; determining resource capacity needs according to demand, determining constraints and usage patterns over time; forecasting future capacity needs; and communicating resource capacity requirements in time.

In the context of capacity planning, consideration is given to all input requirements, conversion processes and outputs.

Align Resource Capacity Planning Goals

Process Identifier: 1.5.14.2

Hierarchy Level: 3

Brief Description:

Align Resource Capacity Planning Goals business activity correlates Resource Capacity Plans with business demand.

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Extended Description:

Align Resource Capacity Planning Goals business activity correlates Resource Capacity Plans with business demand.

Establish Resource Capacity GAP

Process Identifier: 1.5.14.3

Hierarchy Level: 3

Brief Description:

Establish Resource Capacity GAP business activity identifies any disparity between resource capacity goals and business objectives.

Extended Description:

Establish Resource Capacity GAP business activity identifies any disparity between resource capacity goals and business objectives.

Forecast Resource Capacity Need

Process Identifier: 1.5.14.4

Hierarchy Level: 3

Brief Description:

Forecast Resource Capacity Need business activity develops projections that predict resource capacity growth, by understanding the impact on business.

Extended Description:

Forecast Resource Capacity Need business activity develops projections that predict resource capacity growth, by understanding the impact on business.

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Define Resource Capacity Requirement

Process Identifier: 1.5.14.4.1

Hierarchy Level: 4

Brief Description:

Define Resource Capacity Requirement business activity establishes, measures, and adjusts conditions that impact on planning resource capacity.

Extended Description:

Define Resource Capacity Requirement business activity establishes, measures, and adjusts conditions that impact on planning resource capacity.

Establish Resource Capacity Availability Timing Requirement

Process Identifier: 1.5.14.4.2

Hierarchy Level: 4

Brief Description:

Establish Resource Capacity Availability Timing Requirement business activity identify scheduling needs for resource capacity changes.

Extended Description:

Establish Resource Capacity Availability Timing Requirement business activity identify scheduling needs for resource capacity changes.

Implement Resource Capacity Plan

Process Identifier: 1.5.14.5

Hierarchy Level: 3

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Brief Description:

Implement Resource Capacity Plan business activity puts into effect the steps to achieve Resource Capacity Plans, with defined resource capacity target and resource capacity management course of action.

Extended Description:

Implement Resource Capacity Plan business activity puts into effect the steps to achieve Resource Capacity Plans, with defined resource capacity target and resource capacity management course of action.

Specify Required Resource Capacity

Process Identifier: 1.5.14.5.1

Hierarchy Level: 4

Brief Description:

Specify Required Resource Capacity business activity supports implementing resource capacity plans by identifying clearly, and definitely, resource capacity expectations.

Extended Description:

Specify Required Resource Capacity business activity supports implementing resource capacity plans by identifying clearly, and definitely, resource capacity expectations.

Specify Resource Capacity Course of Action

Process Identifier: 1.5.14.5.2

Hierarchy Level: 4

Brief Description:

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Specify Resource Capacity Course of Action business activity identifies clearly, and definitely, the sequence of activities that must be followed to support the 'Resource Capacity Plan' expectations.

Extended Description:

Specify Resource Capacity Course of Action business activity identifies clearly, and definitely, the sequence of activities that must be followed to support the 'Resource Capacity Plan' expectations.

Analyze Resource Capacity

Process Identifier: 1.5.14.6

Hierarchy Level: 3

Brief Description:

Analyze Resource Capacity business activity establishes, and models insights, patterns and changes to resource capacity by assessing potential capacity need and actual capacity that a resource currently achieves.

Extended Description:

Analyze Resource Capacity business activity establishes, and models insights, patterns and changes to resource capacity by assessing potential capacity need and actual capacity that a resource currently achieves.

This business activity covers collection of resource-related data (usage data, performance data, operational data etc.) to support Resource Managers to identify patterns and insights relating to a resource.

Analyze Resource Capacity supports Resource Capacity Management by providing relevant and timely insights, including available inventory holding capacity, resource "time and material" needs, etc. and how capacity utilization (allocated, in-use, unused) can be optimized to increase business value.

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Identify Resource Capacity Optimization Need

Process Identifier: 1.5.14.6.1

Hierarchy Level: 4

Brief Description:

Identify Resource Capacity Optimization Need business activity establish the best, or most effective, use of resource capacity.

Extended Description:

Identify Resource Capacity Optimization Need business activity establish the best, or most effective, use of resource capacity.

Assess Available Resource Capacity

Process Identifier: 1.5.14.6.2

Hierarchy Level: 4

Brief Description:

Assess Available Resource Capacity business activity evaluates unused resource capacity in relation to resource capacity plans and resource utilization trends.

Extended Description:

Assess Available Resource Capacity business activity evaluates unused resource capacity in relation to resource capacity plans and resource utilization trends.

Model Resource Capacity

Process Identifier: 1.5.14.6.3

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Hierarchy Level: 4

Brief Description:

Model Resource Capacity business activity establishes a representation of Resource Capacity based on demand and defined business properties.

Extended Description:

Model Resource Capacity business activity establishes a representation of Resource Capacity based on demand and defined business properties.

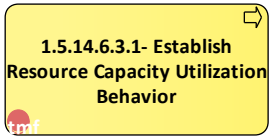

Figure 26: Model Resource Capacity Children

Establish Resource Capacity Utilization Behavior

Process Identifier: 1.5.14.6.3.1

Hierarchy Level: 5

Brief Description:

Establish Resource Capacity Utilization Behavior business activity develops Resource Capacity usage models.

Extended Description:

Establish Resource Capacity Utilization Behavior business activity develops Resource Capacity usage models.

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Perform Resource Capacity and Capacity Demand Trend Analysis

Process Identifier: 1.5.14.6.4

Hierarchy Level: 4

Brief Description:

Perform Resource Capacity and Capacity Demand Trend Analysis business activity analysis trends between available resource capacity and resource capacity demand according to business and ongoing resource usage.

Extended Description:

Perform Resource Capacity and Capacity Demand Trend Analysis business activity analysis trends between available resource capacity and resource capacity demand according to business and ongoing resource usage.

Optimize Resource Capacity

Process Identifier: 1.5.14.7

Hierarchy Level: 3

Brief Description:

Optimize Resource Capacity business activity is in charge of improving resource capacity usage by managing underlying and related dependencies and constraints.

Extended Description:

Optimize Resource Capacity business activity is in charge of improving resource capacity usage by managing underlying and related dependencies and constraints.

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For Resource Capacity Management, Optimize Resource Capacity can enable reclaim unused or available Resource capacities, and also allocate additional capacity just in time to meet demand. It can also help to repurpose resources based on their capacity utilization patterns, and support configuration management of all constraints and dependencies.

Manage Resource Capacity Thresholds

Process Identifier: 1.5.14.7.2

Hierarchy Level: 4

Brief Description:

Manage Resource Capacity Thresholds business activity is in charge of administering resource capacity thresholds based on operational conditions and concerns.

Extended Description:

Manage Resource Capacity Thresholds business activity is in charge of administering resource capacity thresholds based on operational conditions and concerns.

Manage Resource Capacity Utilization

Process Identifier: 1.5.14.7.1

Hierarchy Level: 4

Brief Description:

Manage Resource Capacity Utilization business activity is in charge of fulfilling and assuring the effective use of resource capacity.

Extended Description:

Manage Resource Capacity Utilization business activity is in charge of fulfilling and assuring the effective use of resource capacity.

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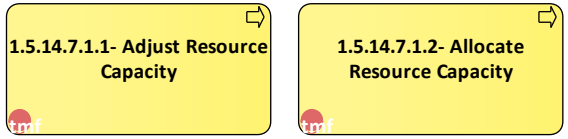

Figure 27: Manage Resource Capacity Utilization Children

Adjust Resource Capacity

Process Identifier: 1.5.14.7.1.1

Hierarchy Level: 5

Brief Description:

Adjust Resource Capacity business activity is in charge of fulfilling changes to Resource Capacity.

Extended Description:

Adjust Resource Capacity business activity is in charge of fulfilling changes to Resource Capacity.

Allocate Resource Capacity

Process Identifier: 1.5.14.7.1.2

Hierarchy Level: 5

Brief Description:

Allocate Resource Capacity business activity assign the amount in capacity that a resource must support.

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Extended Description:

Allocate Resource Capacity business activity assign the amount in capacity that a resource must support.

Monitor Resource Capacity

Process Identifier: 1.5.14.8

Hierarchy Level: 3

Brief Description:

Monitor Resource Capacity business activity identifies relevant resource capacity attributes/properties, and monitors and tracks all attributes, their use, and changes.

Extended Description:

Monitor Resource Capacity business activity identifies relevant resource capacity attributes/properties, and monitors and tracks all attributes, their use, and changes.

Monitor Resource Capacity supports all activities of Resource Capacity Management by providing timely visibility for all concerns around Resource Capacity

Report Resource Capacity

Process Identifier: 1.5.14.9

Hierarchy Level: 3

Brief Description:

Report Resource Capacity business activity prepares and presents detailed accounts of all relevant resource capacity information, in a timely manner that is relevant and reliable for use by business.

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Extended Description:

Report Resource Capacity business activity prepares and presents detailed accounts of all relevant resource capacity information, in a timely manner that is relevant and reliable for use by business.

Different business functions with interest for Resource Capacity may require specific reports with specific information. This business activity ensures reports for Resource Capacity are collated, organized and presented in ways that are useful, and reliable to the needs of business functions in a timely manner.

Resource Catalog Lifecycle Management

Process Identifier: 1.5.15

Hierarchy Level: 2

Brief Description:

Catalog Lifecycle Management business process covers a set of business activities that enable manage the lifecycle of an organizations catalog from design to build according to defined requirements.

Extended Description:

Catalog Lifecycle Management business process covers a set of business activities that enable manage the lifecycle of an organizations catalog from design to build according to defined requirements.

Catalog Lifecycle Management proves the overarching governance to manage all the stages in the realization and operationalization of the Product/Service/Resource Catalog in support of the organizations business goals.

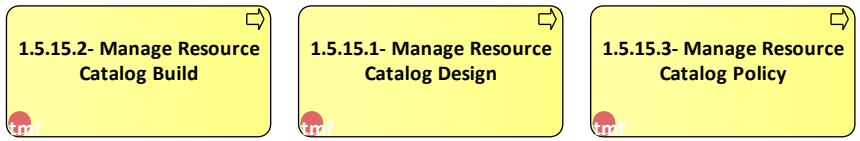

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Figure 28: Resource Catalog Lifecycle Management Children

Manage Resource Catalog Build

Process Identifier: 1.5.15.2

Hierarchy Level: 3

Brief Description:

Manage Catalog Design business activity covers the design of the Catalog in accordance to the Catalog Management Lifecycle.

Extended Description:

Manage Catalog Design business activity covers the design of the Catalog in accordance to the Catalog Management Lifecycle.

This business activity leverages input from the Catalog Design to construct the product/service/resource Catalog in order to meet the business operations goals.

Manage Resource Catalog Design

Process Identifier: 1.5.15.1

Hierarchy Level: 3

Brief Description:

Manage Catalog Design business activity cover the design of the Catalog in accordance to the Catalog Management Lifecycle.

Extended Description:

Manage Catalog Design business activity cover the design of the Catalog in accordance to the Catalog Management Lifecycle.

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Manage Catalog Design business activity will leverage Catalog Specification requirement for all information (features, attributes etc.) to be managed by the catalog in order to enable support the Catalog Lifecycle Management

Manage Resource Catalog Policy

Process Identifier: 1.5.15.3

Hierarchy Level: 3

Brief Description:

Manage Catalog Build business activity establishes, manages and administers the governing of the catalog for operations readiness and operations.

Extended Description:

Manage Catalog Build business activity establishes, manages and administers the governing of the catalog for operations readiness and operations.

This business activity leverages support of the Enterprise governance processes for product/service/resource catalog lifecycle management.

Resource Catalog Operational Readiness Management

Process Identifier: 1.5.16

Hierarchy Level: 2

Brief Description:

Resource Catalog Operational Readiness Management business process establishes and administers the support needed to operationalize Resource catalogs for ongoing day-to-day business needs.

Extended Description:

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Resource Catalog Operational Readiness Management business process establishes and administers the support needed to operationalize Resource catalogs for ongoing day-to-day business needs.

These business activities implement the Resource Catalog through Release and Deploy business activities.

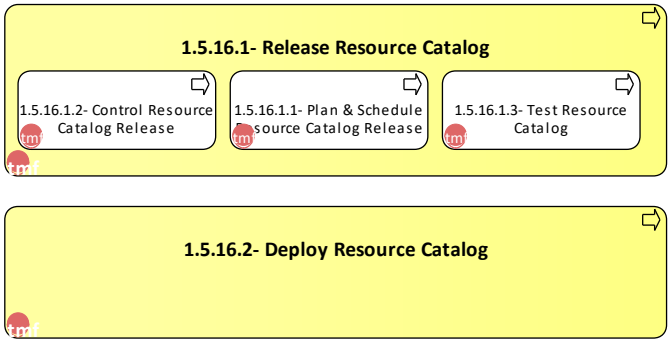

Figure 29: Resource Catalog Operational Readiness Management Children

Deploy Resource Catalog

Process Identifier: 1.5.16.2

Hierarchy Level: 3

Brief Description:

Deploy Resource Catalog business activity manage the availability and use of the Resource catalog in operations.

Extended Description:

Deploy Resource Catalog business activity manage the availability and use of the Resource catalog in operations.

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Deploy Resource Catalog business activity includes identifying stakeholders, creating catalog category hierarchies, catalog integration, and catalog federation for any scenario, such as with business partners and between business functions.

Release Resource Catalog

Process Identifier: 1.5.16.1

Hierarchy Level: 3

Brief Description:

Release Resource Catalog business activity ensure all cross-functional activities needed to support catalog maintenance and operations, such as training and updating the support of the catalog are in place.

Extended Description:

Release Resource Catalog business activity ensure all cross-functional activities needed to support catalog maintenance and operations, such as training and updating the support of the catalog are in place.

Release Resource Catalog business activity includes identifying stakeholders, catalog integration, catalog federation etc. for any scenario in support of the organizations business goals, including Release conditions that support users, customers and business partners.

Control Resource Catalog Release

Process Identifier: 1.5.16.1.2

Hierarchy Level: 4

Brief Description:

Control Resource Catalog Release business activity is used to support the various catalog release tasks.

Extended Description:

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Control Resource Catalog Release business activity is used to support the various catalog release tasks.

Plan & Schedule Resource Catalog Release

Process Identifier: 1.5.16.1.1

Hierarchy Level: 4

Brief Description:

Plan & Schedule Resource Catalog Release business activity is used to handle the release plan for the Resource Catalog.

Extended Description:

Plan & Schedule Resource Catalog Release business activity is used to handle the release plan for the Resource Catalog.

Test Resource Catalog

Process Identifier: 1.5.16.1.3

Hierarchy Level: 4

Brief Description:

Test Resource Catalog business activity is used to establish the quality, reliability and performance requirement of the catalog according to the Resource management needs.

Extended Description:

Test Resource Catalog business activity is used to establish the quality, reliability and performance requirement of the catalog according to the Resource management needs.

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Resource Catalog Content Management

Process Identifier: 1.5.17

Hierarchy Level: 2

Brief Description:

Resource Catalog Content Management business process define and provide the business activities that support the day-to-day operations of Resource Catalogs in order to realize the business operations goals.

Extended Description:

Resource Catalog Content Management business process define and provide the business activities that support the day-to-day operations of Resource Catalogs in order to realize the business operations goals.

Resource Catalog Content Management business processes include administering the Resource Catalog instance in production, maintaining catalog entries, assuring catalogs, managing catalog access, managing entry lifecycle through versioning, handling catalog entity entry and changes, supporting distribution of catalogs as needed, and supporting user-facing activities.

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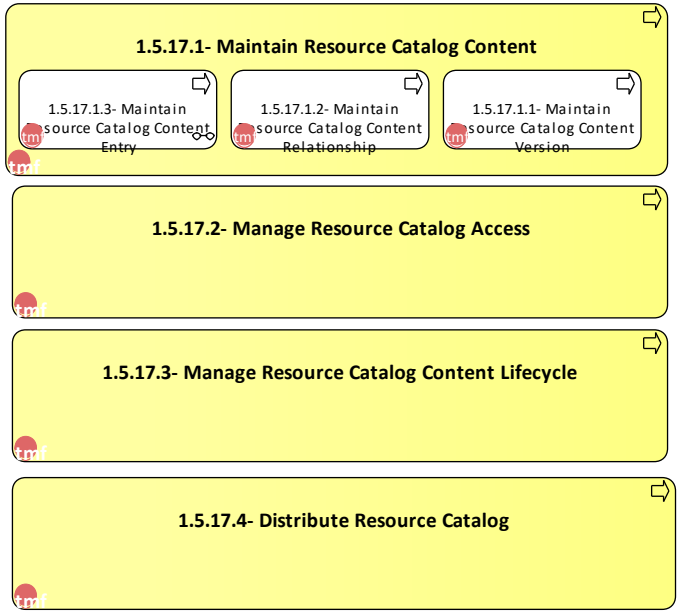

Figure 30: Resource Catalog Content Management Children

Maintain Resource Catalog Content

Process Identifier: 1.5.17.1

Hierarchy Level: 3

Brief Description:

Maintain Resource Catalog Content business activity handles catalog content entry for adding or updating the operational Catalog by following the established Catalog Content Management operations standards and policies.

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Extended Description:

Maintain Resource Catalog Content business activity handles catalog content entry through adding or updating the Catalog by following the established Catalog Content Management standards and policies.

Manage Catalog Entry support the following activities and sub-activities:

Maintaining Catalog Content Versions

Maintain Catalog Content Relationship

Maintain Catalog Content taxonomy

Organize Catalog Entry

Classify Catalog Entry

Maintain Resource Catalog Content Version

Process Identifier: 1.5.17.1.1

Hierarchy Level: 4

Brief Description:

Maintain Resource Catalog Content Version business activity handles tracking and controlling changes for Resource catalog entries.

Extended Description:

Maintain Resource Catalog Content Version business activity handles tracking and controlling changes for Resource catalog entries.

Maintain Resource Catalog Content Relationship

Process Identifier: 1.5.17.1.2

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Hierarchy Level: 4

Brief Description:

Maintain Resource Catalog Content Relationship business activity handles associating Resource catalog entries.

Extended Description:

Maintain Resource Catalog Content Relationship business activity handles associating Resource catalog entries.

Catalog Content Relationship for Resource catalog entries can be internal and external to the Product, Service or Resource domains.

Maintain Resource Catalog Content Entry

Process Identifier: 1.5.17.1.3

Hierarchy Level: 4

Brief Description:

Maintain Resource Catalog Content Entry business activity handles content entry, content organization, content classification in the Resource Catalog according to the Resource Catalog model.

Extended Description:

Maintain Resource Catalog Content Entry business activity handles content entry, content organization, content classification in the Resource Catalog according to the Resource Catalog model.

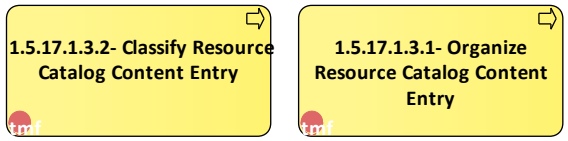

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Figure 31: Maintain Resource Catalog Content Entry Children

Classify Resource Catalog Content Entry

Process Identifier: 1.5.17.1.3.2

Hierarchy Level: 5

Brief Description:

Classify Resource Catalog Content Entry business activity specifically handles Catalog Content organization according to business classification characteristics for the Resource Catalog.

Extended Description:

Classify Resource Catalog Content Entry business activity specifically handles Catalog Content organization according to business classification characteristics for the Resource Catalog.

Classify Resource Catalog Content Entry is specific to shared qualities or characteristics defined by the organization.

Organize Resource Catalog Content Entry

Process Identifier: 1.5.17.1.3.1

Hierarchy Level: 5

Brief Description:

Organize Resource Catalog Content Entry business activity handles Catalog Content organization according to the Resource Catalog model.

Extended Description:

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Organize Resource Catalog Content Entry business activity handles Catalog Content organization according to the Resource Catalog model.

Organize Catalog Content Entry business activities can include ordering, arranging, sorting, assembling, and grouping.

Manage Resource Catalog Access

Process Identifier: 1.5.17.2

Hierarchy Level: 3

Brief Description:

Manage Resource Catalog Access business activity handles access to the catalog according to the Catalog Content Management standards and policies.

Extended Description:

Manage Resource Catalog Access business activity handles access to the catalog according to the Catalog Content Management standards and policies.

Manage Resource Catalog Access will handle assigning catalog managers and granting privileges.

Manage Resource Catalog Content Lifecycle

Process Identifier: 1.5.17.3

Hierarchy Level: 3

Brief Description:

Manage Resource Catalog Content Lifecycle business activity handles changes to catalog content according to the Catalog Content Management standards and policies.

Extended Description:

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Manage Resource Catalog Content Lifecycle business activity handles changes to catalog content according to the Catalog Content Management standards and policies.

Distribute Resource Catalog

Process Identifier: 1.5.17.4

Hierarchy Level: 3

Brief Description:

Distribute Resource Catalog business activity handles exchange and distribution of catalogs according to the Catalog Content Management standards and policies.

Extended Description:

Distribute Resource Catalog business activity handles exchange and distribution of catalogs according to the Catalog Content Management standards and policies.

Resource Catalog Planning Management

Process Identifier: 1.5.18

Hierarchy Level: 2

Brief Description:

Resource Catalog Planning Management business process covers a set of business activities that understand and enable establish the plan to define, design and operationalize a catalog in order to meet the needs and objectives of Resource cataloging.

Extended Description:

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Resource Catalog Planning Management business process covers a set of business activities that understand and enable establish the plan to define, design and operationalize a catalog in order to meet the needs and objectives of Resource cataloging.

The Resource Catalog Planning Management business process ensure that the organization is able to identify the most appropriate scheme and goal for it catalog. It includes designing the Catalog plan and developing the specification according to Resource management requirement.

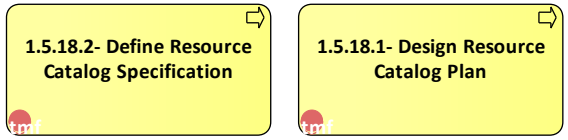

Figure 32: Resource Catalog Planning Management Children

Define Resource Catalog Specification

Process Identifier: 1.5.18.2

Hierarchy Level: 3

Brief Description:

Define Catalog Specification business activity establishes the Catalog specifications based on the product/service/resource strategy to support Catalog Planning Management.

Extended Description:

Define Catalog Specification business activity establishes the Catalog specifications based on the product/service/resource strategy to support Catalog Planning Management.

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Define Catalog Specification leverages Specification Management business activities to establish the mandatory and optional design requirements, as well as catalog entry fields according to product/service/resource management needs. This process will be governed by the Enterprise Business Process for Catalog Management.

Design Resource Catalog Plan

Process Identifier: 1.5.18.1

Hierarchy Level: 3

Brief Description:

Design Catalog Plan business activity understands requirements and capabilities demanded from the business to establish the Catalog plan based on the Resource strategy for Resource Catalog Planning Management.

Extended Description:

Resource Design Catalog Plan business activity understands requirements and capabilities demanded from the business to establish the Catalog plan based on the Resource strategy for Resource Catalog Planning Management.

Resource Specification Management

Process Identifier: 1.5.19

Hierarchy Level: 2

Brief Description:

Resource Specification Management business process leverages captured resource requirements to develop, master, analyze, and update documented standard conditions that must be satisfied by a resource design and/or delivery.

Extended Description:

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Resource Specification Management business process leverages captured resource requirements to develop, master, analyze, and update documented standard conditions that must be satisfied by a resource design and/or delivery.

Resource Specifications Management can result in establishing in a centralized way, technical (know-how) standards. Such standards provide the organization with a means to control and approve the values and inputs of specification through structure, review, approval and distribution processes to stakeholders and suppliers.

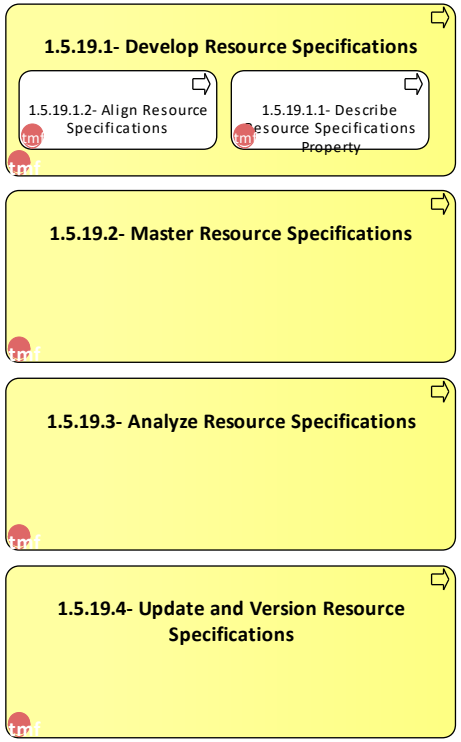

Figure 33: Resource Specification Management Children

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Develop Resource Specifications

Process Identifier: 1.5.19.1

Hierarchy Level: 3

Brief Description:

Develop Resource Specification business activity define and elaborate on the conditions for meeting resource requirement for consistency, quality, functionality and fit for purpose.

Extended Description:

Develop Resource Specification business activity define and elaborate on the conditions for meeting resource requirement for consistency, quality, functionality and fit for purpose.

This activity leverages captured resource requirements and resource preferences and builds in tolerances according to stakeholders for suppliers to adhere to.

Align Resource Specifications

Process Identifier: 1.5.19.1.2

Hierarchy Level: 4

Brief Description:

Align Resource Specifications business activity handle the alignment of resource specification to functional and nonfunctional requirements.

Extended Description:

Align Resource Specifications business activity handle the alignment of resources specification to functional and nonfunctional requirements.

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The alignment of resource specifications can be to other resources as well as realization of the specific resource according to business concerns (e.g. cost, other accounting and non-accounting requirements etc.).

Describe Resource Specifications Property

Process Identifier: 1.5.19.1.1

Hierarchy Level: 4

Brief Description:

Describe Resource Specifications Property set out and develop the frames, rules, constraints, models and the set of principles on which the resource specifications can be modelled according to resource requirements.

Extended Description:

Describe Resource Specifications Property set out and develop the frames, rules, constraints, models and the set of principles on which the resource specifications can be modelled according to resource requirements.

Master Resource Specifications

Process Identifier: 1.5.19.2

Hierarchy Level: 3

Brief Description:

Master Resource Specifications business activity learn and understand the collection of resource specifications in order to manage improvement or evolution of standards that will used for design and construction of resource.

Extended Description:

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Master Resource Specifications business activity learn and understand the collection of resource specifications in order to manage improvement or evolution of standards that will used for design and construction of resource.

This business activity is used by organizations to establish a collection of standardized, pre-written specifications or guide specifications that can be used by resource designers, resource architects, and other resource construction entities.

Analyze Resource Specifications

Process Identifier: 1.5.19.3

Hierarchy Level: 3

Brief Description:

Analyze Resource Specifications business activity research, assess and evaluate performance criteria and factors to develop, master, control/update resource specifications for resource requirement adherence.

Extended Description:

Analyze Resource Specifications business activity research, assess and evaluate performance criteria and factors to develop, master, control/update resource specifications for resource requirement adherence.

This business activity can include multiple analyzes of resources specifications, including standard analysis (based on empirical research) or custom analysis (based on other standards). Analysis of Resource Specification include testing Resource specifications and the ability to meet business requirements, as well as conform to operational mandates.

Update and Version Resource Specifications

Process Identifier: 1.5.19.4

Hierarchy Level: 3

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Brief Description:

Update and Version Resource Specifications business activity update and track changes to existing resource specifications according to versioning policies to support requirement management.

Extended Description:

Update and Version Resource Specifications business activity update and track changes to existing resource specifications according to versioning policies to support requirement management.

This business activity manages all changes to resource specifications over their lifecycle. Resource specification versioning is an effective means to communicate changes to resource specifications, so that resource planners and consumers know what to expect. It includes communicating when changes are made to resource specification properties.

Resource Anomaly Lifecycle Management

Process Identifier: 1.5.20

Hierarchy Level: 2

Brief Description:

Resource Anomaly Lifecycle Management business process establishes and controls all activities that are involved in overseeing, directing, administering, controlling and organizing the definition, detection/prediction, mitigation and learnings related to Resource Anomaly Management.

Extended Description:

Resource Anomaly Lifecycle Management business process establishes and controls all activities that are involved in overseeing, directing, administering, controlling and organizing the definition, detection/prediction, mitigation and learnings related to Resource Anomaly Management.

Resource Anomaly Lifecycle Management cover the lifecycle activities, including:

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Establishing “what is normal” for Resource in order to inform “deviations” or aberrations.

Orchestrating Resource Anomaly Management activities to meet business needs.

Managing the ability to acquire and apply knowledge and skills from past and present Resource anomalies.

Managing actions that make the best or effective use of Resource Management Closed Loops.

Monitoring feedback to and from Resource Management to support Resource Anomaly Management activities.

Reporting on Resource Anomaly Management Closed Loops.

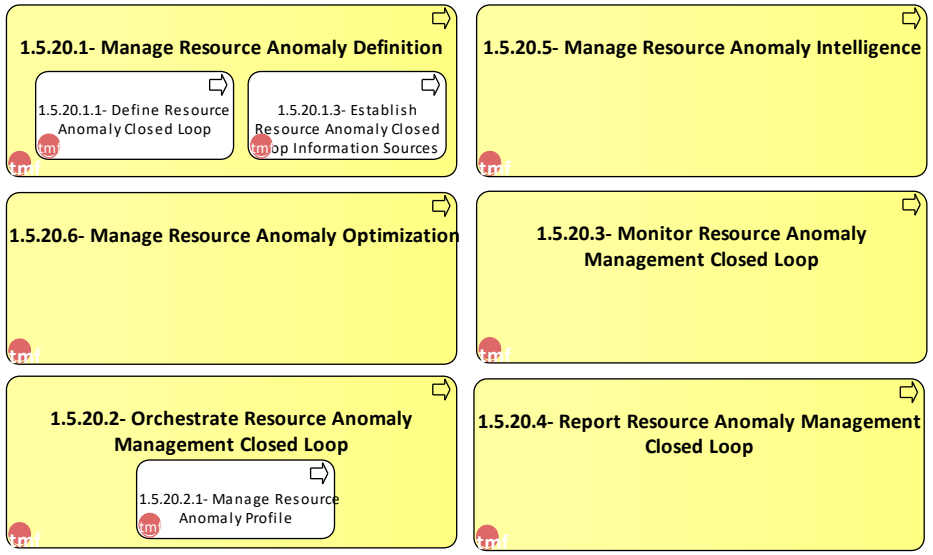

Figure 34: Resource Anomaly Lifecycle Management Children

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Manage Resource Anomaly Definition

Process Identifier: 1.5.20.1

Hierarchy Level: 3

Brief Description:

Manage Resource Anomaly Definition business activity is in charge of defining and describing the information about data which is used in Resource Anomaly Management.

Extended Description:

Manage Resource Anomaly Definition business activity is in charge of defining and describing the information about data which is used in Resource Anomaly Management.

Manage Resource Anomaly Definition business activity covers activities that describe the criteria, characteristics, and expectations of information required for Resource Anomaly Detection. All descriptions and definitions of the information about data use in Resource Anomaly Management business activities may include, but not be limited to "normal pattern data", "constraints data", "goals data" etc.

Define Resource Anomaly Closed Loop

Process Identifier: 1.5.20.1.1

Hierarchy Level: 4

Brief Description:

Define Resource Anomaly Closed Loop business activity state and describe the Resource Anomaly Closed Loops along with associated information that lead to identifying, detecting or predicting a Resource Anomaly.

Extended Description:

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Define Resource Anomaly Closed Loop business activity state and describe the Resource Anomaly Closed Loops along with associated information that lead to identifying, detecting or predicting a Resource Anomaly.

This business activity supports Anomaly definition process and the Closed Loops associated, including all associated inputs that reasonably affect the management of an Anomaly Lifecycle. This activity also ensures that the organization understands, and has well-defined operational workflows underpinning Anomaly Management.

Establish Resource Anomaly Closed Loop Information Sources

Process Identifier: 1.5.20.1.3

Hierarchy Level: 4

Brief Description:

Establish Resource Anomaly Closed Loop Information Sources business activity identifies all information sources and the requisite data that must be managed to support Resource Anomaly Management Closed Loops.

Extended Description:

Establish Resource Anomaly Closed Loop Information Sources business activity identifies all information sources and the requisite data that must be managed to support Resource Anomaly Management Closed Loops.

Establish Resource Anomaly Closed Loop Information Sources help to identify the information (not limited to business and operations rules), business logic, levels of classification (by importance, by period etc.), impact and urgency etc. that can help detect Resource Anomaly. This process supports Resource Anomaly Management during day-to-day resource delivery.

Orchestrate Resource Anomaly Management Closed Loop

Process Identifier: 1.5.20.2

Hierarchy Level: 3

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Brief Description:

Orchestrate Resource Anomaly Management Closed Loop business activity is in charge of organizing, arranging and coordinating changes that control Resource Anomaly Management Closed Loops.

Extended Description:

Orchestrate Resource Anomaly Management Closed Loop business activity is in charge of organizing, arranging and coordinating changes that control Resource Anomaly Management Closed Loops.

This business activity will cover:

Managing Resource Anomaly Detection activities - initiating, suspending, maintaining, and terminating.

Organizing Resource Anomaly Detection activities - including arranging the sequence for Closed Loop activities.

Coordinating changes to a Resource Anomaly Management Closed Loop.

Manage Resource Anomaly Profile

Process Identifier: 1.5.20.2.1

Hierarchy Level: 4

Brief Description:

Manage Resource Anomaly Profile business activity is in charge of recording and analyzing Resource Anomaly Management frameworks in order to enable assess and predict Resource Anomaly Management capabilities according to well-specified resource management models.

Extended Description:

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Manage Resource Anomaly Profile business activity is in charge of recording and analyzing Resource Anomaly Management frameworks in order to enable assess and predict Resource Anomaly Management capabilities according to well-specified resource management models.

Manage Resource Anomaly Profile supports Resource Anomaly Management operations, as well as capturing the business intents that need to be associated with Resource Anomaly Closed Loops and Resource Anomaly Management Lifecycle.

Monitor Resource Anomaly Management Closed Loop

Process Identifier: 1.5.20.3

Hierarchy Level: 3

Brief Description:

Monitor Resource Anomaly Management Closed Loop business activity provides the capability to observe Resource Anomaly Management Closed Loops.

Extended Description:

Monitor Resource Anomaly Management Closed Loop business activity provides the capability to observe Resource Anomaly Management Closed Loops.

This business activity helps to scan; watch; monitor; and gain awareness of Resource Anomaly Management activities that provide feedback between Resource Anomaly Management Lifecycle(s) stages.

Report Resource Anomaly Management Closed Loop

Process Identifier: 1.5.20.4

Hierarchy Level: 3

Brief Description:

Report Resource Anomaly Management Closed Loop business activity documents information organized to deliver an account of any, and all Resource Anomaly Closed Loops.

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Extended Description:

Report Resource Anomaly Management Closed Loop business activity documents information organized to deliver an account of any, and all Resource Anomaly Closed Loops.

Anomaly Closed Loops may fall into either or across any of the four stages in any of the overarching Closed Loop pattern - Observe, Orient, Decide and Act - or other variations.

Manage Resource Anomaly Intelligence

Process Identifier: 1.5.20.5

Hierarchy Level: 3

Brief Description:

Manage Resource Anomaly Intelligence business activity is in charge of acquiring and applying new learning, knowledge and skills to Resource Anomaly Management.

Extended Description:

Manage Resource Anomaly Intelligence business activity is in charge of acquiring and applying new learning, knowledge and skills to Resource Anomaly Management.

This business activity supports improving, optimizing, or evolving Resource Anomaly Management activities. Manage Resource Anomaly Intelligence helps to incorporate new knowledge and skills into Resource Anomaly Management processes. This is as well to enable improvement and optimization of all Anomaly Management in a continuum.

Manage Resource Anomaly Optimization

Process Identifier: 1.5.20.6

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Hierarchy Level: 3

Brief Description:

Manage Resource Anomaly Optimization business activity is in charge of actions that make the best and most effective use of Resource Anomaly Management activities to improve Resource Anomaly Management.

Extended Description:

Manage Resource Anomaly Optimization business activity is in charge of actions that make the best and most effective use of Resource Anomaly Management activities to improve Resource Anomaly Management and ultimately Resource Management.

Manage Resource Anomaly Optimization business activity can manage the business logic and business rules associated with Resource Anomaly Detection, Resource Anomaly Assessment, Resource Anomaly Forecasting and Prediction, Resource Anomaly Mitigation, as well as underlying Resource Knowledge Management.

This business activity also includes fine-tuning of business logic, operations logic, business and operations rules that improve efficiency of Anomaly Management as a whole.

Resource Anomaly Management

Process Identifier: 1.5.21

Hierarchy Level: 2

Brief Description:

Resource Anomaly Management business processes establish actions that predict and detect aberrations or outlier events/activities, assess them for their impact, mitigate them, and record them before they ever become Resource Problem Management concerns.

Extended Description:

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Resource Anomaly Management business processes establish actions that predict and detect aberrations or outlier events/activities, assess them for their impact, mitigate them, and record them before they ever become Resource Problem Management concerns.

By establishing that an action or event is abnormal (based on known patterns), Resource Anomaly Management helps to assess them through a set of activities that may triage, plan detailed assessment, classify them and provide mitigatory actions for them. Through Resource Anomaly Management, assurance of Products, based on abnormal events or activities can be well categorized, prioritized and actioned on. Resource Anomaly Management is different from Resource Problem Management as the later addresses known issues, faults or problems.

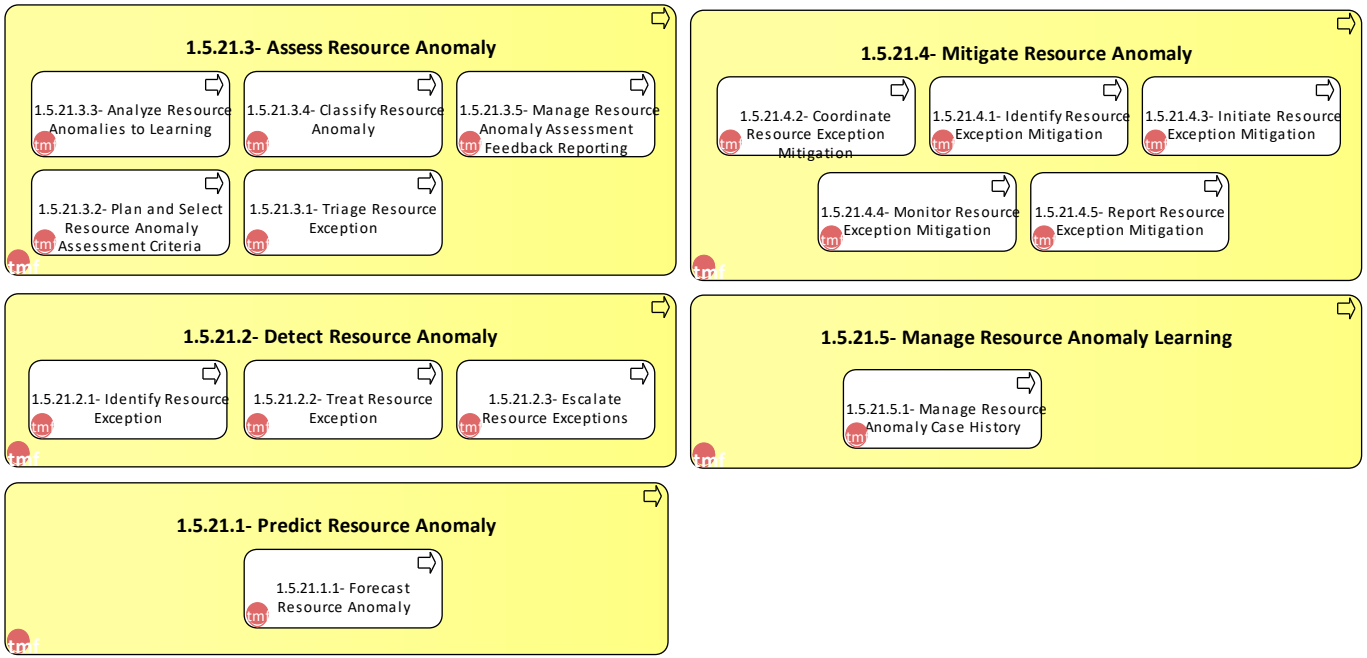

Figure 35: Resource Anomaly Management Children

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Predict Resource Anomaly

Process Identifier: 1.5.21.1

Hierarchy Level: 3

Brief Description:

Predict Resource Anomaly business activity is in charge of declaring or indicating in advance (or foretells on the basis of resource delivery observations, resource use experience or resource management reasons) upcoming exceptions, as well as trends that can lead to outlier activities / events or resource problems.

Extended Description:

Predict Resource Anomaly business activity is in charge of declaring or indicating in advance (or foretells on the basis of resource delivery observations, resource use experience or resource management reasons) upcoming exceptions, as well as trends that can lead to outlier activities / events or resource problems.

Resource Anomaly Prediction business activity can help to detect and interpret early activities or events with their impending exceptions while suggesting causes.

Forecast Resource Anomaly

Process Identifier: 1.5.21.1.1

Hierarchy Level: 4

Brief Description:

Forecast Resource Anomaly business activity predicts future Resource Anomaly along with implications using historical, and /or ongoing patterns observed through exceptions.

Extended Description:

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Forecast Resource Anomaly business activity predicts future Resource anomaly along with implications using historical, and /or ongoing patterns observed through exceptions.

Forecast Resource Anomaly business activity may include what-if analysis, or impact analysis of the anomalous exceptions. Forecast Resource Anomaly business activity supports the identification of potential risks of an exception, and the course of action needed to avoid potential risks.

Detect Resource Anomaly

Process Identifier: 1.5.21.2

Hierarchy Level: 3

Brief Description:

Detect Resource Anomaly business activity identify Resource delivery activities that are a deviation (aberrations or abnormal actions/events) from well-define norms or expectations.

Extended Description:

Detect Resource Anomaly business activity identify Resource delivery activities that are a deviation (aberrations or abnormal actions/events) from well-define norms or expectations.

Resource Anomaly Detection finds patterns in Resource Management activities, as well as with engaged parties given a deviation from planned, or expected business objectives or goals.

Identify Resource Exception

Process Identifier: 1.5.21.2.1

Hierarchy Level: 4

Brief Description:

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Identify Resource Exceptions business activity supports the detection of Resource Anomaly by investigating (recognizing and establishing) particular activity or event as an exception.

Extended Description:

Identify Resource Exception business activity supports the detection of Resource Anomaly by investigating (recognizing and establishing) particular activity or event as an exception.

Identify Resource Exception business activity will support all Resource Anomaly Management business activities - Assess Resource Anomaly, Predict Resource Anomaly, Mitigate Resource Anomaly and Learn Resource Anomaly.

Treat Resource Exception

Process Identifier: 1.5.21.2.2

Hierarchy Level: 4

Brief Description:

Treat Resource Exception business activity provides the set of actions that support managing a products' anomalous exception.

Extended Description:

Treat Resource Exception business activity provides the set of actions that support managing a products' anomalous exception.

This covers actions that may be:

Proactive: Where the treatment of the conditions leading from the exception are controlled;

Reactive: Where the treatment is directly in response to the exception.

Treat Resource Exceptions business activity helps to outline a series of actions that can expedite decisions for mitigation.

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Escalate Resource Exceptions

Process Identifier: 1.5.21.2.3

Hierarchy Level: 4

Brief Description:

Escalate Resource Exception business activity is in charge of increasing the attention given to handling of a Resource Exception.

Extended Description:

Escalate Resource Exception business activity is in charge of increasing the attention given to handling of a Resource Exception.

Escalating a Product's exception may be based on several factors, including defined business rules of escalation, or the lack thereof for handling a product's exception.

Assess Resource Anomaly

Process Identifier: 1.5.21.3

Hierarchy Level: 3

Brief Description:

Assess Resource Anomaly business activity check, estimate, appraise and evaluate Resource anomaly observations to determine implications and follow-on treatment.

Extended Description:

Assess Resource Anomaly business activity check, estimate, appraise and evaluate Resource anomaly observations to determine implications and follow-on treatment.

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Assess Resource Anomaly ensures that the business can ascertain understanding of a product's anomalous activity, its extent/scale, cause and effect, and provide evidence-based information to develop courses of action.

Triage Resource Exception

Process Identifier: 1.5.21.3.1

Hierarchy Level: 4

Brief Description:

Triage Resource Exception business activity is in charge of identifying and assigning the degrees of urgency, and the order of treatment of products' anomalous exceptions.

Extended Description:

Triage Resource Exception business activity is in charge of identifying and assigning the degrees of urgency, and the order of treatment of products' anomalous exceptions.

Triage Resource Exceptions sub activities may include prioritizing exceptions and underlying Resource anomaly events while examining causes that enable apply an initial set of mediatory actions.

Plan and Select Resource Anomaly Assessment Criteria

Process Identifier: 1.5.21.3.2

Hierarchy Level: 4

Brief Description:

Plan and Select Resource Anomaly Assessment Criteria business activity define the course of action to assess a Resource anomaly, along with organizing the product's anomaly assessment criteria in order to best facilitate detection and mitigation of anomaly

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Extended Description:

Plan and Select Resource Anomaly Assessment Criteria business activity define the course of action to assess a Resource anomaly, along with organizing the product's anomaly assessment criteria in order to best facilitate detection and mitigation of anomalies.

The 'Plan and Select Resource Assessment Criteria' process helps the organization to optimize the approaches to assess Resource anomaly criteria in order to continually avoid impact of false positives and/or false negatives in the detection of Resource anomalies.

Analyze Resource Anomalies to Learning

Process Identifier: 1.5.21.3.3

Hierarchy Level: 4

Brief Description:

Analyze Resource Anomalies to Learning business activity establish actions to discover and correlate anomalous behavior to 'learned normal' or 'expected behavior'.

Extended Description:

Analyze Resource Anomalies to Learning business activity establish actions to discover and correlate anomalous behavior to 'learned normal' or 'expected behavior'.

Analyze Resource Anomalies to Learning sub activities help to enrich the organization's Anomaly Detection activities altogether, which can result in a reduction in the time to detect, triage and treat Resource anomalies.

Classify Resource Anomaly

Process Identifier: 1.5.21.3.4

Hierarchy Level: 4

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Brief Description:

Classify Resource Anomaly business activity establish ways to organize Resource anomalies into established categories according to business priorities and objectives.

Extended Description:

Classify Resource Anomaly business activity establish ways to organize Resource anomalies into established categories according to business priorities and Resource management objectives.

Classify Resource Anomaly includes establishing robust classification models that help to triage anomalies, identify communication needs, determine and measure impact, and report on Resource anomalies.

Manage Resource Anomaly Assessment Feedback Reporting

Process Identifier: 1.5.21.3.5

Hierarchy Level: 4

Brief Description:

Manage Resource Anomaly Assessment Feedback Reporting business activity is in charge of establishing and handling the activities that prepare and report the feedback based of Resource Anomaly Assessment activity.

Extended Description:

Manage Resource Anomaly Assessment Feedback Reporting business activity is in charge of establishing and handling the activities that prepare and report the feedback based of Resource Anomaly Assessment activity.

Manage Resource Anomaly Assessment Feedback business activity covers tasks that prepare and report feedback of an Anomaly Assessment activity. This business activity is also helpful in supporting the optimization of the Anomaly Detection business activity, such as fine-tuning know-how for Resource anomaly detection.

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Reports generated from this business activity can be utilized for Resource Anomaly Mitigation and can help to select the consequent actions required to mitigate a Resource anomaly.

Mitigate Resource Anomaly

Process Identifier: 1.5.21.4

Hierarchy Level: 3

Brief Description:

Mitigate Resource Anomaly business activity is in charge of allaying or alleviate the impact of a product's anomalous exception.

Extended Description:

Mitigate Resource Anomaly business activity is in charge of allaying or alleviate the impact of a product's anomalous exception.

Mitigate Resource Anomaly identify and coordinate mitigatory actions required for product's anomalous exceptions, including the initiation of remedial actions, and providing support to monitor and report on the mitigatory action.

Mitigate Resource Anomaly can ensure the effect of a detected anomalous exception is minimized or resolved through appropriate actions. The aim of this business activity is to reduce Resource Problem Management occurrence.

Coordinate Resource Exception Mitigation

Process Identifier: 1.5.21.4.2

Hierarchy Level: 4

Brief Description:

Coordinate Resource Exception Mitigation business activity is in charge of managing the different activities of an exception mitigation plan all together to successfully realize a mitigation plan.

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Extended Description:

Coordinate Resource Exception Mitigation business activity is in charge of managing the different activities of an exception mitigation plan all together to successfully realize a mitigation plan.

Identify Resource Exception Mitigation

Process Identifier: 1.5.21.4.1

Hierarchy Level: 4

Brief Description:

Identify Resource Exception Mitigation business activity helps to map product's anomalous exceptions with actions that will reduce impact to business.

Extended Description:

Identify Resource Exception Mitigation business activity helps to map product's anomalous exceptions with actions that will reduce impact to business.

Identifying Resource exception and their mitigatory action involves tasks that may leverage established business policies and constraints, and/or be based on knowledge and skills related to context.

This business activity helps to incorporate the right treatment based on the importance, impact or risk factors involved in a products operation. Any treatment of this nature can be static, based on pre-defined business rules, or dynamic based on skills and knowledge derived from context.

Initiate Resource Exception Mitigation

Process Identifier: 1.5.21.4.3

Hierarchy Level: 4

Brief Description:

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Initiate Resource Exception Mitigation business activity is in charge of starting the correct activities that mitigate Resource anomaly exceptions, cognizant of business requirement and goals.

Extended Description:

Initiate Resource Exception Mitigation business activity is in charge of starting the correct activities that mitigate Resource anomaly exceptions, cognizant of business requirement and goals.

Initiate Resource Exception Mitigation actions include recommendations for course of actions, fostering dynamic execution of actions, and/or scheduling actions based on predetermined business conditions and policies.

Monitor Resource Exception Mitigation

Process Identifier: 1.5.21.4.4

Hierarchy Level: 4

Brief Description:

Monitor Resource Exception Mitigation business activity is in charge of observing, checking and keeping a continuous record of mitigatory actions that support Resource Anomaly Management.

Extended Description:

Monitor Resource Exception Mitigation business activity is in charge of observing, checking and keeping a continuous record of mitigatory actions that support Resource Anomaly Management.

Monitor Resource Exception Mitigation covers the action and data that are involved (including collection and analysis on predefined set of information – e.g. SLOs, SLAs, SLI etc.) in Resource Anomaly Management business activities, tasks and workflows.

Report Resource Exception Mitigation

Process Identifier: 1.5.21.4.5

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Hierarchy Level: 4

Brief Description:

Report Resource Exception Mitigation business activity is in charge of providing an account (documented or otherwise) of mitigatory actions taken to lessen or reduce the impact of a Resource anomaly.

Extended Description:

Report Resource Exception Mitigation business activity is in charge of providing an account (documented or otherwise) of mitigatory actions taken to lessen or reduce the impact of a Resource anomaly.

Reporting on Resource Exception Mitigation may follow well-defined business and operations criteria, or predefined set of metrics (e.g. SLOs, SLAs, SLIs, OLAs etc.) that are underlying to mitigate Resource Anomaly Management business processes.

Manage Resource Anomaly Learning

Process Identifier: 1.5.21.5

Hierarchy Level: 3

Brief Description:

Manage Resource Anomaly Learning business activity is in charge of capturing, or acquiring knowledge and skill related to handled anomalous exceptions (from detection, through assessment to mitigation).

Extended Description:

Manage Resource Anomaly Learning business activity is in charge of capturing, or acquiring knowledge and skill related to handled anomalous exceptions (from detection, through assessment to mitigation).

Manage Anomaly Learning business activity ensure that knowledge and skills gained from Resource anomalies can aide future Resource Anomaly Management activities.

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Manage Resource Anomaly Case History

Process Identifier: 1.5.21.5.1

Hierarchy Level: 4

Brief Description:

Manage Resource Anomaly Case History business activity records all past information on Resource Anomaly events, Resource Anomaly Management activities and data underlying, and representing Resource Anomaly cases.

Extended Description:

Manage Resource Anomaly Case History business activity records all past information on Resource Anomaly events, Resource Anomaly Management activities and data underlying, and representing Resource Anomaly cases.

Manage Resource Anomaly Case History includes the maintenance of past and present records, all artifacts, knowledge-base (e.g. the specifications, profiles, exception patterns, action patterns etc.) and versioned information.

This business activity supports the organization to maintain skills and learning repository to identify and predict Anomalous Exception patterns and mitigatory actions.

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